Modular Drone Project

System Design and Development Document

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Project Overview

This document as a whole covers the system design and development of our robotics capstone project throughout the fall semester of 2023 and spring semester of 2024. This section of the document covers the introduction, problem statement, motivation, and a detailed description of our proposed solution.

Introduction

Our system, referred to as a modular drone platform or AIR (Aero-Intelligent Robot), is designed to serve as a versatile, single adaptable platform capable of efficiently executing a diverse range of tasks. Through the utilization of quick swappable drone attachments, referred to as modules, users can seamlessly swap tools, optimizing the drone's functionality for specific needs.

Problem Statement

Recognizing the limitations in the existing drone market, specifically the absence of a versatile, commercially available modular drone platform, we aim to fill this void. In a market projected to grow at a compound annual rate of 13.9% from 2023 to 2030 [1], we see a substantial opportunity for improvement in this sector. Our goal is to develop a system that gives users the flexibility to customize both the drone's software as well as the hardware and software of its modules. This approach directly addresses the prevailing shortcomings in the market, where users are often compelled to buy drones tailored for a singular purpose.

Motivation

Our project is motivated by three main visions. Firstly, we aim to establish a drone as a versatile research platform, enabling users to customize both hardware and software for testing new features, addressing the current lack of robust commercially available options. Secondly, we envision drones as everyday tools. We want to integrate them into practical tasks beyond their conventional industrial or hobbyist uses. Our goal is to bridge the gap between very specialized and general-purpose drones, unlocking their potential for a broader range of everyday functions. Lastly, we would like to mitigate the limitations of "unitaskers" which dominate the current drone market. We feel that there are many drones that are designed for singular purposes such as cinematography drones, and feel that this is a lack of a drone’s utilization. With the drone market anticipated to grow substantially in the coming years, we see room for improvement in this methodology in drone technology.

System Description

The modular drone platform is designed to replace weighty multipurpose drones or an array of specialized drones by allowing users to swap modules based on task requirements. The proposed modules, including the ARM Module, Science Payload, and SkyHook Module, offer a range of functionalities from robotic manipulation to environmental data collection. The system's baseline functionality includes object detection/tracking, GPS, SLAM, and takeoff/landing safety, ensuring consistent performance across modules.

Figure 1: System CAD for initial system visualization.

System Dimensions and Portability

The drone's physical dimensions are engineered to meet specific criteria, ensuring portability and versatility. It adheres to size constraints with a weight limit of 5 pounds and a compact frame measuring smaller than a 1-meter radius and 0.5 meters in height. This design focus on portability enables the drone to fit comfortably within a standard trunk with dimensions of approximately 1.2 meters. Additionally, the modular frame allows for easy assembly and battery replacement, enhancing the system's adaptability and user convenience.

Payload Capacity and Mission Diversity

The platform exhibits a comparatively large payload capacity, capable of carrying up to 10 pounds, including the attached modules or additional equipment. The purpose of the changeable modulus is to allow for adaptability to a diverse range of missions, showcasing the platform's multifunctional nature. Among the array of modules available, examples include the ARM (Aerial Robotic Manipulator) Module, empowering the platform to manipulate objects in remote or elevated locations, the SkyHook Module, facilitating controlled object raising and lowering, and specialized scientific modules capable of gathering essential data in various environmental settings. The adaptability afforded by these modules also extends to a multitude of practical tasks: one for example could be the efficient dispersal of seeds in agricultural fields through a seed dropping module. With a versatile assortment of modules, the platform can be configured to address a wide range of real-world challenges, allowing for success in diverse industries.

User Interface and Control System

The user interface of the platform is designed to offer an intuitive and simple experience. It consists of a screen-based device that allows users to interact with the system for mission planning, control, and monitoring. Depending on the mission requirements, users can switch between autonomous mode and manual flight mode. In autonomous mode, the platform can follow geographic fences specified by latitude and longitude coordinates, search designated areas, or navigate via predefined waypoints. In manual flight mode, operators can use a standard drone remote controller for direct control over the drone. This user-centric approach ensures that both experienced operators and new users can efficiently utilize the system for their specific needs.

Safety and Fail-Safe Mechanisms

Safety is paramount, and the system is equipped with several essential safety features. For instance, it incorporates a "Return to Home" function, which, in the event of signal loss or emergency situations, guides the drone back to its takeoff location. There is an “Emergency Landing” mode in which if “Return to Home” is not an option the drone will land as soon as possible. Additionally, an "Emergency Stop" mechanism is readily accessible, allowing operators to halt all drone functions instantly. Fail-safe protocols are in place to address unexpected contingencies, such as low battery levels or communication disruptions, ensuring the safe return and operation of the drone.

Development Environment and Customization

We will allow users to connect to the autonomy computer with the ability to adapt code within the robot. It is expected to provide users with the flexibility to write their own code, creating custom tasks and extending the system's functionality as needed. This open development approach enables users to adapt the platform to suit unique mission requirements, fostering innovation and adaptation.

Versatile Applications:

The platform's adaptability allows it to excel in a multitude of applications. Beyond the mentioned modules, it can undertake tasks such as environmental monitoring, infrastructure inspection, and disaster response. By extending the system's capabilities through module swapping and custom coding, users can address a wide array of real-world challenges across various industries, making this platform a valuable tool for a diverse range of applications. Some of the tasks we aim to complete with the drone include: localization, navigation, and delivery, which currently have much more expensive alternatives that we hope to address.

User Demographics

The main user groups for this product are the developers of the product and the end customers. There is no additional maintenance or installation support required. The developer team is comprised of the four authors of this document. The customers of the system will likely be experienced drone users. However, our system should be designed intuitively so that new users are also able to use the system without too much technical study.

System Requirements

This portion of the document covers the fundamental assumptions, requirements, scenarios, and use cases crucial for the successful implementation of our system. Throughout this section, we aim to establish a comprehensive foundation that guides the development and operation of the system.

Assumptions

System Assumptions

- The system will not require maintenance for at least a year after production.

Module Assumptions

- The system will be able to support module swapping without damaging the overall system.
- The system will be powered down before attempting to swap the modules.
- When using the ARM Module, the application will be non-hazardous to surrounding objects and people.
- When using the SkyHook Module, there will be human ground support to attach the payload to the drone.
- When using the Science Module, the sensors must be compatible with the rest of the system and relay information separately.

User Assumptions

- The user has used a technical system before and is able to understand the included Instruction Manual without additional support.
- The user is not using the system with malicious intent.
- The user will not need more than one module at a time.
- The user will not be frequently (loosely defined as more than 1 time per use session) swapping the modules.
- The user properly registers the system with the local government as required.
- The user will not use the drone in hazardous environments that would cause harm to the system.

Technical Requirements

The Technical Requirements are split into those for the Main Robot and the individual modules: ARM Module, SkyHook Module, Science Payload. The subsections and the requirements are sorted in order from highest to lowest priority.

Main Robot

R1 Robot must fly
- R1.1 Vertical takeoff
- R1.2 Rotate in place
- R1.3 Hover for 20 mins
- R1.4 200 meter range
- R1.5 5lb payload

R2 Robot must sense
- R2.1 Front, Above, and Below occupancy map
  - R2.1.1 Depth map update at 30 fps (2m range)
- R2.2 360° obstacle detection (10m range)
- R2.3 Map creation
- R2.4 Global position (1.5m accuracy)

R3 Robot must be capable of autonomous operation
- R3.1 Autonomously stabilize with all accessories
- R3.2 Autonomously hold position
  - R3.2.1 Height
  - R3.2.2 Orientation
  - R3.2.3 Position relative to local environment
- R3.3 Fly to set position
  - R3.3.1 Return to start
- R3.4 Avoid obstacles

R4 Robot must plan
- R4.1 Path planning with real time obstacle avoidance

R5 Robot must communicate with operator
- R5.1 Serial radio link
  - R5.1.1 Emergency stop (Return to Land)
  - R5.1.2 Manual flight commands
  - R5.1.3 Returns basic status information
- R5.2 Wifi connection
  - R5.2.1 Receive waypoint coordinates
  - R5.2.2 Advanced flight commands
  - R5.2.3 Returns advanced status, maps, logs
- R5.3 <1 hr operator training time

R6 Robot must respond quickly

- R6.1 >30hz depth map updat
- R6.2 >100mbps wifi communication
- R6.3 >5hz gps update
- R6.4 >30mph flight speed
- R6.5 <3m stopping distance
- R6.6 >5hz path planning
- R6.7 >1hz map update

R7 Robot must have modular accessory interface

- R7.1 Provide power
- R7.2 Provide control commands for specific actuators
- R7.3 Save log / scientific data
- R7.4 Provide mounting provisions

R8 Robot must comply with federal and local regulations

R9 Robot must perform in described weather conditions

- R9.1 Temp (-5°C, 40°C)
- R9.2 Wind <15 mph
- R9.3 Wet ground but no precipitation
- R9.4 Daylight/external light (30-100,000 lux)

R10 Robot must not disturb environment

- R10.1 Less than 65db of noise at 20ft away
- R10.2 Does not damage surroundings

R11 Robot must be portable

- R11.1 Weight less than 3000 g
- R11.2 Smaller than 1 m diameter and .5 m tall
- R11.3 Frame must be modular
- R11.4 Swappable battery

R12 Robot must save map and log information locally

ARM Module

R13 Must be directly controlled by robot computer

R14 Must be able to grasp a variety of objects

- R14.1 Various Objects up to 80mm
- R14.2 Cylinders up to 80mm
- R14.3 Up to 2 lbs

R15 Must extend .5m outside of robot frame

R16 Must accurately position

- R16.1 <10mm error from set position relative to robot frame

R17 Arm must respond quickly

- R17.1 <2s travel time to stable position from initial command

SkyHook Module

R18 **Must be directly controlled by robot computer**
R19 **Must be able to lower 100ft**
R20 **Must be able to position with 1m**
R21 **Must be able to hold objects 3.5lbs**
R22 **Requires user to clip hook**

Science Payload

R23 **Robot must measure from an external source**
R24 **Must integrate with robot computer**

System Requirements

Similar to the Technical Requirements, the System Requirements are organized from highest to lowest priority.

Maintainability Requirements

SR1 Necessary training and drone operation best practices is required
SR2 General assembly practices known for minor repairs
- SR2.1 Drone is able to be dis- and re- assembled using standard tools
- SR2.2 Spare parts for "minor repairs" can be easily acquired by users

SR3 Company is ready to maintain and repair drones for large scale operations.
- SR3.1 Critical components are stored in surplus in house for ready and quick access
- SR3.2 For major repairs, maximum downtime is 2 weeks

SR4 Drones are monitored frequently by users
- SR4.1 Drones have a self-diagnosis procedure to help report this to users
  - SR4.1.1 This includes a battery health monitor specifically, among other monitors

SR5 Drone software is easy to upload, and the company supplies updates to continuously optimize drone processes.

Reliability Requirements

SR6 Including hardware replacements and overall maintenance, system will be expected to last 2 years
- SR6.1 Failure Modes are executed as soon as problems are detected during operation
  - SR6.1.1 Issues are also reported to user

SR7 Users follow drone best practices to prolong drone lifespan and operation duration
SR8 Over 4 weeks of company imposed downtime is unacceptable
SR9 Drone monitors and logs system health for review and failure anticipation
- SR9.1 Users can opt to automatically report failure logs to cloud, for company wide drone improvement

Scenarios: System Modes and States

Startup Mode

- **Description**: This is the initial mode when the system is powered on.
- **States**:
  - **Initialization**: The system initializes essential components, such as sensors, communication systems, flight controllers, and motors.
  - **Power-on Self-Test**: State that allows the system to perform diagnostic tests to ensure all components are functioning correctly and together.
  - **Standby**: The system is ready for user input but is not in flight
  - **Start-up Error**: The system is not ready for user input and requires action before any operation can be performed.

Flight Modes

- **Description**: These modes define the various states of in-flight operation.
- **States**:
  - **Take-Off**: The system initializes and will lift to a designated height if safe to do so.
  - **Manual Flight**: User-piloted mode where the user had full control over the drone's movement.
  - **Autonomous Flight**: The system operates autonomously, following pre-set flight paths, or responding to mission commands.
  - **Hover Mode**: The system remains at a fixed position and altitude
  - **Emergency Obstacle Avoidance**: The system maneuvers as immediate action is necessary to avoid a collision.
  - **Return to Home**: The system returns to the takeoff position either by user choice or in case of emergency.
  - **Safety Hold**: When safety thresholds are exceeded (e.g., high winds), the system maintains stability and lowers to the ground.
  - **Landing**: The system brings itself safely to the ground and lands within a designated area.

Module Modes

- **Description**: States related to module configuration on the system.
- **States**:
  - **Module Attachment**: The system is prepared to accept a new module, providing power and communication connections.
  - **Module Detachment**: The system releases a module, disconnecting power and communication.

- **Module Self-Test**: State that allows the system to perform diagnostic tests to ensure the module is functioning and communicating with the system.
- **Manuel Module Operation**: User-controlled mode where the user had full control over the system's module.
- **Autonomous Module Operation**: The system operates autonomously, following preset actions, or responding to mission commands.

Maintenance and Diagnostics Mode

- **Description**: States for maintenance, troubleshooting, and checks.
- **States**:
  - **Diagnostic Check:** The system runs internal diagnostics to identify any hardware or software issues.
  - **Maintenance Mode:** The system enters a mode where users can perform routine maintenance, including checking and replacing components.

Emergency Modes

- **Description**: States activated in response to critical situations
- **States**:
  - **Emergency Landing**: The system initiates a controlled descent and landing in case of severe issues.
  - **Emergency Stop**: An immediate halt to all system functions in response to a critical safety concern.\

Shutdown Mode

- **Description**: The final mode when the drone is powered down.
- **States**:
  - **Safe Shutdown:** The drone powers down all systems in a controlled manner, ensuring safety and data integrity.

Use Case Diagram

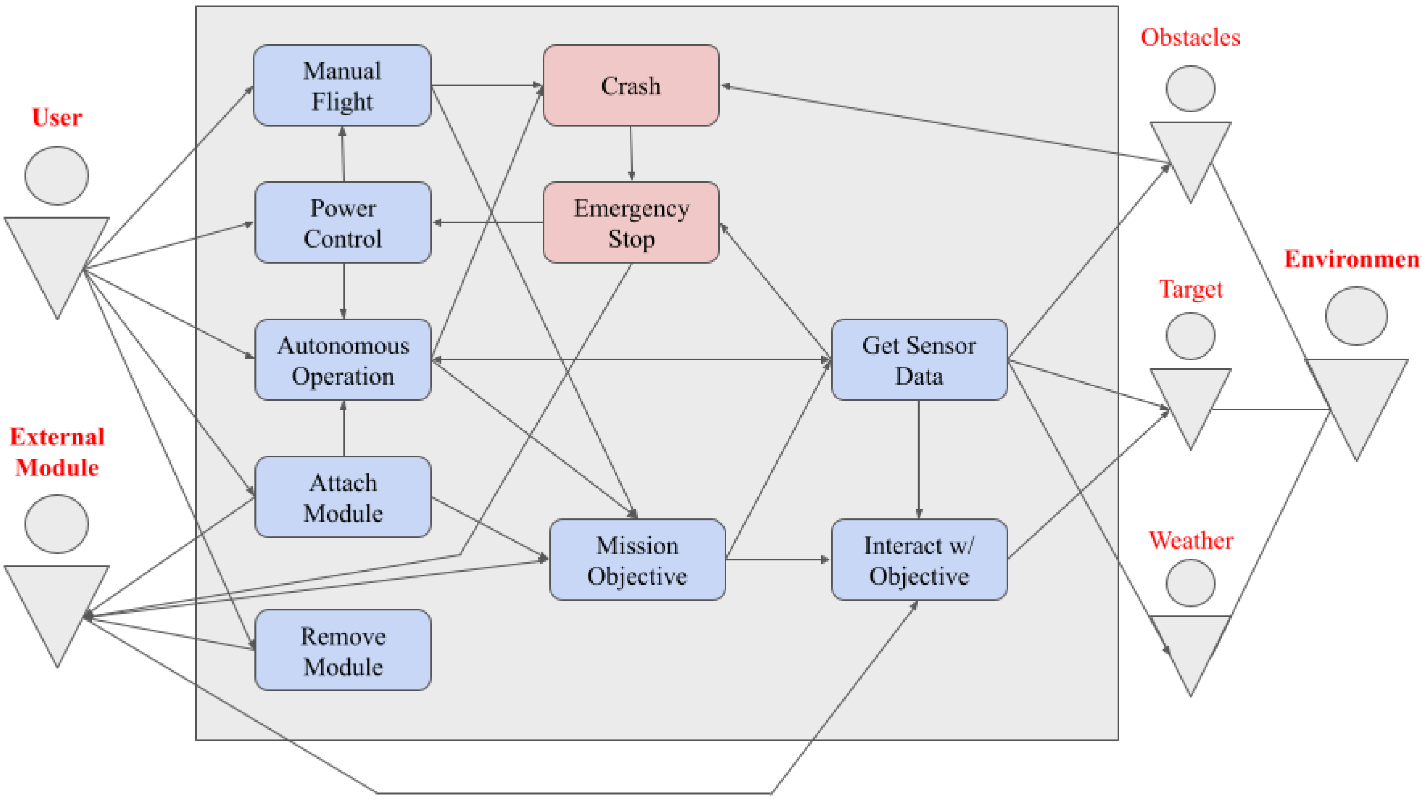


Figure 2: Use Case Diagram for System*.

*Blue boxes are typical use cases where red boxes are negative use cases.

Concept Design

This segment of the document serves to create a conceptual framework for the development of our modular drone platform. Through the concept design, the following topics will be explored: the physical form through computer-aided design (CAD), the concept of operation, subsystems, software architecture, operational scenarios, alternative considerations, and prototyping timeline.

Physical Form CAD

Scaled Drawings

Below are scaled drawings created in CAD for the drone and its modules. This is to provide a more detailed view of function and form of each module and the base.

Drone w/ ARM Module

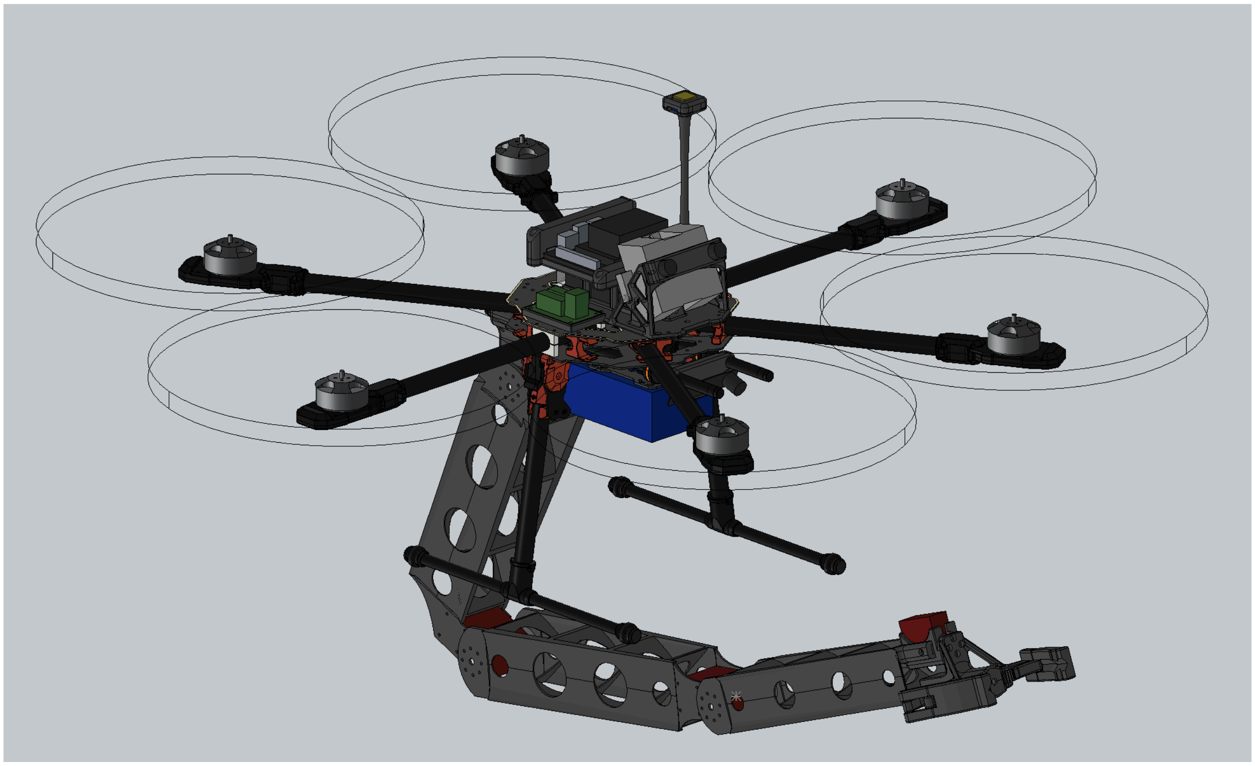


Figure 3: Drone with ARM Module extended to complete a mission task.

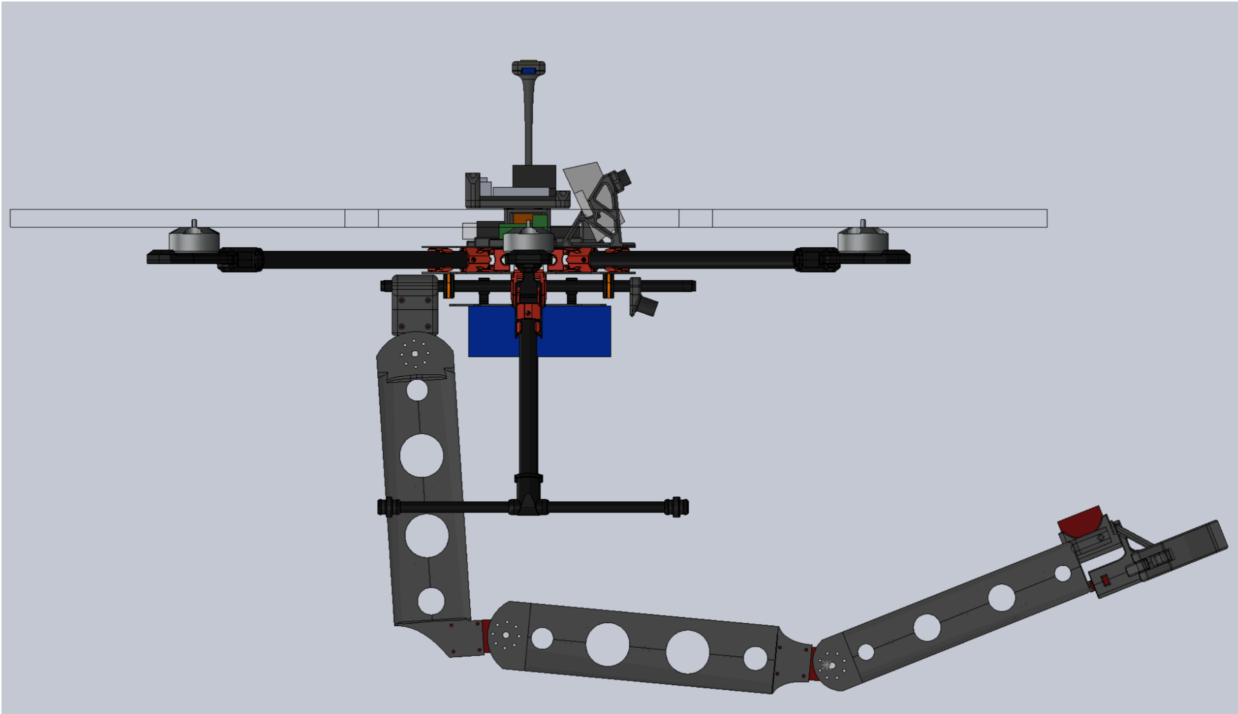


Figure 4: Side-view of Drone with ARM Module extended to complete a mission task.

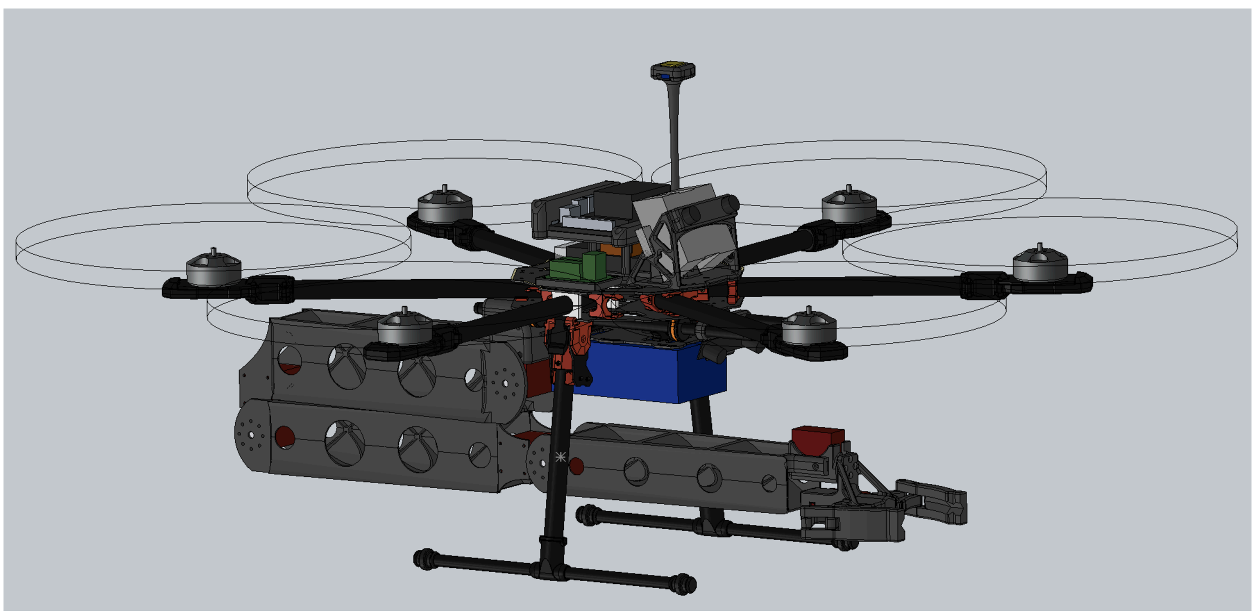


Figure 5: Drone with ARM Module positioned in the in-transit configuration.

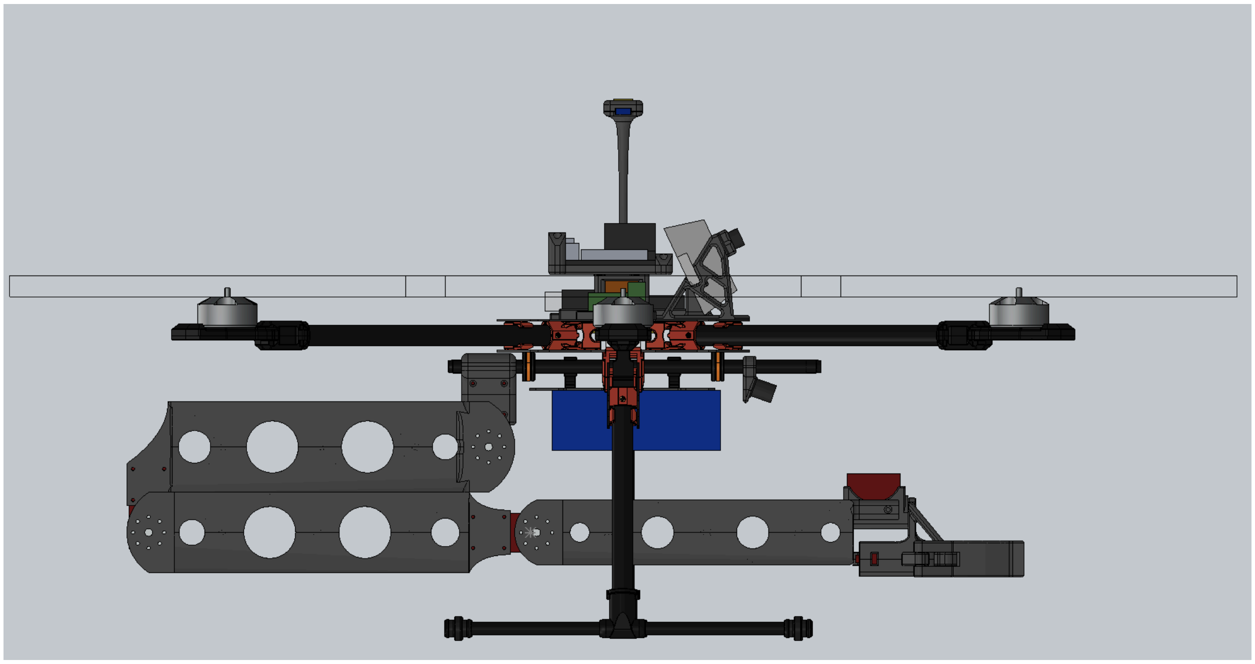


Figure 6: Side-view Drone with ARM Module positioned in the in-transit configuration.

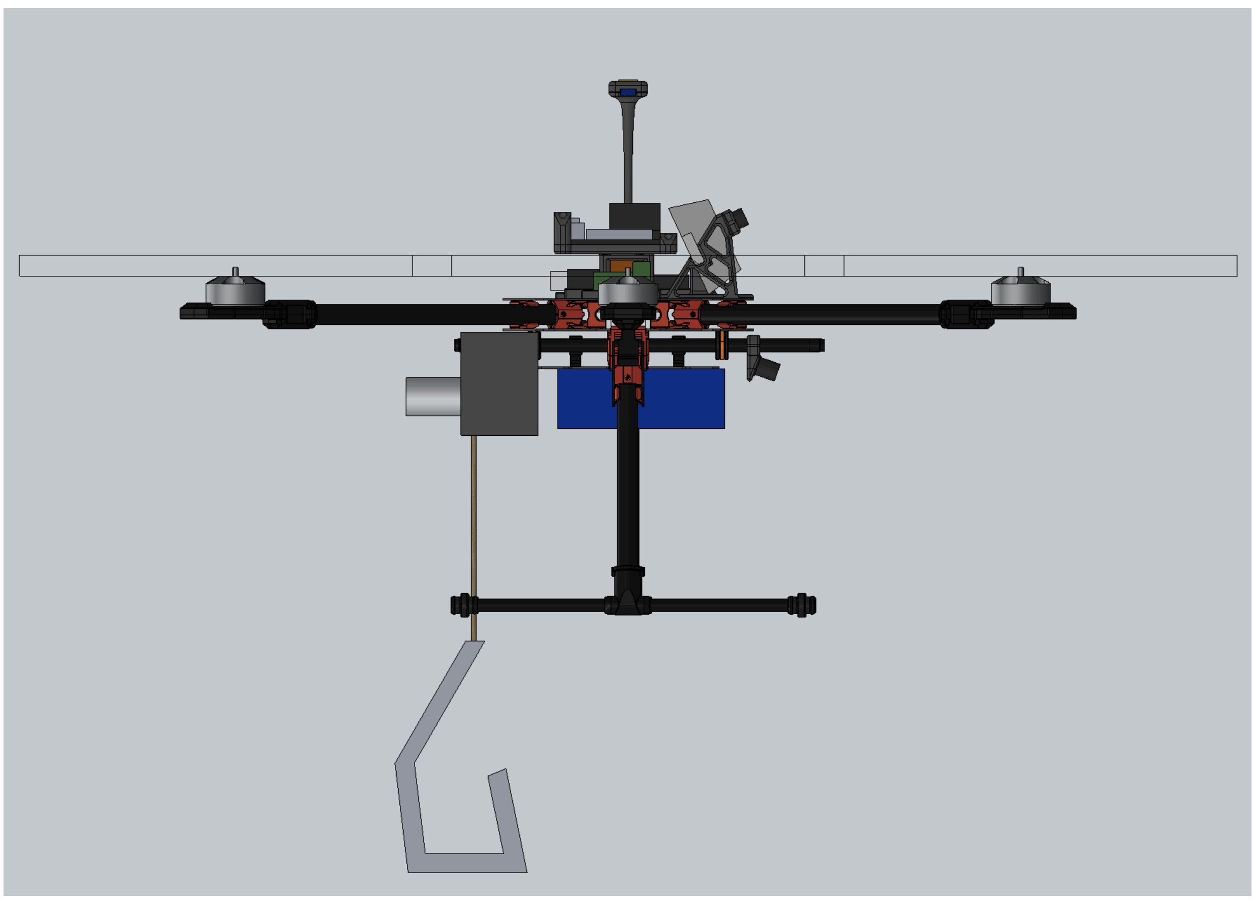


Figure 7: Side-view Drone with Sky Hook Module

Concept of Operation

The operation of our modular drone platform is based on its natural flexibility and easy-to-use design. The system is engineered to offer a versatile solution for users, catering to a broad array of tasks while ensuring an efficient and straightforward workflow.

At the core of its operation is the concept of module-based adaptability. Users can swap modules to tailor the system's functionality to their specific mission requirements. Whether it's a task that demands precise object manipulation, data collection, or delivery services, the user can easily select the suitable module, attach it to the drone, and complete the task.

The operation of the platform can be delineated into several key steps:

Module Selection

Users choose the module that best suits the intended task. Modules range from the Robotic Arm Module for hands-on tasks to specialized scientific modules for data collection.

This design decision can be traced to R7.

Attachment

The selected module is securely attached to the drone's modular frame. The system's design allows for simple and secure attachment, making it accessible to a wide range of operators.

These design decisions can be traced to R6, R7, and R8.

Mission Planning

Users can plan their missions using the platform's user interface. Depending on the task, they can specify geographic boundaries, waypoints, or specific commands to achieve the mission objectives.

These design decisions can be traced to R1-R6 and R8-R12.

Deployment

The user will deploy the drone powering on both the transmitter and drone. The user will also ensure that the rotors are secured to the drone, the drone has a stable connection to the transmitter, and the system diagnostic and flight readiness tests are positive.

These design decisions can be traced to R1-R6 and R8-R12

Autonomous or Manual Operation

The user interface allows for two operation modes. In autonomous mode, the platform can follow predefined mission plans, waypoints, and navigate within specified geographic boundaries. In manual mode, users can take direct control using a standard drone remote controller. Once one of these is selected the drone is ready for takeoff and mission completion.

This design decision can be traced to R3 and R5.

Safety and Monitoring

Safety is a top priority, and the system incorporates features like "Return to Home" for safe navigation in case of signal loss or emergencies. Users can also monitor the platform's status and mission progress in real-time through the user interface.

This design decision traces back to R1 and R5. The System Requirements on maintainability (SR1-SR5) and reliability (SR6-SR9) can also be traced from this design because safety is one of our primary concerns.

Mission Completion

The platform executes the mission as per the user's instructions, leveraging the chosen module's capabilities. Once the mission is complete, users can safely detach the module for future use.

The concept of operation is user-centric and adaptable, enabling operators with varying levels of expertise to efficiently utilize the system for their specific tasks. The modular approach provides flexibility, allowing users to configure the platform for a diverse range of real-world applications. Whether it's performing hands-on work with the robotic arm, gathering data with scientific modules, or conducting precision deliveries, the system offers a comprehensive solution tailored to the needs of multiple industries and mission types.

Subsystems

Subsystem Connectivity

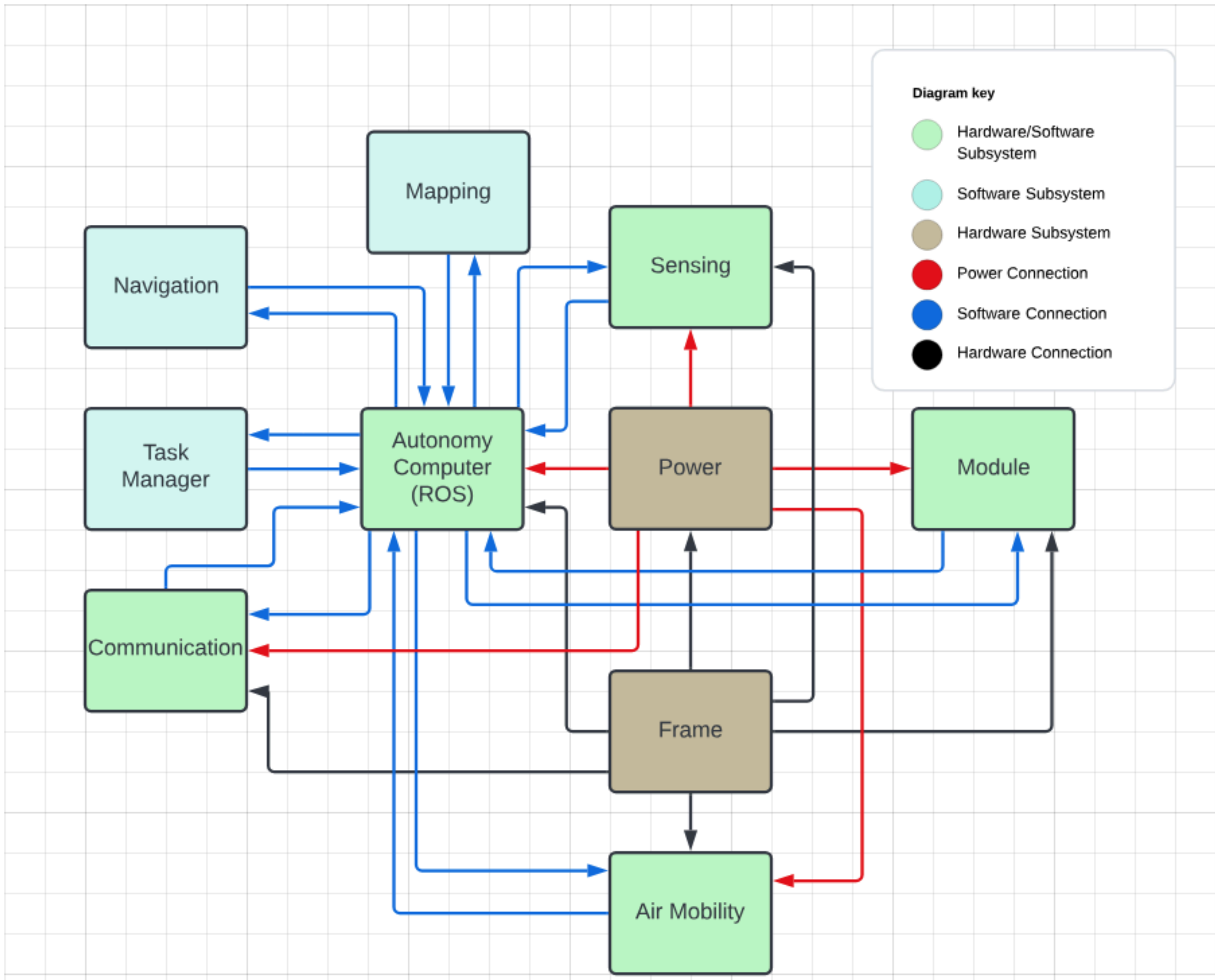


Figure 8: Subsystem Connectivity Diagram

Frame (Hardware)

The frame subsystem performs the task of holding all hardware components together. Other important tasks of the frame subsystem include:

- Regulating temperature of components
- Wire routing/ management
- Damping oscillations for important components

Parts List:

- Tarot 680 Drone Frame Kit
- GPS mount (3D Printed)
- Autonomy Computer mount (3D Printed)
- Flight Computer Mount (3D Printed)
- Serial Receiver Mount (3D Printed)
- Electronic Speed Controller (ESC) Mount x 6 (3D Printed)

Mounting (Hardware)

The mounting subsystem, now a new subsystem, performs the task of securing the module, battery or batteries, and lower stereo camera to the drone. This system is vital to the success of our system being that COM issues have been raised and the need for quick swappable modules are paramount. These parts mount directly to the base plate of the drone and will replace the existing mounting provisions of carbon fiber tubes and mounts.

Parts List:

- Center Module Mount (3D Printed)
- Rear Module Mount (3D Printed)
- Forward Module and Stereo Mount (3D Printed)
- Reusable Cinch Straps 1" x 12"
- M2.5 Nuts, Bolts, and Washers

Power (Hardware)

The power subsystem performs the task of supplying power to all components. It is responsible for power source (battery) as well as power regulation, filtering, and distribution to all components.

Parts List:

- 24v-12v Power Supply
- Flight controller / 24v-12v Power Supply
- 12V Power Distribution Board
- 6s 10000mah Lion Battery

Air Mobility (Hardware / Software)

The Air Mobility subsystem performs the task of providing lift and manipulating the orientation of the drone. Air mobility achieves these tasks using a dedicated flight computer. The flight computer takes in angle and throttle commands from the autonomy computer. The flight computer then sends signals to electronic speed controllers (esc) that control the speed of each motor. Consequently the motors spin propellers and exert forces on the system. The flight controller also takes in data from internal accelerometers (IMU) and gyroscopes in order to run a feedback loop and ensure the subsystem reaches the desired specifications.

Parts List:

- Cube Orange Flight Controller
- MAD 4008 Motor x6
- Electronic Speed Controller (ESC0 x 6
- Proplers x 6

Module (Hardware / Software)

The Module subsystem performs a different task depending on what module is being used. In general the module will receive requests from the autonomy computer and will need to perform actuations and/or return data to the autonomy computer. It should be consistent that the module is responsible for the majority of the control implementation allowing the autonomy computer to send simple high level requests.

The ARM Module subsystem performs the tasks of moving to specific locations and grabbing and placing objects. This module will receive goal position commands along with a grab or place action from the autonomy computer. The module will compute all required inverse kinematics and motion trajectories to position the arm. The module will update the autonomy computer with its position and end effector state.

The SkyHook Module subsystem performs the task of deploying and raising the hook to pick up objects with the help of a user. The module takes in simple commands of raise and lower.

The science instrument module performs the task of reporting the acquired data to the autonomy computer when requested.

Parts List:

- ARM Servo Motor x 3
- ARM links x 3
- ARM End Effector
- SkyHook Motor
- SkyHook Winch and cable
- Science Payload

Autonomy Computer / ROS (Hardware/ Software)

The autonomy computer subsystem performs the task of providing power and I/O for sensors and auxiliary devices and connecting software modules using ROS.

Parts List:

- Jetson Orin Nano
- 500gb NVME SSD
- USB to TTL signal converter

Communication (Hardware / Software)

The communication subsystem performs the task of supplying pathways for the user to send and receive data from the drone. One main component is a wifi card and antennas that allow for the streaming of video such as RViz outputs. In addition the communication subsystem includes a wireless serial link between the user and drone. Both the autonomy computer and flight computer are attached to this computer resulting in a variety of high or low level control and safety options for the operator.

Parts List:

- Serial Radio Receiver
    - Connects the joystick transmitter to the flight controller
- Wifi Router
    - For the base station to connect to the drone

Sensing (Hardware / Software)

The sensing subsystem performs the task of recording processing data from various sensors to be used by other subsystems. This subsystem is responsible for providing depth information for all directions of the drone. A majority of the depth information will be gathered from depth maps generated by stereo cameras. In addition, sensing will provide gps information.

Parts List:

- ELP Stereo Camera x 2
- SAM M10Q GPS

Mapping (Software)

The mapping subsystem performs the task of fusing sensing data as well as imu and accelerometer data from the flight computer to perform SLAM. The outcome is a live map of the drones position and the surrounding point cloud from all current and past sensor readings.

Navigation (Software)

The navigation subsystem performs the task of generating a path to a given waypoint and calculating commands for the air mobility subsystem. This subsystem takes in a goal position from the task manager subsystem as well as the current local and global maps and plans a path that can be executed. The system is also responsible for converting the path into a series of motion and angle commands that can be sent to the flight controller for motion.

Task Manager (Software)

The task manager subsystem performs the task of linking subsystems together and synchronizing operations. This subsystem has a direct input from the communication subsystem and can then direct messages to navigation, module, or flight controller.

Software Architecture

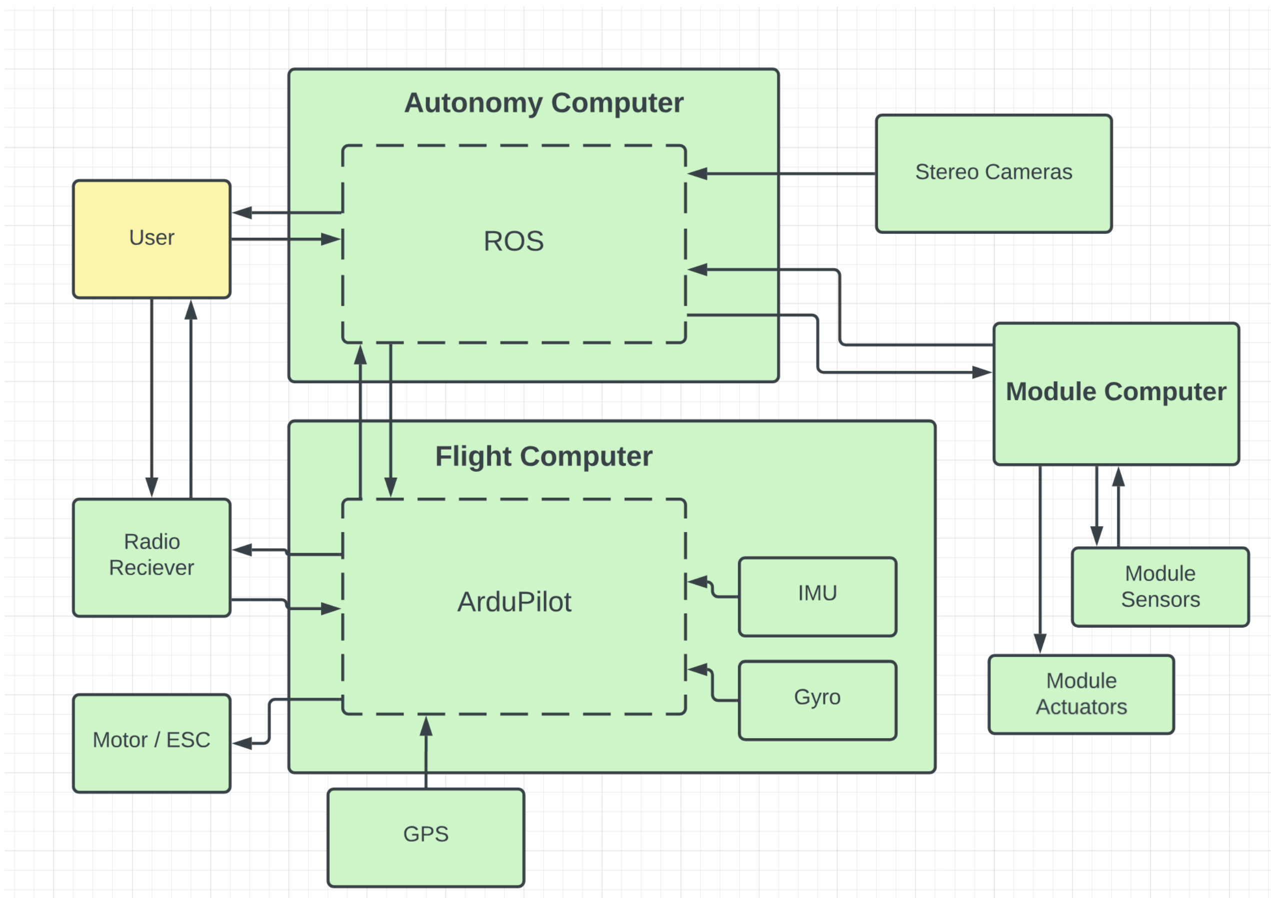


Figure 9: High Level Software Architecture Diagram

Autonomy Computer

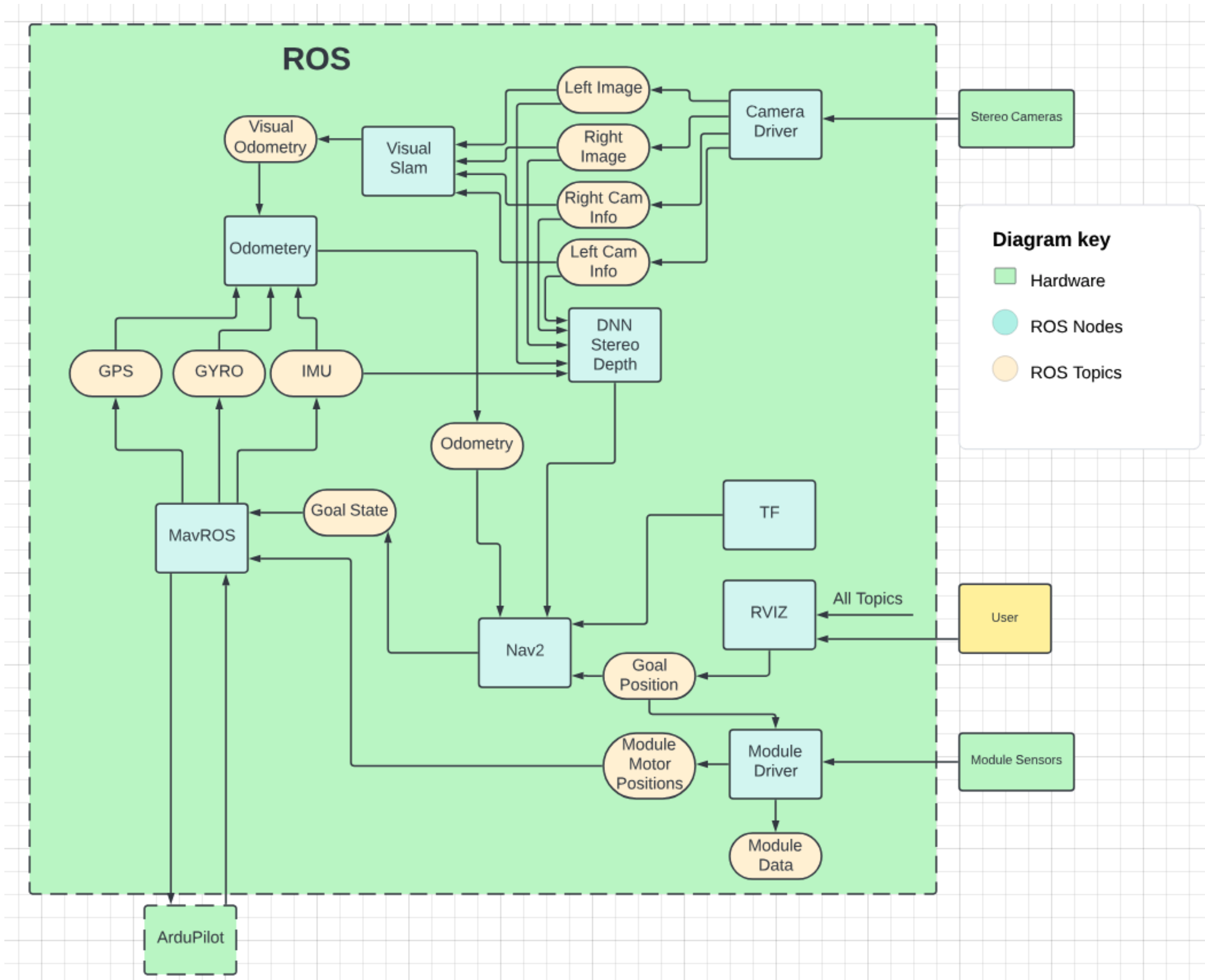


Figure 10: ROS Software Diagram on Autonomy Computer

The Autonomy computer software includes the following Subsystems:

Autonomy Computer (ROS) which links all the Nodes and Topics. The Sensing subsystem which includes the Camera Driver, Visual Slam, and DNN Stereo Depth Nodes. The communication subsystem which includes the RVIZ Node. The Mapping subsystem which includes the Odometry Node. The navigation subsystem which includes the Nav2 Node. The Task Manager subsystem which includes RVIS, MavROS, and Module Driver Nodes. And finally the module subsystem which includes the Module Driver Node.

ROS Software will be developed and compiled directly on device. The device has a wifi module so either ssh or an external monitor/mouse/keyboard can be used.

Flight Computer

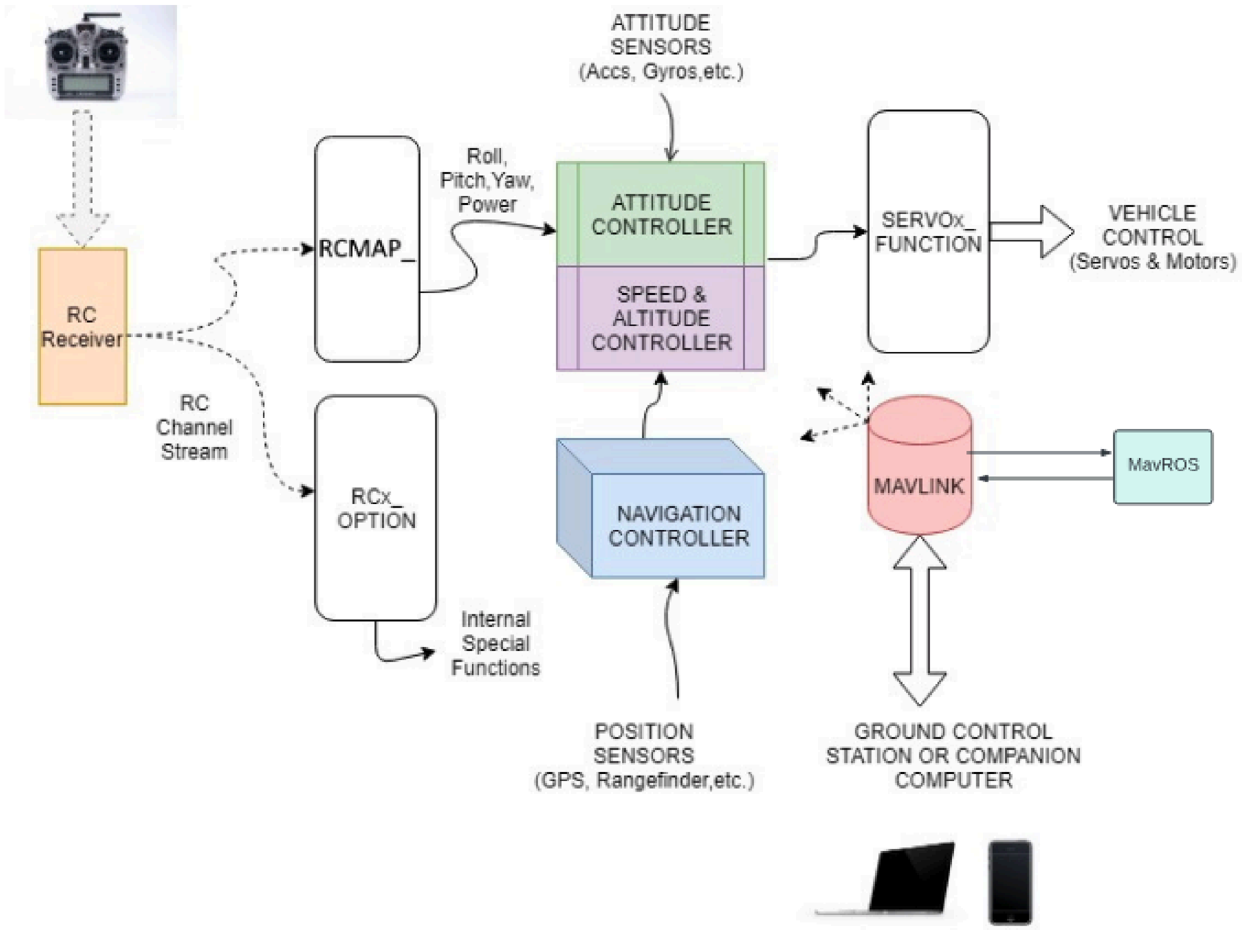


Figure 11: Ardupilot Software Architecture on Flight Computer.

The Flight Computer software is the main component of the Air Mobility Sub-System. It is built around the Ardupilot software. The Ardupilot software uses a standard messaging protocol called Mavlink that can be interfaced with ROS directly. In addition we will be implementing custom attitude and speed controllers to better compensate for the loads caused by the modules.

Flight Controller software will be written and compiled on personal computers and then flashed to the computer with a USB cable.

Operation Scenarios

Start Up:

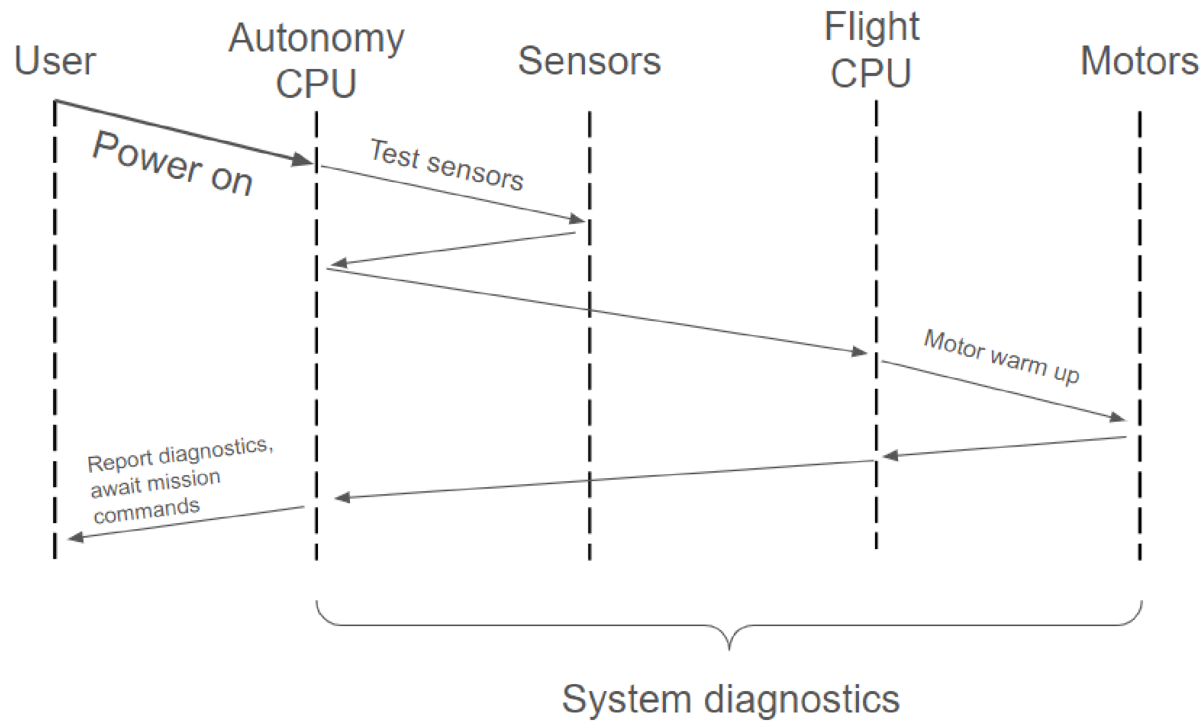


Figure 12: Scenario Diagram for System Start Up

After the user powers the drone on, the autonomy computer tests sensors and flight computer. The flight computer begins warming up motors assuming nothing wrong is detected, and initial diagnostics are reported to the operator as the drone awaits mission commands.

Autonomous Flight:

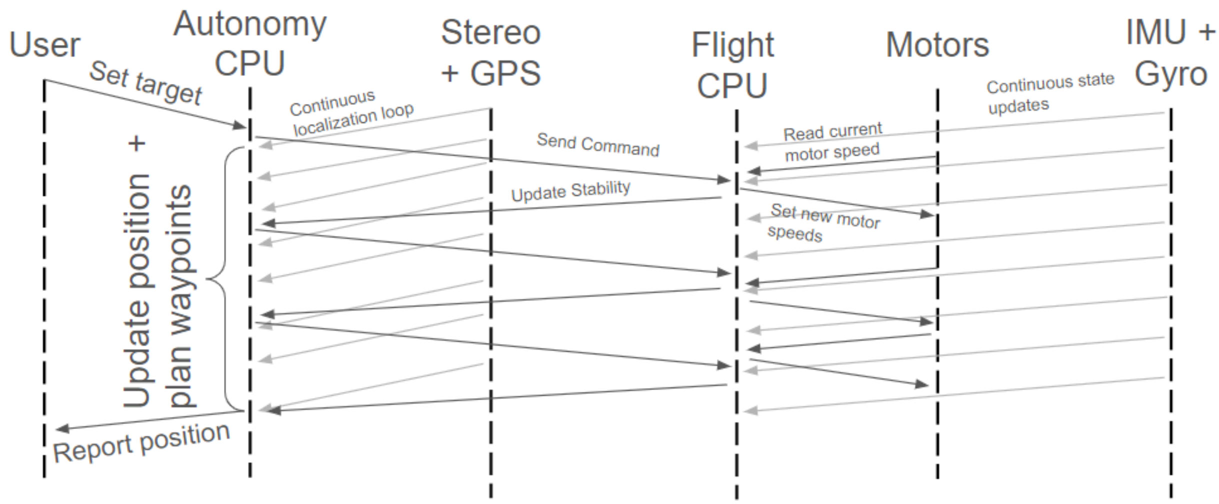


Figure 13: Scenario Diagram for Autonomous Flight Process

After the operator sets a target for the drone, the autonomy computer uses the stream of localization information to calculate the current location and plan waypoints for the drone. The next waypoint is sent to the flight controller which continuously intakes motor speeds and IMU and gyro readings to update its states, and adjusts them to output new motor speeds. The drone's stability is then reported back to the autonomy computer.

ARM Module Operation:

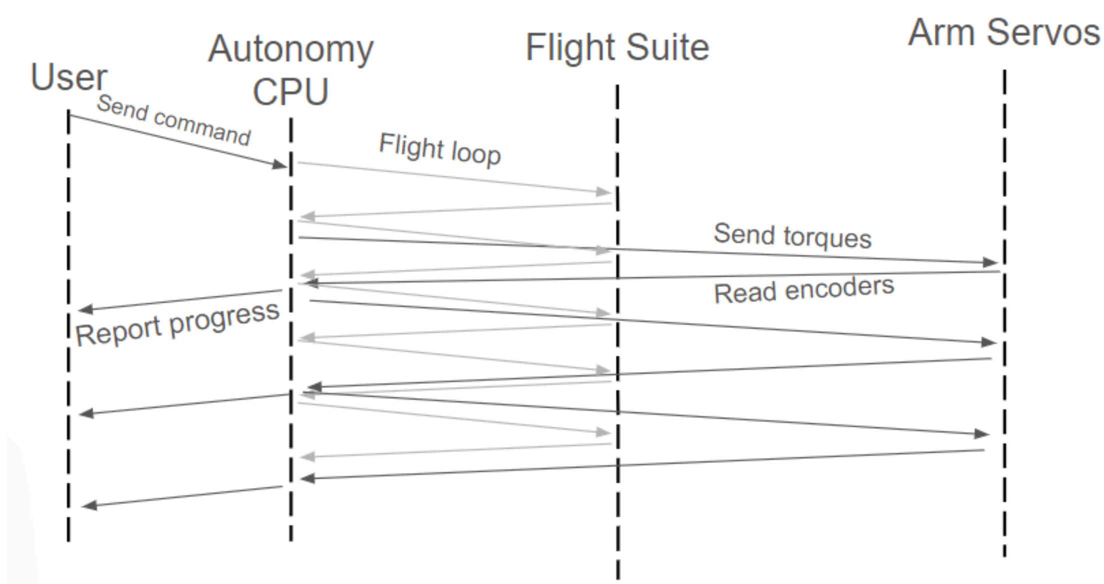


Figure 14: Scenario Diagram for ARM Module Operation

While the autonomy computer is in communication with the flight suite of computers, sensors and motors, the operator can send a command to pick up an object. The autonomy computer computes the inverse kinematics required to pick up the object and sends the appropriate torques to the ARM servos, who's encoder readings are reported back to the computer.

Alternatives Considerations

The drone has a number of design considerations we needed to address before proceeding with the drone design, most of which hinged on the question of deciding whether to have an indoor or outdoor functioning drone.

Propellers

After considering between 4 and 6 propellers, we went with 6 because of the increased carrying capacity, and overall controllability as well as the benefit of having some built in redundancy if one of the motors failed.

This design decision can be traced to R1.

Module Mounting

There are several potential ways to mount the module onto the drone in a manner that is both secure but also makes the module interchangeable. We initially pondered bolting the module to the base, but ultimately decided on using rails to slide the module onto the drone with set screws to hold it in place.

This design decision can be traced to R7.

Sensor Type

There were a variety of sensors that we considered using for localization and obstacle detection, but we mainly considered Lidar vs a stereo camera suite. After drafting the following trade study, we settled on the stereo camera suite.

This design decision can be traced to R2.

Table 1: Trade study to identify appropriate sensor for the system.

Sensor Type Trade Study				
	Weights	**Lidar**	**Stereo**	**Time of Flight**
Cost	-5	9	9	1
FOV	10	3	9	1
Weight	-2	3	1	1

Software Complexity	-4	1	9	1
Resolution	8	3	9	1
FPS	7	1	3	3
Documentation	2	3	3	3
Range	6	9	3	1
Accuracy	10	3	3	3
Mounting	1	9	3	9
		105	157	79

Computation Options

There were several computation alternatives we chose between, including a singular computer executing all tasks, two computers to offload processing power needed for flight specifically, or adding an additional purely module computer to handle all module operations. We ultimately settled on the two computer options as the module computation can be handled by the shared computation between the autonomy and flight computers.

This design decision can be traced to R2-R7.

Prototyping Timeline

In order to have a functional product by the end of the production timeline, we will create several larger prototypes accompanied by some subsystem level prototypes as needed.

Subsystem Level Prototypes

There are several subsystems that need prototypes to confirm their functionality before they can be integrated into a larger system. Having a proof of concept will allow us to check and scale efficiency and correctness as needed to meet our overall requirements.

We plan to build prototypes for the following subsystems within the next few months. We will primarily use Jira to track the progress of each prototype.

Table 2: Outline of key subsystems with their associated expected completion dates.

Subsystem	Expected Complete Date	Completed?
Module Interface Node	12/15/2023	Y
Module: ARM	3/10/2024	N
Module: SkyHook	3/10/2024	N

Mapping	2/15/2024	N
Navigation	2/25/2024	N

System Level Prototypes

Once the subsystem level prototypes are validated via the technical and system requirements, we will integrate several subsystems at a time into our system to make sure the system is robust. The rough timeline order of our assembly is as follows:

- AIR_v0: Initial frame with Power, Air Mobility, and Modular Interface node. All operations can be teleoperated at this stage. The Communications subsystem should be integrated at this point so that testers can read raw output. Science Payload can be attached as a simple module.
- AIR_v1: Drone should include autonomous operation through integration of Mapping, Navigation and Sensing subsystems.
- AIR_v2: Drone should be able to meet all technical requirements with the ARM Module attached.
- AIR_v3: Drone should be able to meet all technical requirements with the SkyHook Module attached.
- AIR_v4: Drone should be ready for the final stage of production.

Work Breakdown Structure and Schedule

Our team utilized JIRA as the software to organize the work breakdown structure of our project. We have populated our system via level 1 - 4 tasks, assigning an estimated amount of time to complete the task, an expected start and end date of the task, and an assignee to the task. We attempted to structure these in an intelligent way to allow us to be done early in the case that there are unforeseen delays in any portion of our project. The below figures will show our Jira interface and Gantt Chart Schedule which will help our team be successful and efficient with our work and time. A detailed version of the charts are detailed in the Appendix.

Work Breakdown Structure

The work breakdown structure (WBS) is decomposed into higher level subsystems which are broadly assigned to each group member to monitor. The original estimate represented the total time to complete each subsystem based on the time estimates of each task within the subsystem.

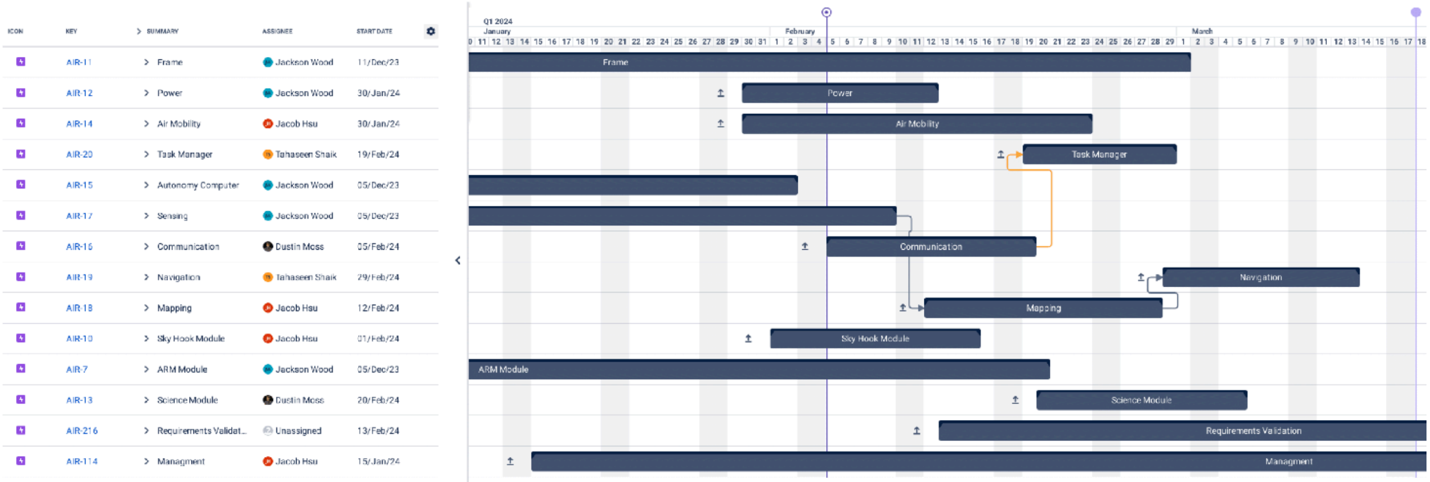
ICON	KEY	SUMMARY	STATUS [CHILDREN STAT...	ASSIGNEE	EPIC LINK	ORIGINAL ESTIMATE
	AIR-11	Frame	22.2% 22.2% 55.6%	Jackson Wood		1w 4d
	AIR-12	Power	40% 0% 60%	Jackson Wood		1w
	AIR-14	Air Mobility	55.6% 0% 44.4%	Jacob Hsu		2w
	AIR-20	Task Manager	100% 0% 0%	Tahaseen Shaik		1w
	AIR-15	Autonomy Computer	0% 0% 100%	Jackson Wood		3d
	AIR-17	Sensing	44.4% 11.1% 44.4%	Jackson Wood		1w 4d
	AIR-16	Communication	100% 0% 0%	Dustin Moss		1w 3d
	AIR-19	Navigation	100% 0% 0%	Tahaseen Shaik		1w 1d
	AIR-18	Mapping	100% 0% 0%	Jacob Hsu		2w
	AIR-10	Sky Hook Module	75% 25% 0%	Jacob Hsu		1w
	AIR-7	ARM Module	38.5% 53.8% 7.7%	Jackson Wood		3w
	AIR-13	Science Module	100% 0% 0%	Dustin Moss		1w 3d
	AIR-216	Requirements Validation	100% 0% 0%	Unassigned		3w 3d
	AIR-114	Managment	100% 0% 0%	Jacob Hsu		1w 1d

Shown in the appendix is a more detailed WBS, in which each subsystem (or epic) contains broad tasks or phases associated with each subsystem such as its design, build or implementation, and testing, each of which are filled with lower level tasks assigned to each group member. For example, the WBS for the Frame subsystem is depicted below:

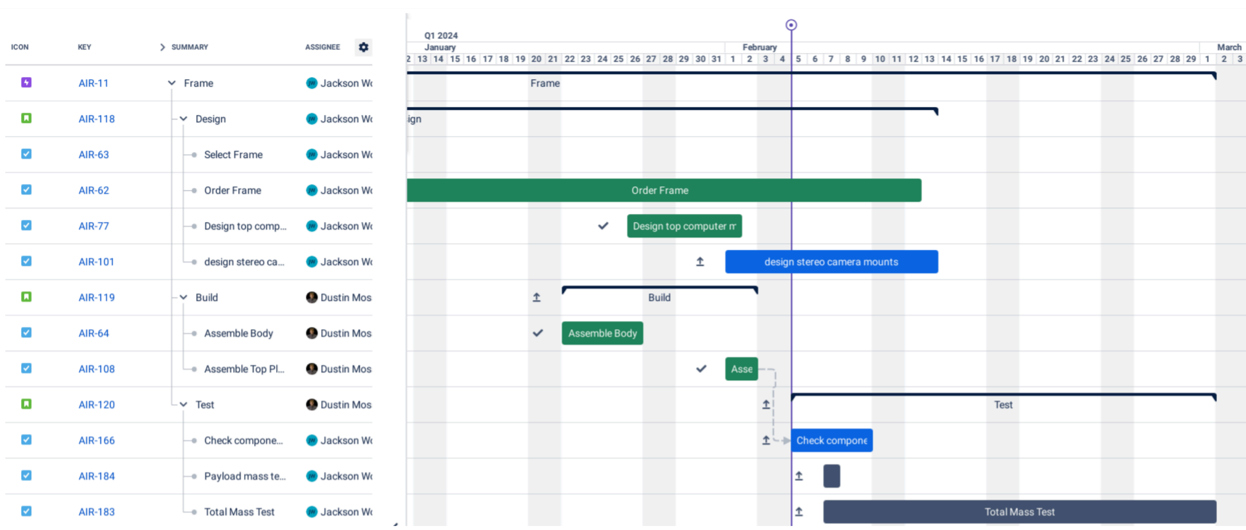
ICON	KEY	SUMMARY	STATUS [CHILDREN STAT...	ASSIGNEE	EPIC LINK	ORIGINAL ESTIMATE
	AIR-11	Frame	22.2% 22.2% 55.6%	Jackson Wood		1w 4d
	AIR-118	Design	0% 25% 75%	Jackson Wood	FRAME	1w
	AIR-63	Select Frame	DONE	Jackson Wood	FRAME	1d
	AIR-62	Order Frame	DONE	Jackson Wood	FRAME	1d
	AIR-77	Design top computer mo...	DONE	Jackson Wood	FRAME	2d
	AIR-101	design stereo camera m...	IN PROGRESS	Jackson Wood	FRAME	1d
	AIR-119	Build	0% 0% 100%	Dustin Moss	FRAME	2d
	AIR-64	Assemble Body	DONE	Dustin Moss	FRAME	1d
	AIR-108	Assemble Top Plate	DONE	Dustin Moss	FRAME	1d
	AIR-120	Test	66.7% 33.3% 0%	Dustin Moss	FRAME	1d
	AIR-166	Check component interf...	IN PROGRESS	Jackson Wood	FRAME	1d
	AIR-184	Payload mass test	TO DO	Jackson Wood	FRAME	1h
	AIR-183	Total Mass Test	TO DO	Jackson Wood	FRAME	1h

Schedule

From the WBS, we then created a Gantt chart based on the estimated time to complete each task and general dependencies and deadlines of the project.



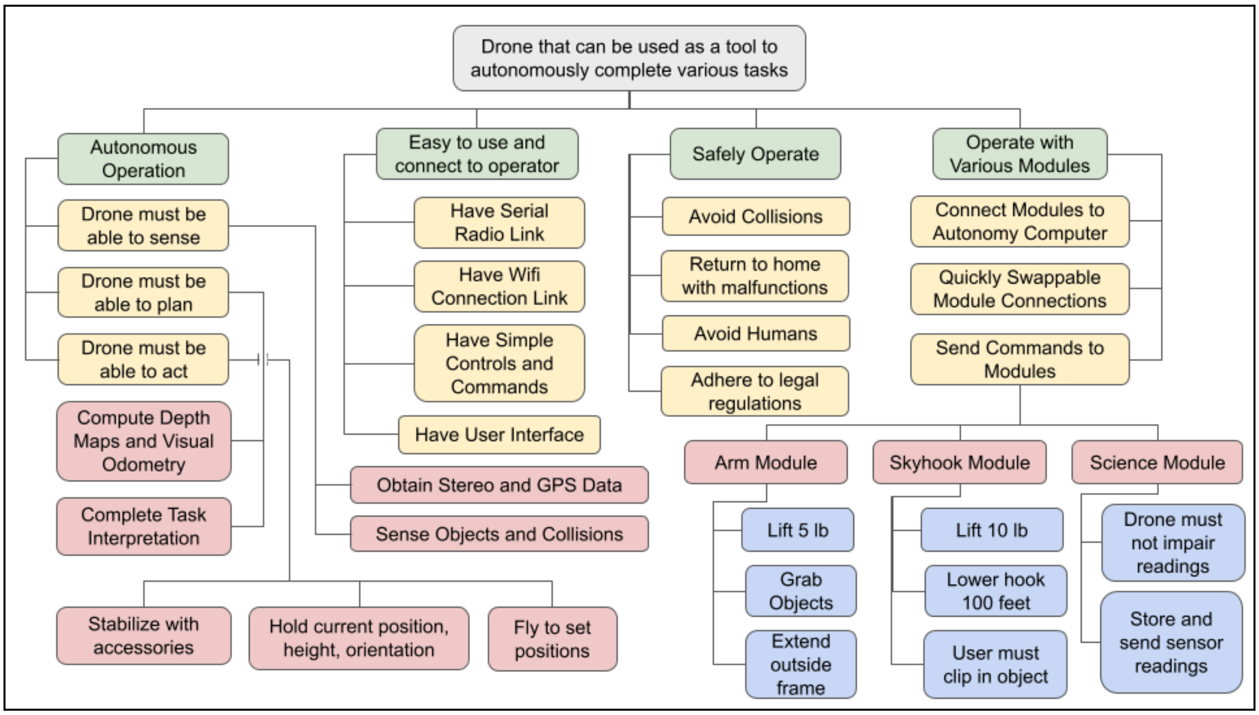
This chart is a high level depiction of the general outline for working on each subsystem, where the subsystem's duration and start times are based on the estimated time to complete individual tasks as shown in the schedule for the Frame subsystem below:



The tasks are color coded such that green represents completed tasks, blue represents tasks in progress and gray represents unstarted tasks. Again, a full schedule is shown in the Appendix.

Objective Tree

Below is the representation of our objective tree. We simplified our main objective to be a: “Drone that can be used as a tool to automount complete various tasks” This was split off into 4 main sub-objective which then continued to split off until we believe we reached verifiable goals. This tree will aid in answering questions such as what needs will the system meet as well as what is its purpose.



Overall we want our drone to work as an efficient tool that will allow the operator to complete complex or hard tasks with ease. The idea is that we want to showcase how a single reliable drone platform with Autonomy, CV, Path Planning, etc. is capable of completing a diverse set of tasks successfully.

Detailed Design

This section outlines the key aspects of the Detailed Design for our modular drone platform. This will cover trade studies, part lists, system processes, concept of operation, operation plan, and usage and safety guidelines. This section forms a foundational framework for subsequent development phases.

Trade Studies

Our first trade study was on what flight controller to use. We found the Pixhawk 6x and Cube Orange + to be very similar systems in most regards. The biggest decider was the larger amount of documentation available for the Cube. In addition the Cube has a much larger community that has developed many software packages that should make software setup easier.

Table 3: Trade study to determine the flight controller [2].

Flight Controller Trade Study				
	Weights	**Holybro F7**	**Pixhawk 6x**	**Cube Orange +**
Cost	-5	1	9	9
Clock Rate	2	1	9	9
Ports	4	1	9	9
Software integration	8	1	3	9
hardware Integration	6	1	3	3
Sensor Accuracy	9	3	9	9
Sensor Redundancy	3	3	9	9
Documentation	10	1	3	9
		61	189	297

Our next trade study was for the autonomy computer. We found the Jetson Orin Nano to be the best choice mainly due to its significantly superior computing power. This power is essential for the stereo image processing we are planning to implement and offsets the additional cost.

Table 4: Trade study to determine the autonomy computer [2].

Autonomy Computer Trade Study				
	Weights	**Raspberry Pi**	**Jetson Nano**	**Jetson Orin Nano**
Cost	-5	1	3	9
Clock Rate	9	1	3	9

Ports	4	3	3	3
Hardware Integration	6	1	3	3
Available Packages	7	3	3	9
Development Ease	5	1	3	9
Documentation	10	1	9	3
		58	168	204

Next we wanted to perform a trade study on what type of sensor we should predominantly use. We decided on using stereo cameras. The main downside of stereo cameras is the significantly greater software complexity and compute required. However stereo produces the best resolution and has the greatest field of view (FOV). Based on our assessment these benefits outweighed the costs and made stereo cameras the best sensor for our project.

Table 5: Trade study to determine the sensor type.

Sensor Type Trade Study				
	Weights	**Lidar**	**Stereo**	**Time of Flight**
Cost	-5	9	9	1
FOV	10	3	9	1
Weight	-2	3	1	1
Software Complexity	-4	1	9	1
Resolution	8	3	9	1
FPS	7	1	3	3
Documentation	2	3	3	3
Range	6	9	3	1
Accuracy	10	3	3	3
Mounting	1	9	3	9
		105	157	79

Now that we have decided to use stereo cameras we need to now select a particular model. We ultimately decided on using the ELP Stereo camera that outputs synchronized stereo pairs over USB. The Oak-d Lite had the advantage of doing some of the vision processing on board but since we already are going to be using a fast autonomy computer these features were not worth the extra cost. We also considered using 2 individual Arducam but this presented many issues with synchronization as well as limited us to only using one stereo pair given the amount of video parts we have on the Jetson Orin Nano.

Table 6: Trade study to determine the stereo camera.

Stereo Camera Trade Study				
	Weights	**Oak-d Lite**	**ELP Stereo**	**2x Arducam**
Cost	-3	9	1	9
Weight	-1	9	3	1
Software Complexity	-4	1	3	9
Resolution	4	3	1	9
FPS	8	1	3	3
Documentation	2	3	3	3
Mounting	1	9	3	1
		-5	19	3

Our next trade study was to determine what type of batter to use. The first question was if we preferred LiPo or LION battery type. It seems that our system will run sufficiently efficiently that we will not need a high discharge rate and therefore Lion batteries are preferred due to their higher power density. Next we had to decide between making our own batteries versus buying them. We decided that the extra dev time was worth saving money especially because we are interested in the process of making our own batteries.

Table 7: Trade study to determine the battery.

Battery Trade Study				
	Weights	**LION**	**LIPO**	**Custom LION**
Cost	-4	9	3	1
Weight	-6	3	9	3
Discharge Rate	2	1	3	1
Size	-4	3	3	3
Dev Time	-4	1	1	9
Configurability	1	1	1	9
		-67	-75	-59

We performed a trade study to determine the optimal motor for our system. This trade study mostly boiled down to what motor met our thrust requirements as efficiently as possible. This resulted in the MAD 4008 motor being selected

Table 8: Trade study to determine the motors.

Motor Trade Study				
	Weights	**2806.5**	**4008**	**5008**
Cost	-6	1	3	9
Weight	-2	1	3	9
Thrust	7	1	3	9
Amp Draw	-5	9	1	3
Prop limiting	-8	1	0	0
		-54	-8	-24

We looked at three propeller types. The props ended up having very similar performance so we chose the Eolo because it was the cheapest, lightest and was designed for reduced noise.

Table 9: Trade study to determine the sensor type.

Propeller Trade Study				
	Weights	**FIUZER 1304.4**	**Eolo 1305**	**T-motor 1302**
Cost	-3	9	1	1
Weight	-2	3	1	3
Max Load	4	1	1	1
Durability	5	1	3	1
Efficiency	-4	1	1	9
Noise	-6	3	1	3
		-46	4	-54

When looking at electronic speed controllers (ESC) we wanted a device with the best update rate that was capable of handling our loads and was the cheapest. The luminier blheli_32 30 Amp seems the best option given these criteria.

Table 10: Trade study to determine the ESCs.

ESC Trade Study				
	Weights	**blheli s**	**lum_32 30Amp**	**Iflight_32 80Amp**
Cost	-5	1	3	9
Weight	-3	1	3	9

Curent	4	1	3	9
Telemetry	2	1	3	1
Speed	10	1	9	9
		8	84	56

We chose between 3 GPS modules. The M8N and M8P modules have been released for a long time and have cases and mounting options built for them. In addition the M8P provides the potential for RTK functionality by using an additional gps base station which adds significant complexity. We ended up choosing the M10Q which is the latest GPS module from U-blox. We believe the increase in accuracy from the newer model as well as the much lower price point makes this device the best option for our system.

Table 11: Trade study to determine the GPS.

GPS Trade Study				
	Weights	**M10Q**	**M8N**	**M8P + RTK Base**
Cost	-3	1	3	9
Weight	-2	1	3	9
Accuracy	4	3	1	9
Mounting	5	1	1	3
Power	-4	1	1	3
Setup Complexity	-6	1	1	3
		2	-16	-24

We chose between two sizes of a commercial frame as well as building a custom frame. We found that a custom frame was not worth the extra development time and would likely be more expensive than an off the shelf frame anyways. We then chose the smaller of the two frames as it was much cheaper and still provided our required lift parameters. In addition this specific frame has a built in power distribution board (PDB) that makes electrical setup easier.

Table 12: Trade study to determine the system frame.

Frame Trade Study				
	Weights	**Tarot 680**	**Tarot 960**	**Custom**
Cost	-3	1	9	3
Weight	-2	1	9	3
Portability	4	1	3	3
Component Mounting	5	1	1	3
Power	4	9	1	1

Module Mounting	4	1	1	9
Max Thrust	3	1	9	1
Dev Time	-8	1	1	9
		39	-1	-17

Parts List

Table 13: List of parts and associated costs for the system.

		Quantity	Mass (g)	Cost ($)	Voltage (V)	Amps (A)	Total Watts (W)	Total Mass (g)	Total Cost ($)
Frame Sub-System									
	Tarot 680 Frame	1	600	150			0	600	150
	GPS Stand	1	20	1			0	20	1
	Top Mount Plate	1	64	1			0	64	1
	Cube Retaining Plate	1	23.2	1			0	23.2	1
	Nano Retaining Bars	1	18.1	1			0	18.1	1
	Stereo Cover	2	5	1			0	10	2
	Bottom Stereo Mount	1	15	1			0	15	1
	Assorted Hardware	1	20				0	20	0
Air Mobility Sub-System									
	Cube Orange +	1	73	350	5	2.5	12.5	73	350
	Tarot 4108 Motor	6	93	37.25	24	68	9792	558	223.5
	Luminear 32A ESC	6	6	20	24	0.34	48.96	36	120
	Eolo Propellers	6	14	7			0	84	42
	Servo Wire	1	5	15.8			0	5	15.8
	Servo Wire Pins	1	1	12.69			0	1	12.69
ARM Module Sub-System									
	Annimos 150kg Servo	3	193	55	12	8	288	579	165
	Annimos 25Kg Servo	2	70	30	5	2.1	21	140	60
	ARM End Effector	1	100	5			0	100	5
	ARM link 1	1	114	5			0	114	5
	ARM link 2	1	105	5			0	105	5
	ARM link 3	1	95	5			0	95	5
	Seeduino Controller	1	10	10			0	10	10
	Current Sensor	1	5	4			0	5	4
Power Sub-System									
	Battery	2	1108	137.5	24	0.01	0.48	2216	275
	24v-12v dc converter	1	258	23	12	0.6	7.2	258	23
	24v-5v dc converter	1	5	5	5	0.4	2	5	5
	cube orange supply	1	14	0			0	14	0
	Addition Motor Wire	1	30	16			0	30	16
	12v PDB	1	29	10			0	29	10
Communication Sub-System									
	ELRS Radio Reciever	1	10	23	5	0.005	0.025	10	23
Sensing Sub-System									
	ELP Stereo Camera	2	15	100	5	0.25	2.5	30	200
	SAM M10Q GPS	1	8	41	5	0.013	0.065	8	41
Autonomy Sub-System									
	Jetson Orin Nano	1	176	500	12	1.25	15	176	500
	NVME Storage	1	4.5	45	12	0.5	6	4.5	45
	USB to TTL	1	6	10			0	6	10
TOTAL							10195.73	5455.8	2317.99

System Processes

Fabrication Process

The main body of the robot is the drone frame. We have decided to purchase the Tarot T680 drone frame kit. In order to attach components to the frame we will be 3D printing custom mounting plates for all the critical components. Many of these 3D printed components will require post processing including sanding and inserting heat set thread inserts.

Assembly Instructions

1. Electrical

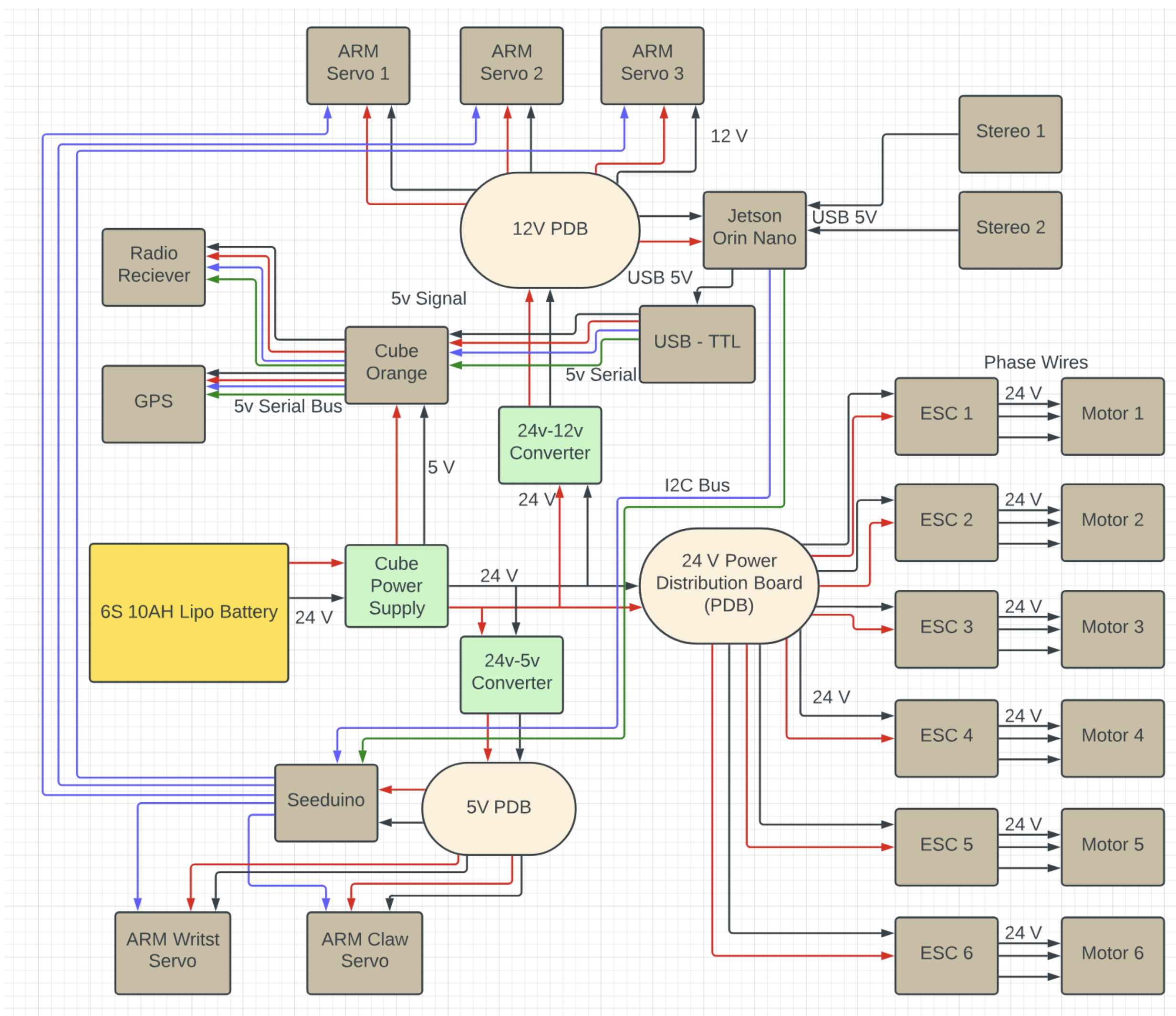


Figure 15: Electrical system diagram.

1.1. Prepare the top plate of the Tarot 680 frame kit for use as the 24v power distribution board (PDB).
1.2. Solder the main power input lead to the input on the PDB. Also solder the power wires for each electronic speed controller (ESC) to the PDB spot for the respective motor. Solder the motor wires to the ESC pads.
1.3. Solder the input to the 24v-12v dc converter to the main power pads. Connect the output to the 12v PDB. Connect the cube orange power supply XT60 in line with the main power input lead. Connect the output to the power 1 port on the Cube board.
1.4. Connect the jetson nano to the 12v PDB. Connect the main ARM servo motors to the 12v PDB.
1.5. Plug the both stereo camera usb to the jetson nano. Plug the USB to TTL adapter to the Jetson usb. Connect the TTL output to the Cube Telem 2 port.
1.6. Connect the FS-iA6B Radio Receiver to the RC In port on the Cube
1.7. Connect the ARM motor wires to the Aux out ports on the Cube.
1.8. Battery Build
    1.8.1. Use the Battery Holders to place the batteries in vertically in 2 rows and 6 columns
    1.8.2. Use the nickel strips and a spot welder to attach pairs of batteries in series.
    1.8.3. Solder a balance wire to each pair.
    1.8.4. Connect all 6 pairs in parallel.
    1.8.5. Solder the main power wire to the output terminals of the parallel pack.
    1.8.6. Wrap the battery with the wrapper and heat shrink to fit.

2. Mechanical

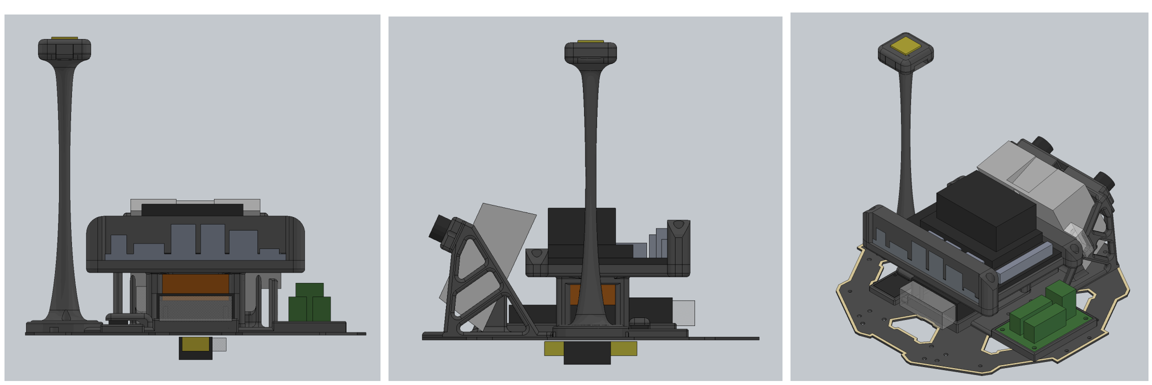


Figure 16: Top Mount Plate With Components

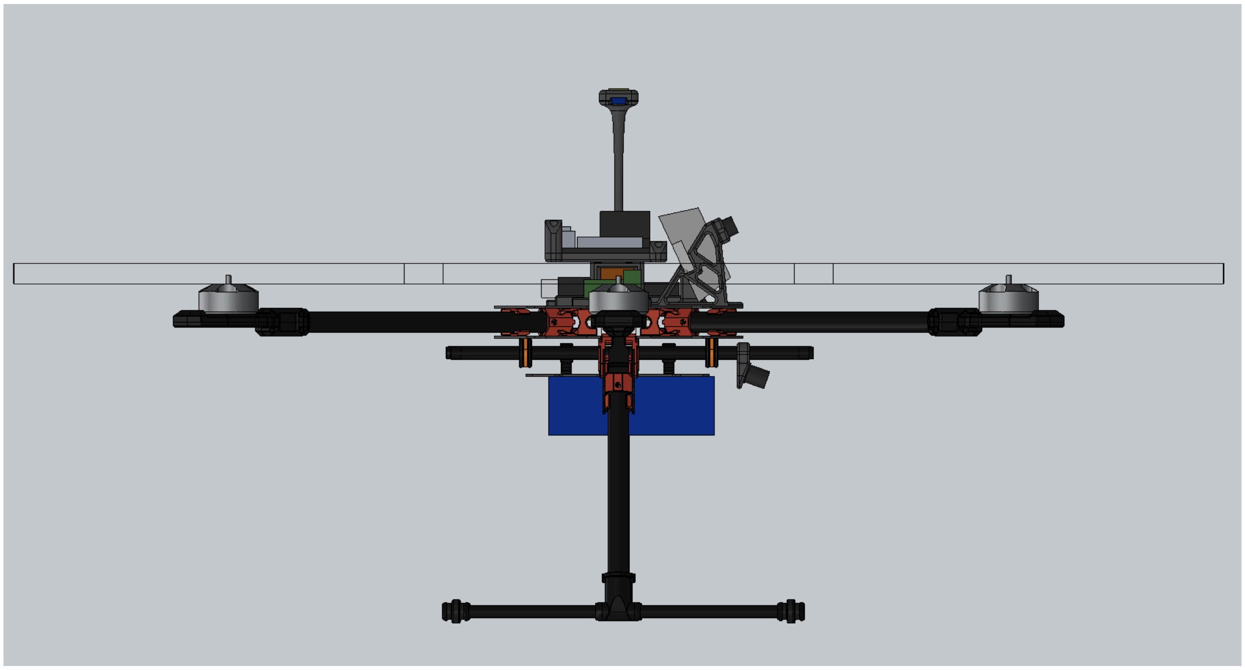


Figure 17: Mechanical system diagram side view.

2.1. Assemble the Tarot 680 drone frame according to instructions. Take care when installing the top plate to not pinch any wires.
2.2. Mount the 24v-12v dc converter to the top of the top plate.
2.3. Mount the motors to the frame using the main mounting holes at the end of each arm. Mount the ESC to the underside of each arm using the 3D printed esc mount clips.
2.4. Mount the Cube to the top plate
2.5. Mount the Jetson to the top plate
2.6. Mount the 24-12v to the top plate
2.7. Mount the 12v PDB to the top plate
2.8. Mount the 24-5v to top plate
2.9. Mount the SAM M10Q gps to the top of the GPS post and mount the GPS post to the top plate
2.10. Mount the FS-iA6B Radio Receiver to the bottom plate using the receiver mount part. Heat shrink the antennas to the receive mount arms.
2.11. Mount the two stereo cameras using the stereo mount plates for the top and bottom frame plates.
2.12. Attach the battery to the battery plate using the battery velcro straps.
2.13. Slide ARM attachment to the Module rails and secure in place using the set screws.

Software Modules & Interactions

Our software architecture is distributed between two different computers and is outlined in Figure 18. The autonomy computer handles all high level control. This includes stereo image processing, mapping, path planning, and navigation. The flight controller handles all lower level control such as stabilization and all motor control. The flight controller also provides GPS, accelerometer, and gyro data. The two computers are connected by a serial bus allowing for real time data transfer.

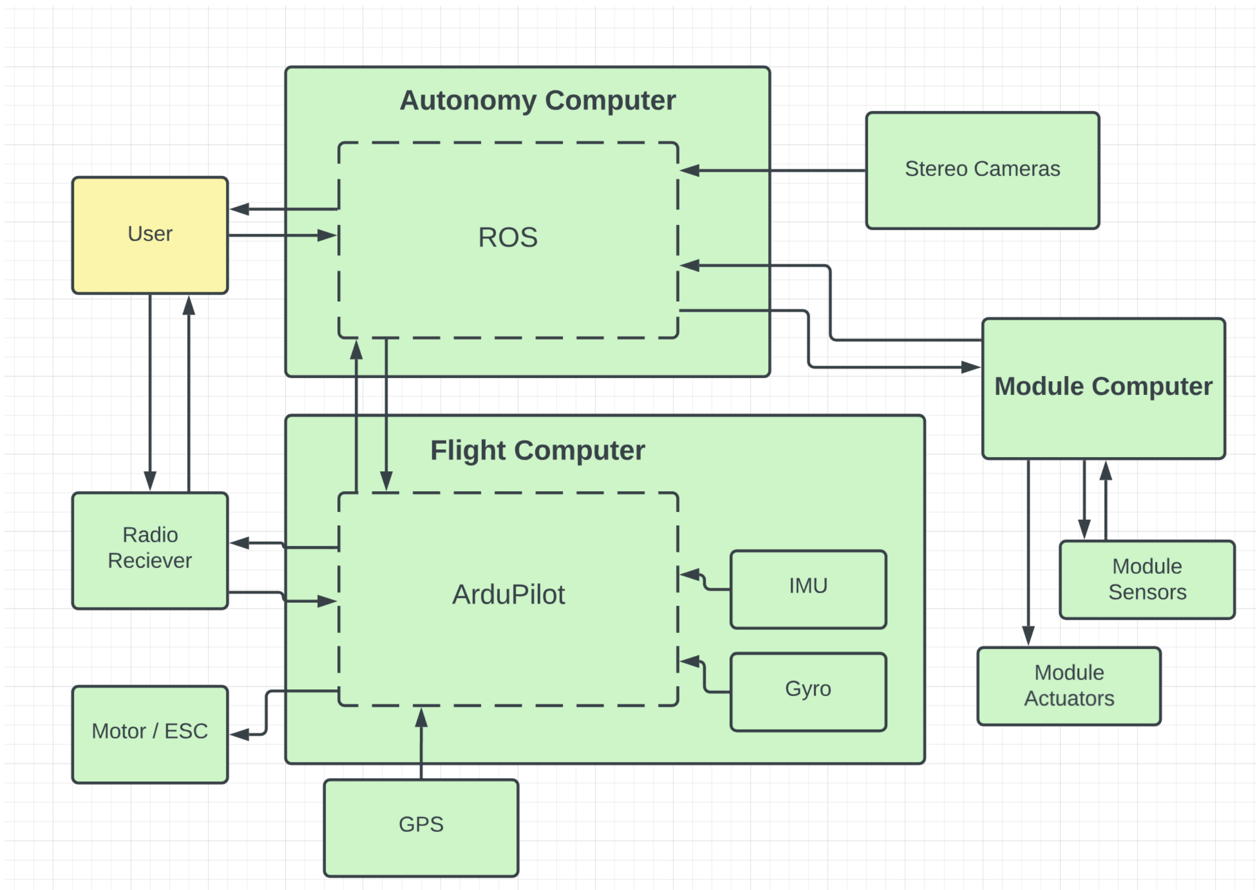


Figure 18: High level software architecture.

Autonomy Computer

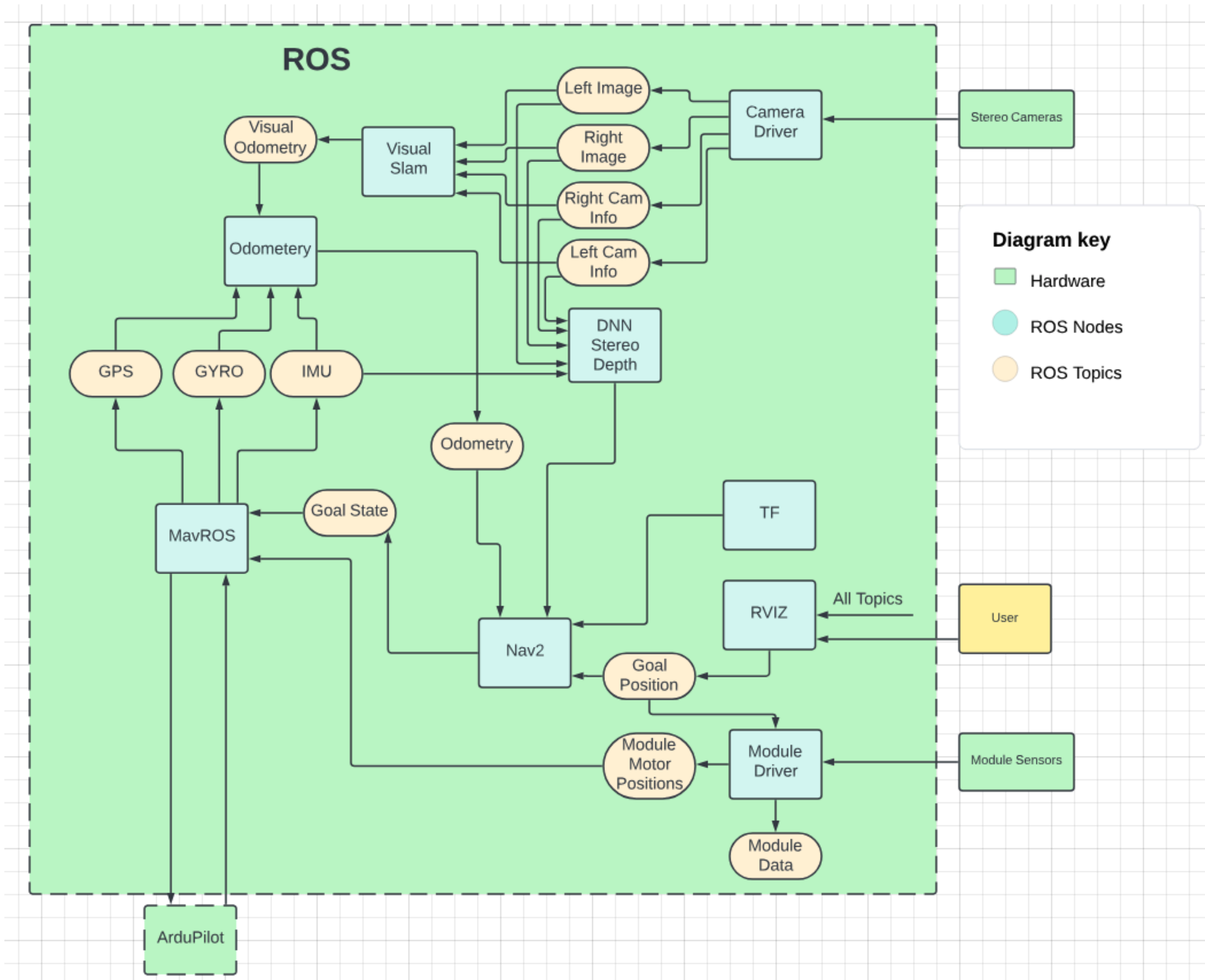


Figure 19: Detailed view of autonomy computer.

All software on the Autonomy Computer will be part of a ROS workspace. This will allow faster development by using pre built packages. Specifically we will be using the Isaac ROS 2 Humble build which will give us access to packages designed for GPU acceleration on the Nvidia Jetson Orin Nano. Some important packages from this build are DNN Stereo Depth, Visual SLAM, Nav2, and MavROS.

ROS Nodes

- DNN Stereo Depth
  - Takes in a left and right stereo image pair as well as camera properties and outputs a depth map.
- Visual SLAM
  - Takes in a left and right stereo image pair, camera properties, and accelerometer data, and outputs relative odometry positions

- Nav2
  - Takes in odometry, global map, local map, system model and outputs goal positions
- RVIZ
  - Used to visualize ros topics for development. Can also be used to set waypoints.
- MavROS
  - Used to converter between ROS topics and Mavlink messages in both directions
- ELP Camera Driver (Custom)
  - Reads in stereo camera frames. Separates the image into left and right camera frames. Formats the data for a ros image message. Also publishes a camera info topic for each camera.
- Odometry node (Custom)
  - Fuses all odometry sources (visual, gps, accelerometer, gyroscope) using a Kalman filter.
- Module Driver (Custom)
  - Customizable mode for every different module
  - ARM
    - Takes in goal arm positions and computes inverse kinematics. Outputs motor positions.
  - Sky Hook
    - Take in goal height. Outputs motor position
  - Science Module
    - Takes in data from the module. Publishes to module data topic. Logs data

Flight Computer

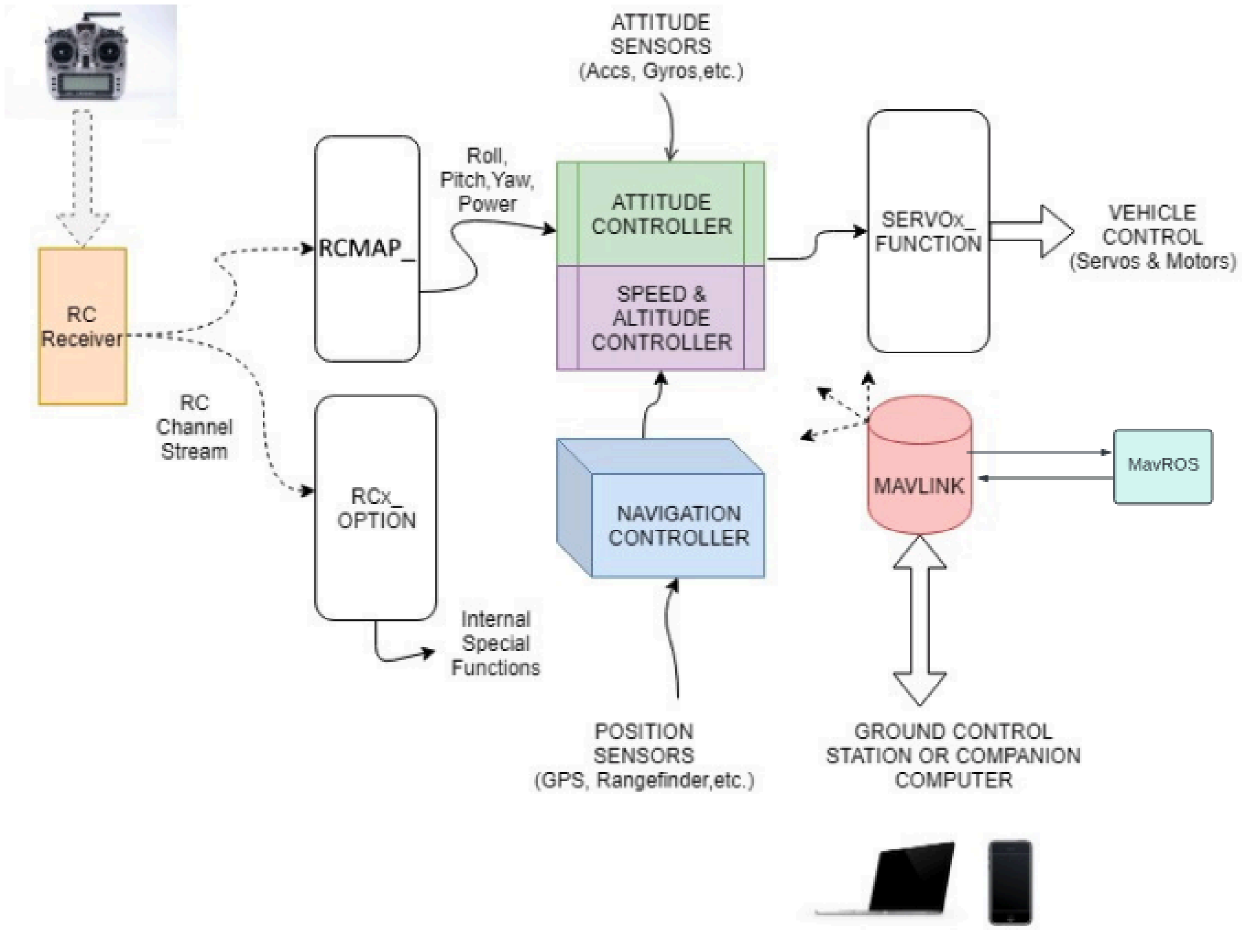


Figure 20: Detailed view of flight computer processes. [9]

Software for the Flight Computer will be majority from the open source software called ArduPilot with some small modifications. ArduPilot has built in support for sensor reading and transmission to the Autonomy Computer. The base ArduPilot build also handles all stabilization and serial radio connection with the operator. However we plan to implement our custom stabilization software that accounts for the extra loads induced by the modules. In addition we will be writing code for custom module control. For the ARM module we will be creating a program that can receive motor position commands from the Autonomy Computer and control the servo motors to the commanded positions.

Software Deployment

Software Deployment for our system is fairly simple. All ROS software for the autonomy computer will be written directly on the machine. Alternatively code can be written using a virtual connection to the machine if a desk setup is not available. After development is complete the software will be set up to automatically run on startup. In order to deploy code to the flight computer the firmware flashing program STMCUBE32 will be used. Code can be written on any computer and then flashed to the Flight Computer using STM. In addition we will be using the application QGroundControl to interface with the Flight Computer. This software is already built to be used with ArduPilot and has been tested to work right out of the box.

Concept of Operations

System States

1. Standby State
    a. Description: The Drone is powered on and ready for deployment but is in a state of low activity, awaiting further instructions.
    b. Conditions: Power is supplied to the Drone, and it is not actively engaged in any mission or operation.
2. Module Swap State
    a. Description: The Drone is powered on but the modular node is powered off for safe module swapping. Upon connection, the module should be recognized in software.
    b. Conditions: Power is supplied to the Drone, and it is not actively engaged in any mission or operation.
3. Initialization State
    a. Description: The Drone is going through the startup and self-check procedures, ensuring all systems are functional before moving to an active state.
    b. Conditions: The Drone is in the process of initializing sensors, calibrating systems, and conducting pre-flight checks.
4. Active State
    a. Description: The Drone is actively engaged in a mission, whether it be inspection, payload delivery, or any other predefined task by the user.
    b. Conditions: The Drone is in flight, executing mission-specific tasks based on user commands or pre-programmed instructions.
5. Emergency State
    a. Description: The Drone has encountered a critical issue or has received an emergency command, triggering immediate response actions.
    b. Conditions: This state includes scenarios such as critical system failures, loss of communication, or a manual emergency stop command.
6. Emergency Landing State
    a. Description: The Drone initiates a controlled emergency landing due to factors such as adverse weather conditions or unexpected obstacles.
    b. Conditions: Activated when predefined safety criteria for emergency landing are met.
7. Return to Home State

a. Description: The Drone initiates an automatic return to its home location, typically in response to low battery levels or a manual command.
b. Conditions: Activated when specific criteria, such as low battery or user command, are met, prompting the Drone to safely return to its take-off point.

8. Charging State
   a. Description: The Drone is connected to a charging station, replenishing its battery power for future missions.
   b. Conditions: Activated when the Drone is docked or connected to a charging source.

Operational Modes

1. Manual Mode
   - Description: The Drone is under direct manual control by the operator, allowing for precise pilot input.
   - Conditions: This mode is selected when the operator is controlling the Drone using a remote controller or other manual input devices.
2. Autonomous Mode
   - Description: The Drone operates independently based on pre-programmed instructions or real-time data analysis.
   - Conditions: Activated when the Drone is executing a predefined mission plan, using sensors and algorithms to navigate and make decisions autonomously.
3. Task-Specific Modes (e.g., Surveillance Mode, Inspection Mode)
   - Description: The Drone adapts its behavior and sensor configurations based on the specific task or mission it is assigned.
   - Conditions: Activated when the Drone enters a mission-specific mode, optimizing its sensors and behaviors for the assigned task.
4. ARM Mode
   - Description: The Drone configures itself to accommodate the ARM module in use and establishes the specific positions needed for the particular mission. This is particularly important for the initial, in-flight and in-use configurations.
   - Conditions: Activated when the ARM module is attached, allowing the Drone to utilize the capabilities of the ARM.
5. SkyHook Mode
   - Description: The Drone configures itself to accommodate the SkyHook module in use, adapting its behavior to support the functionality of the module. Once again, the appropriate initial, in-flight, and in-use configurations are established.
   - Conditions: Activated when a module is attached, allowing the Drone to utilize the capabilities of the SkyHook module.
6. Science Module Mode

- Description: The Drone configures itself to accommodate the Science module in use, adapting its behavior to focus on data collection.
- Conditions: Activated when a module is attached, allowing the Drone to utilize the capabilities of the Science module.

7. User Defined Module Mode
   - Description: The Drone configures itself to accommodate the specific module in use, adapting its behavior to support the functionality of that module.
   - Conditions: Activated when a module is attached, allowing the Drone to utilize the capabilities of that particular module.
8. Maintenance Mode
   - Description: The Drone enters a state where it is undergoing routine maintenance, calibration, or system checks.
   - Conditions: Activated during scheduled maintenance intervals or as part of a proactive system health management strategy.
9. Mapping Mode
   - Description: The Drone optimizes its sensors and flight patterns for aerial mapping and surveying tasks.
   - Conditions: Activated when the Drone is executing mapping missions, utilizing specialized sensors and cameras.

Temporal Aspects

The diagram below visualizes the various temporal aspects of the system and represents any concurrent modes. This is representative of the base lifecycle of standby to maintenance and repeat.

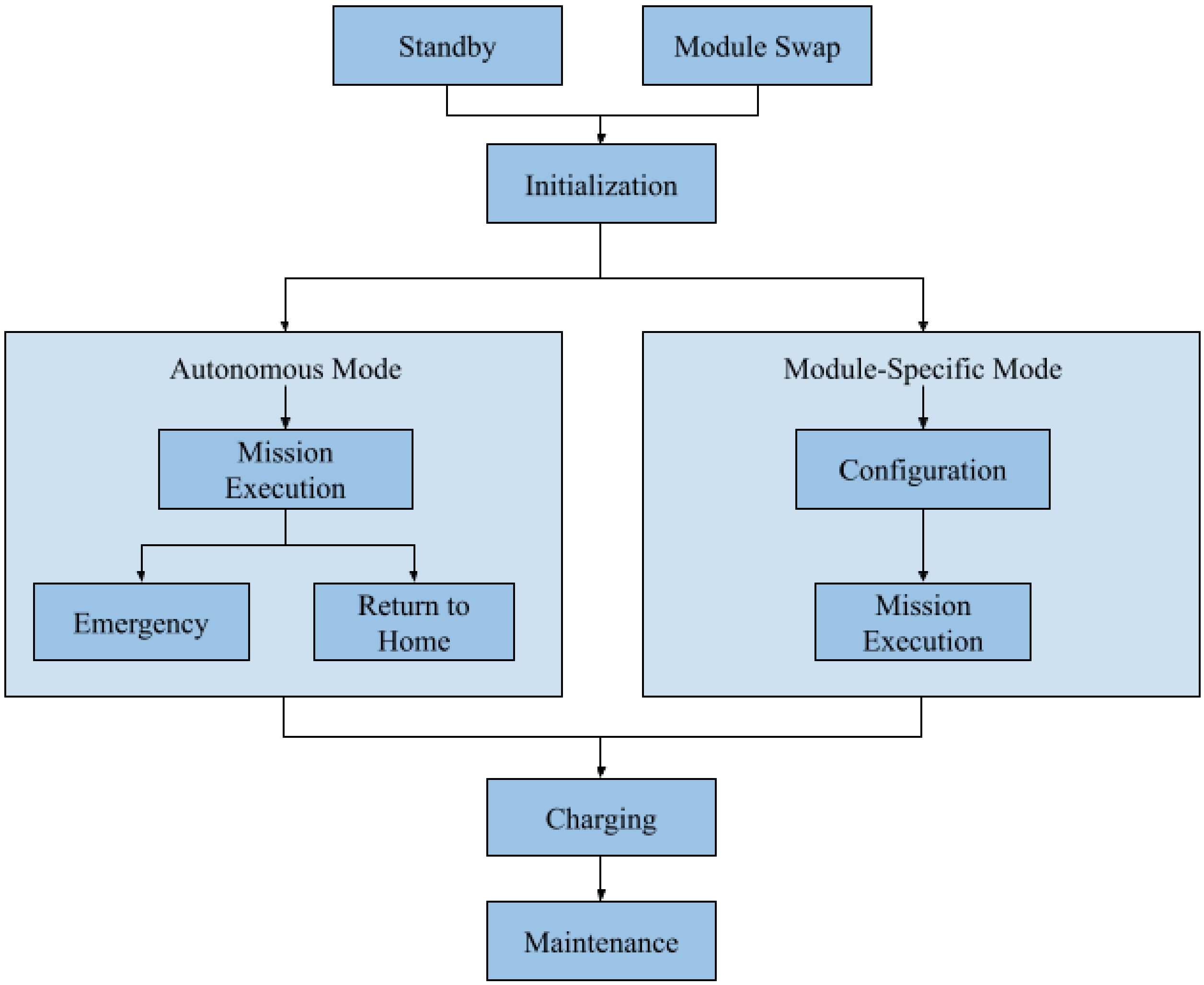


Figure 21: Diagram of temporal aspects of states and operational modes in a basic operation plan.

Fault Recovery

1. Logging and Fault Detection
    a. Description: Drone performance and health are continuously monitored by data logger, and triggers Emergency State should significant errors arise.
    b. Conditions: Logger identifies significant unanticipated changes in system states.
2. Fault Correction and Degradation

   a. Description: Based on type of failure decides whether to trigger an Emergency Landing State or Return to Home State. Additionally activates degraded states with limited functionality, again depending on emergency severity.
   b. Conditions: Triggered when a significant fault is detected.
3. Recovery and Reconfiguration
   a. Description: Drone is collected after failure and fixed as necessary by the operator, who has access to data logs and reported errors.
   b. Conditions: Drone is recovered after a failure is detected.

Operations Plan

1. Introduction

This section of the document aims to describe the systematic deployment, preparation, and start-up procedures for safe and efficient drone operations. The following procedures should be completed by all users of the AIR drone, no matter how familiar they might be with the system, ensuring safe and reliable operation.

The following procedures are listed in the order users should complete them, from pre to post deployment operations.

2. Deployment Process

2.1. Pre-Deployment

Verify mission objective and area of operation, ensuring the drone will remain within line of sight during all points of operation. The area should be cleared of obstacles so the drone can maneuver safely. Also verify all usage and safety guidelines are followed as described below. Additionally, the weather should be clear, without precipitation, for best drone usage.

2.2. Pre-Flight Preparation

Before turning on the drone, the operator should check the drone for any kinds of critical defects including but not limited to the frame, propellers, sensors, and ground equipment like a remote control. One of the most important aspects to verify is the battery level, which should be high enough to operate the drone for the duration of the mission. The drone firmware should also be up to date, to ensure the most updated versions of software is available for operation.

2.3. Start-Up Process

Drone is powered on and system checks are initiated, any inconsistencies or issues should be addressed immediately before flight begins. Before operating the drone, a short test flight should be conducted to verify stability and responsiveness.

2.4. Operation

2.4.1. In-Flight Operation

During operation the drone should stay within visual line of sight at all times, as well as following altitude limits and other safety protocols as detailed below. During any kind of flight operation, the battery should be heavily monitored such that the drone will always have enough to return to a designated "home."

2.4.2. Emergency Operation

Should an emergency occur, the drone will either attempt to return home or make an emergency landing. If an emergency occurs, the operator should be prepared to retrieve the drone from any landing position, and also inspect the drone itself and drone logs to determine operating failures and damage to the drone.

2.5. Post-Operation

Operators should always review drone diagnostics and logs after operation to ensure proper usage, and to detect any inconsistencies before the next operation. Properly stowing the drone away is also paramount to avoid unintended stresses that could compromise the integrity of the drone for future use.

3. Conclusion

With this section we hope to emphasize the proper operation techniques for safe and efficient drone usage. By following these procedures, an operator guarantees the smoothest possible operation and prolonged lifespan of the drone.

Usage and Safety Guidelines

1. Introduction

This section of the document covers the Drone Usage and Safety Guidelines. This portion of the document is designed to ensure safe and responsible operation of our autonomous drone system. Whether you are an experienced drone operator or a first time user, it is critical to familiarize yourself with these guidelines to maximize the efficiency of operation while prioritizing safety.

This section covers a range of topics, including user responsibilities, pre-flight checklists, emergency procedures, and legal considerations. We encourage you to read each section thoroughly and refer back to this document whenever needed. Remember, responsible drone operation not only enhances your experience but also promotes a positive image of drone technology.

Please note the federal laws examined in this document are for the United States of America. If you plan to operate internationally ensure you comply with the laws and regulations set forth by the country you are operating.

2. User Responsibilities

This section outlines the responsibilities of drone operators to ensure safe and responsible usage of unmanned aircraft systems (UAS). Adhering to these responsibilities is crucial for maintaining compliance with regulations, minimizing risks, and ensuring a secure environment for drone operations.

2.1 Awareness of Regulations

Users must familiarize themselves with and stay updated on all relevant federal, state, and local regulations governing drone operations. This includes understanding airspace restrictions, operational limitations, and specific rules based on the nature of the drone use (recreational or commercial).

2.2 Compliance with No Drone Zones

No Drone Zones are designated areas where the operation of drones is strictly prohibited. These restrictions may be implemented at the federal, state, or local levels, including regulations set by municipalities or townships. Operators should use tools such as the B4UFly which is both a website [8] as well as a smartphone app to identify and respect No Drone Zones. Flying in restricted areas, such as near airports, critical infrastructure, or emergency response activities, poses serious safety risks and is strictly prohibited.

2.3 Responsible Operation

Drone operators are required to conduct flights responsibly, adhering to guidelines that prioritize safety and mitigate potential risks. Responsible operation involves refraining from engaging in reckless or dangerous behaviors, such as flying over people, moving vehicles, or in a manner that could impede emergency response activities. Operators must be vigilant about respecting privacy and avoiding unnecessary noise disturbances, demonstrating a commitment to ethical and considerate drone use. Additionally, flights should be conducted within a maximum altitude of 400 feet, ensuring compliance with airspace regulations and contributing to safe and controlled drone operations.

2.3 Maintaining Visibility

Operators must always keep the drone within their visual line of sight. Losing sight of the drone increases the risk of collisions and compromises overall flight safety. In cases where beyond visual line of sight (BVLOS) operations are necessary, operators should obtain required approvals before operating and comply with specific regulations. To do so a BVLOS waiver must be submitted to the Federal Aviation Administration (FAA) and must be approved. Please note to date, that only 1% of BVLOS waivers submitted to the FAA have been approved.

2.4 Training and Certification

Operational proficiency is crucial for drone users, and obtaining the required training and certification is essential. For recreational drone operators, it is required to complete The Recreational UAS Safety Test (TRUST) to enhance awareness and understanding of safe flying practices. This test is a valuable resource for recreational users, providing essential guidelines for responsible drone operation. The FAA requires operators to carry proof of test passage when flying. For operators planning to utilize this drone platform to engage in commercial drone activities, such as aerial photography or surveying, remote pilot certification under Part 107 is mandatory. This process includes passing an aeronautical knowledge test and adhering to specific age requirements. Further information on both recreational and commercial drone use, including details about training and certification, can be found on the FAA's official website [7] and related resources.

2.5 UAS Registration:

Users are required to register their drones with the FAA if their drone falls within the weight range of 250 grams to 25 kilograms, a criterion met by our system. Registration is facilitated through the FAA DroneZone [6], whether operating under the Exception for Limited Recreational Operations or Part 107. To determine the appropriate category for registration, users can utilize the User Identification Tool [5] or visit the FAA's Getting Started webpage [4].

As of September 16, 2023, all drones must carry a Remote ID broadcast module. Our drone will come equipped with a module installed but the operator must register the serial number attached to this module while completing the registration process.

To complete the registration process, users need to provide essential information such as their physical and mailing addresses, email address, phone number, specific Remote ID serial number, and payment details. Registration fees cost $5, if you fly recreationally, each eligible drone you own will be covered under your registration, if you fly under Part 107, each drone must be registered individually.

This registration is valid for 3 years, and users must renew their drone registration through the FAADroneZone after the registration expires. Please note, registration cannot be transferred between operation types (Part 107 or Exception for Limited Recreational Operations). Following registration, users will receive an FAA registration certificate, a mandatory possession when flying the drone. If someone else operates the drone, they must have the drone registration certificate. All drones must be labeled with the registration number before flight. Failure to register a drone within the specified weight range may result in regulatory and criminal penalties, emphasizing the importance of compliance with FAA registration requirements to ensure safe and legal drone operations. For inquiries or assistance regarding UAS registration, users can contact the UAS Support Center [3].

2.6 Coordination with Authorities:

Operators should coordinate with relevant authorities, such as local airports, when flying within designated proximity. This proactive communication helps ensure the safety of both manned and unmanned aircraft in shared airspace.

2.7 Conclusion

By embracing these user responsibilities, drone operators contribute to the overall safety, professionalism, and positive public perception of unmanned aircraft systems. Adherence to these principles is essential for the continued integration and success of drones in various applications.

These regulations have been compiled as of December 8th, 2023. Federal, state, and local laws and regulations are subject to change at any time. Ensure you have checked all necessary regulations before operating this system.

3. Drone Specifications

3.1 Weight and Payload Capacity:

A current estimate of our drone's base weight is 4.176 kg, excluding any attached modules. The system allows for additional payload capacity, with a maximum weight increase of 4.536 kg when equipped with various modules. Consequently, the maximum possible weight of the system, inclusive of modules or attached payloads, must be under 8.711 kg. Do not exceed this flight weight during operation as the drone will no longer operate safely. This is detailed in system requirements **R1.5** and **R11.1.** It's important to note that this weight falls within the 250g to 25 kg limit established by federal law, as specified in the Code of Federal Regulations PART 107 — Small Unmanned Aircraft Systems.

3.2 Dimensions and Size:

While there are no federal restrictions on the physical dimensions of a drone (excluding mass), this system adheres to the size constraints outlined in our system requirement **R11**. It is designed to be compact and portable, fitting within a 1m radius and standing at a maximum height of 0.5m. This adherence to size restrictions ensures ease of transport and maneuverability, meeting the operational guidelines set forth by AIR.

3.3 Special Features:

Users should be aware that the drone is equipped with the ability to accept a variety of modules, each contributing to the overall functionality of the system. These modules are designed for specific tasks such as robotic manipulation, scientific data collection, aerial retrieval, and more. The modular design allows users to customize the drone's capabilities according to their specific needs, enhancing its versatility and adaptability in various scenarios. It is crucial for users to comply with relevant laws and regulations governing the deployment of specific modules, especially those involving tasks like aerial retrieval, which may have distinct legal impacts. Awareness of legal frameworks ensures responsible and legal utilization of the drone's diverse functionalities. Additionally, users should stay informed about evolving regulations concerning drone technologies and modular attachments to align their operations with current legal standards.

4. Operating Environment

The drone is optimized for efficient and safe operation in various environments, using its modular design to adapt to diverse tasks and scenarios. Operators must adhere to specific guidelines to ensure responsible and effective drone usage in different settings.

4.1 Outdoor Use:

For optimal drone performance and safety, outdoor operation is suitable with consideration for weather conditions and compliance with local regulations. Before initiating flights, a thorough check of weather elements such as temperature, wind speed, precipitation, and lighting is essential. In accordance with system requirement **R9**, the temperature should not surpass 50°C, and wind speed must not exceed 15 mph. While wet ground is acceptable, precipitation during flights is strictly prohibited. Additionally, operations should take place during daylight hours or in the presence of external lighting.

Conduct outdoor flights in large, open, uncrowded areas, ensuring an unobstructed line of sight to the drone. Compliance with airspace regulations and potential obstacles are paramount for the safe outdoor operation of the drone. Strict adherence to designated no-drone zones, as detailed in "2.2 Compliance with No Drone Zones," is necessary. Additionally follow local regulations as detailed in “2.6 Coordination with Authorities”.

4.2 Indoor Use:

Indoor operation of the drone is recommended **only** in vastly spacious environments that can accommodate its size and provide ample clearance for safe flight. Ideal settings of enclosed spaces include large warehouses or stadiums. Prior to indoor flights, operators must ensure proper space for flight, adequate lighting, and the absence of obstacles or objects that could impede the drone's operation. This restriction aims to prevent potential collisions and damage to both the drone and its surroundings.

4.3 Recommended Environments:

Optimal outdoor environments for drone operation include open spaces with minimal obstacles. These settings enhance the drone's navigational capabilities and reduce the risk of collisions. Users are encouraged to choose locations with minimal electromagnetic interference, ensuring reliable communication between the drone and its modules during flight.

4.4 Restricted Environments:

Certain environments pose potential risks and should be avoided during drone operation. Crowded indoor spaces with limited clearance are unsuitable for safe flights. Additionally, operators must refrain from flying near critical infrastructure, airports, or restricted airspace without obtaining the necessary authorization. Compliance with these guidelines ensures responsible drone operation and mitigates potential hazards. These were covered more in depth in sections "2.2 Compliance with No Drone Zones," “2.3 Responsible Operation,” and “2.6 Coordination with Authorities”.

5. Pre-flight Checklist

Mission Objective and Area of Operation:

- [ ] Verify the mission objective and ensure alignment with operational goals.
- [ ] Confirm that the area of operation allows line of sight operation at all times.
- [ ] Clear the area of obstacles to facilitate safe and unobstructed drone maneuvering.

Compliance with Usage and Safety Guidelines:

- [ ] Confirm adherence to federal, state, and local regulations governing drone operations.
- [ ] Check for compliance with No Drone Zones, as identified through tools like B4UFly.
- [ ] Verify that operators have completed required training and certification
- [ ] Confirm registration of the drone, including the presence of a visible registration mark.

Drone Specifications:

- [ ] Check the weight and payload capacity, ensuring they align with R1.5 and R11.1.
- [ ] Verify the presence and proper functioning of any attached modules or payloads.

Operating Environment:

- [ ] Confirm that weather conditions are suitable for drone operation aligning with R9
- [ ] Ensure compliance with recommended environments for use.

Pre-flight Maintenance:

- [ ] Inspect the drone for dirt, debris, or damage, and clean if necessary.
- [ ] Check the battery's charge level, ensuring sufficient level.
- [ ] Conduct routine checks for loose components, damaged propellers, or abnormal sounds.

Module Maintenance:

- [ ] Ensure modules are attached securely.
- [ ] Confirm proper cable connections and recognition by the system.

6. Flight Procedures

6.1 Start-Up Process:

The drone's start-up process is a critical phase to ensure optimal performance and mitigate potential risks. Upon powering on the drone, the system conducts comprehensive checks, identifying any inconsistencies or issues. Operators must address these immediately before initiating the flight. A systematic approach to troubleshooting during the start-up process is essential for a safe and successful operation. Additionally, a short test flight is recommended before the actual mission to verify the drone's stability and responsiveness. This test flight serves as a proactive measure to identify and rectify any issues that may not be apparent during the initial system checks.

6.2 In-Flight Operation:

During in-flight operation, strict adherence to safety protocols is paramount. The drone must remain within visual line of sight at all times, ensuring operators can monitor its trajectory and respond to any unexpected situations promptly. Altitude limits and other safety protocols should be rigorously followed to prevent potential hazards and ensure controlled flight. Continuous monitoring of the battery status is crucial throughout the operation. Operators must ensure the battery maintains sufficient power for a safe return to "home" operation. Additionally users should pay attention to any system warnings, or communication lost and should act as instructed through the warning.

7. Emergency Procedures

Should an emergency occur, the drone will either attempt to return home or make an emergency landing. If an emergency occurs, the operator should be prepared to retrieve the drone from any landing position, and also inspect the drone itself and drone logs to determine operating failures and damage to the drone.

If any harm should occur to someone or damage to someone's property, please address the issues completely. Contact emergency hotlines if needed. This is further explored in section “10. Reporting Incidents”.

8. Maintenance and Care

Ensuring the longevity and optimal performance of the drone requires regular maintenance and proper care. Users are advised to adhere to the following guidelines:

8.1 Overall Cleaning:

Regularly inspect the drone for any dirt, debris, or residue that may accumulate during operations. Clean the exterior surfaces using a soft, lint-free cloth and mild cleaning solutions as needed. Pay particular attention to the sensors, camera lenses, and propellers, ensuring they remain free from obstructions, cracks, or damage.

8.2 Battery Care:

Batteries designated for use in our system must follow the requirements set forth by system requirement **R25**, which states a 10,000 mAh, 6s2p, Lithium-ion (LION) battery must be used. For these batteries, maintaining their health is crucial for optimal performance and safety. Before use, it's recommended to charge the battery while ensuring you avoid overcharging, which can diminish battery life and pose safety risks.

Regular monitoring of the batteries is essential. Users should periodically inspect them for signs of damage, swelling, or any abnormalities. If batteries exhibit wear or reduced performance, it's advisable to replace them promptly.

If the batteries will not be in use for an extended period, proper storage is key. Keep the LION batteries in a cool, dry place and maintain the charge level between 30-60% during storage. This practice helps mitigate issues associated with excessive discharge and ensures the batteries remain in good condition.

If a battery needs to be disposed of, users should follow local regulations and guidelines for the proper disposal of LION batteries. Responsible disposal methods contribute to environmental sustainability. When these batteries reach the end of their lifecycle, contact authorized disposal centers or recycling facilities to ensure a proper and eco-friendly disposal.

8.3 Storage:

When not in use, store the drone in a protective case or a secure location that shields it from dust, humidity, and extreme temperatures. Ensure that the storage area is well-ventilated and free from potential hazards. Store the drone with the battery removed to prevent any discharge over time.

8.4 Routine Checks:

Conduct routine checks before each operation to identify any loose components, damaged propellers, or abnormal sounds. Ensure that all modules are securely attached, and there are no visible signs of wear or damage. If any issues are detected, address them promptly to maintain the drone's reliability.

9. Module Attachment and Removal

Efficient handling of modules is crucial for the drone's optimal functionality. Users should follow these guidelines for attaching and removing modules:

9.1 Attachment:

Before attaching a module, ensure the drone is powered off to prevent accidental activation. Align the attachment points on the module with the rails on the drone's frame and gently secure the module using the set screws. Make necessary cable connections into the drone interface. Confirm that the attachment is secure by checking for any play or movement. Power on the drone and verify that the module is recognized by the system.

9.2 Removal:

Power off the drone before attempting to remove any module to avoid potential hazards. Gently disconnect any cables from the drone's interface. Loosen the set screws and slide the module off of the rails. Store detached modules in a secure and safe location.

10. Reporting Incidents

In case of any accidents, damages, or malfunctions involving the drone, operators are urged to report such incidents promptly. This reporting is crucial for addressing issues, preventing future occurrences, and ensuring the safety of themselves and others. Users should contact local law enforcement, businesses, or relevant authorities to report incidents and seek guidance on resolution if necessary. Providing comprehensive details, such as the nature of the incident, location, time, and any relevant flight data or circumstances, is essential for effective incident resolution and continual improvement of the drone system.

Timely and accurate reporting helps maintain safety standards and contributes to the responsible use of the technology. Users are reminded that they are responsible for any damages caused while operating the drone, and reporting incidents promptly is an integral part of fulfilling this responsibility.

11. Conclusion

These Usage and Safety Guidelines serve as a vital framework for the responsible and safe operation of our autonomous drone system. Operators must prioritize compliance with federal, state, and local regulations, ensuring awareness of airspace restrictions, No Drone Zones, and responsible operation principles outlined in Section 2. Understanding the drone's specifications, features, and adhering to recommended operating environments, as discussed in Sections 3, 4, and 5, respectively, is crucial for informed decision-making during operations. Additionally, strict adherence to pre-flight checklists, flight procedures, and emergency protocols (Sections 5, 6, and 7) enhances operational safety. Maintenance, care, and proper reporting of incidents, as highlighted in Sections 8, 9, and 10, respectively, contribute to sustaining safety standards and continual improvement. Regular review and commitment to these guidelines empower operators to maximize the capabilities of our drone while ensuring its positive integration into various applications.

Testing: Validation and Verification

This section covers our approach to Testing, Validation, and Verification which is crucial for ensuring the effectiveness of our modular drone platform. This portion of the document will cover requirements validation, functionality verification, performance verification, and failure modes.

Requirements Validation

Robot

V1: Validation for R1 (Robot Must Fly)
Objective: Verify that the system meets the specified flying capabilities.

V1.1 Validation Metric for R1.1 (Vertical Takeoff):

a. **Purpose:** Ensure drone's ability for a stable vertical take-off
b. **Tested Subsystems:** Autonomy, Power, Communication, and Sensing
c. **Significance:** Will be a commonly used command.
d. **Measurement Method:** Record the time taken for the drone to achieve a stable vertical takeoff greater than 2m off the ground.
e. **Procedure:** Initiate the vertical takeoff command and measure the time until the drone reaches a stable hover over 2m. Ensure time is under 5 seconds.
f. **Comparison to Ground Truth:** Use measurement tape
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Full Drone System, Stopwatch, and Measuring Tape
i. **Validation Owner:** Dustin Moss
j. **Post Test Actions:** Tune takeoff speed for stability

V1.2 Validation Metric for R1.2 (Rotate in Place):

a. **Purpose:** Ensure drone's ability to rotate in place
b. **What is Tested:** Autonomy Subsystem, Power System, and Sensing
c. **Significance:** Will be a commonly used command.
d. **Measurement Method:** Evaluate the drone's ability to rotate 360 degrees in a confined space.
e. **Procedure:** Command the drone to rotate in place, measuring the angular rotation and ensuring it completes a full circle by video recording where the drone starts facing and seeing where it finishes.
f. **Comparison to Ground Truth:** Review internal logs and compare to video
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Full Drone System, Phone Camera
i. **Validation Owner:** Dustin Moss
j. **Post Test Actions:** Tune compass and other heading identifiers

V1.3 Validation Metric for R1.3 (Hover for 20 mins):

a. **Purpose:** Ensure drone's ability to hover
b. **What is Tested:** Autonomy Subsystem, Power System, and Sensing
c. **Significance:** Test battery life and ability to maintain long flights.
d. **Measurement Method:** Monitor the drone's hover time under normal operating conditions.
e. **Procedure:** Activate the hover command and record the duration until the drone lands or exhibits any deviations. Ensure time is over 20 minutes..
f. **Comparison to Ground Truth:** Timer and video
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Full Drone System, Stopwatch
i. **Validation Owner:** Dustin Moss
j. **Post Test Actions:** Identify inefficiencies with battery usage and potential power bottlenecks

V1.4 Validation Metric for R1.4 (200 meters):

a. **Purpose:** Ensure drone can operate over long distances
b. **What is Tested:** Autonomy Subsystem, Power System, Communication, and Sensing
c. **Significance:** Test ability to maintain operation over large operating distances.
d. **Measurement Method:** Measure the distance covered by the drone during a single flight.
e. **Procedure:** Command the drone to fly in a straight line, recording the gps data of the distance covered until it reaches the maximum range or loses connection. Ensure the range is at least 200 meters.
f. **Comparison to Ground Truth:** Calculate distance from origin using GPS logs
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Full Drone System, GPS logs
i. **Validation Owner:** Dustin Moss
j. **Post Test Actions:** Potentially upgrade communication system or change requirements

V1.5 Validation Metric for R1.5 (5lb Payload):

a. **Purpose:** Ensure the drone can lift necessary mass with an attached module.
b. **What is Tested:** Frame and Power System
c. **Significance:** Test ability to maintain operation while lifting a payload.
d. **Measurement Method:** Confirm the drone's ability to lift and carry a 5 lb payload.
e. **Procedure:** Attach a 5lb weight to the drone and assess its capacity to hover and navigate with the added payload.
f. **Comparison to Ground Truth:** Measure mass, monitor dips in
g. **Meeting Logistics:** 12:00, 3/17/2024, Wightman Park

h. **Equipment:** Full Drone System, Scale, Camera
i. **Validation Owner:** Dustin Moss
j. **Post Test Actions:** Investigate potential causes such as power system limitations or frame integrity, decrease potential power draw, change requirements

V2: Validation for R2 (Robot Must Sense)

Objective: Verify that the system meets the specified sensing capabilities.

V2.1 Validation Metric for R2.1 (Front, Above, and Below Occupancy Map):

a. **Purpose:** Ensure drone can map its surroundings
b. **What is Tested:** Mapping Subsystem
c. **Significance:** Test ability to detect obstacles and identify navigable space
d. **Measurement Method:** Confirm the real-time update of occupancy maps in the specified directions.
e. **Procedure:** Activate sensors to detect obstacles in front, above, and below the drone, ensuring occupancy maps are continuously updated. Ensure update frequency is at least 5 times per second.
f. **Comparison to Ground Truth:** Compare occupancy map to video, read logs to determine update rate
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Sensing and mapping subsystem, environment to model
i. **Validation Owner:** Jacob Hsu
j. **Post Test Actions:** Improve compute power/efficiency for subsystem - see sub-metrics for details

V2.1.1 Validation Metric for R2.1.1 (Depth Map Update at 30 fps - 2m Range):

a. **Purpose:** Ensure drone can detect its surroundings quickly
b. **What is Tested:** Mapping Subsystem
c. **Significance:** Test ability to identify obstacles as they appear, while under full computational load
d. **Measurement Method:** Validate that the depth map updates at a rate of 30 frames per second within a 2m range.
e. **Procedure:** Activate sensors within the 2m range and record the update frequency of the depth map. Ensure rate is at a minimum 30 frames per second.
f. **Comparison to Ground Truth:** Rate measured by logs, obstacles placed over 2m away
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Full Drone System, Logs, Measuring Tape
i. **Validation Owner:** Jacob Hsu
j. **Post Test Actions:** reduce computation inefficiencies that might slow down map creation

V2.2 Validation Metric for R2.2 (360° Obstacle Detection - 10m Range):

a. **Purpose:** Ensure drone can detect an adequate FOV for operations

- **b. What is Tested:** Mapping Subsystem, Sensor layout
- **c. Significance:** Test ability to detect obstacles within expected FOV and operable range
- d. **Measurement Method:** Verify the drone's ability to detect obstacles in a 360° radius with a range of 10m.
- e. **Procedure:** Surround the drone with obstacles and confirm that it successfully detects objects 360° in front of the drone within the 10m range.
- **f. Comparison to Ground Truth:** Measure angle between objects at edges of drone FOV.
- **g. Meeting Logistics:** 1:00 3/24/2024, Wightman Park
- **h. Equipment:** Drone Vision System, markers, compass
- **i. Validation Owner:** Jacob Hsu
- **j. Post Test Actions:** Alter camera locations, or cameras themselves, change requirements

V2.3 Validation Metric for R2.3 (Map Creation):

- **a. Purpose:** Ensure sensors properly integrate with computer to output an accurate map
- **b. What is Tested:** Mapping Subsystem
- **c. Significance:** Test ability to create a map by integrating data from various sensors
- d. **Measurement Method:** Validate the creation of a comprehensive map integrating data from multiple sensors.
- e. **Procedure:** Initiate the map creation process and confirm that data from different sensor streams are fused to generate a complete map. Ensure map accuracy of 50%.
- **f. Comparison to Ground Truth:** compare output map to environment picture or observation
- **g. Meeting Logistics:** 1:00 3/24/2024, Wightman Park
- **h. Equipment:** Drone Vision System
- **i. Validation Owner:** Jacob Hsu
- **j. Post Test Actions:** Scale back sensor fusion, change requirements, divert more tasks to Cube Orange

V2.4 Validation Metric for R2.4 (Global Position - 1.5m Accuracy):

- **a. Purpose:** Ensure drone is accurate in readings for path planning and navigation.
- **b. What is Tested:** Localization, GPS
- **c. Significance:** Tests GPS accuracy
- d. **Measurement Method:** Confirm the accuracy of the drone's global position within a 1.5m margin.
- e. **Procedure:** Obtain the global position coordinates and validate against a known reference point, ensuring the accuracy falls within 1.5m.
- **f. Comparison to Ground Truth:** compare GPS reports against known location

g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** GPS, other map
i. **Validation Owner:** Tahaseen Shaik
j. **Post Test Actions:** Scale back requirements

V3: Validation for R3 (Robot Must be Capable of Autonomous Operation)

Objective: Verify that the system meets the specified capabilities for autonomous operation.

V3.1 Validation Metric for R3.1 (Autonomously Stabilize with All Accessories):

a. **Purpose:** Ensures drone can at least stabilize with expected additions (modules)
b. **What is Tested:** Flight Controller
c. **Significance:** Tests how capable drone is at stabilizing under load
d. **Measurement Method:** Confirm that the drone can autonomously stabilize while equipped with all accessories.
e. **Procedure:** Attach various modules to the drone and assess its ability to autonomously stabilize under normal operating conditions. Ensure capable of stabilization in under 5 seconds.
f. **Comparison to Ground Truth:** Measure mass of additions, record drone in position hold to determine "stability"
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Full drone system, scale, video recorder, timer
i. **Validation Owner:** Jacob Hsu
j. **Post Test Actions:** Reduce drone/module weight, improve power output to rotors

V3.2 Validation Metric for R3.2 (Autonomously Hold Position):

a. **Purpose:** Ensures basic position hold mode works on drone
b. **What is Tested:** Autonomy Computer
c. **Significance:** Many of the drones modules and functions require the ability to hold its position while in flight
d. **Measurement Method:** Validate the drone's capability to autonomously hold its position in terms of height, orientation, and relative position to the local environment.
e. **Procedure:** Initiate the hold position command and verify that the drone maintains stability in terms of height, orientation, and position relative to the local environment. Ensure the drone can hold position with disturbances for at least 5 seconds.
f. **Comparison to Ground Truth:** Time position hold, record to determine position hold stability
g. **Meeting Logistics:** 12:00, 3/17/2024, Wightman Park
h. **Equipment:** Full Drone System, Controller, Timer, Video REcorder
i. **Validation Owner:** Jacob Hsu
j. **Post Test Actions:** Verify Flight Controller working as expected

V3.2.1 Validation Metric for R3.2.1 (Height):

a. **Purpose:** Ensures Drone can operate autonomously at expected heights
b. **What is Tested:** Flight Controller, Localization
c. **Significance:** Common command sent to drone
d. **Measurement Method:** Confirm the drone's ability to autonomously hold a specific height.
e. **Procedure:** Command the drone to hold its height, and verify that it remains stable at the designated altitude. Ensure that deviations are less than +/- 0.5 m.
f. **Comparison to Ground Truth:** Include Measuring Tape in video to confirm accuracy
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Full Drone System, Video recorder, Tape Measure
i. **Validation Owner:** Jacob Hsu
j. **Post Test Actions:**

V3.2.2 Validation Metric for R3.2.2 (Orientation):

a. **Purpose:** Ensure drone can orient itself and change positions to accommodate position
b. **What is Tested:** Compass
c. **Significance:** Common command sent to drone
d. **Measurement Method:** Verify the drone's capability to autonomously hold a specific orientation.
e. **Procedure:** Command the drone to hold a specific orientation, and verify that it maintains stability in the designated position. Ensure that deviations are less than +/- 0.5 m.
f. **Comparison to Ground Truth:** Track heading from logs, and compare to observed heading
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Full drone system, timer
i. **Validation Owner:** Jacob Hsu
j. **Post Test Actions:** tune compass

V3.2.3 Validation Metric for R3.2.3 (Position Relative to Local Environment):

a. **Purpose:** Ensure drone is aware of location relative to environment
b. **What is Tested:** Localization and Sensor Fusion
c. **Significance:** Data will be used often in obstacle avoidance
d. **Measurement Method:** Confirm the drone's ability to autonomously hold a specific position relative to the local environment.
e. **Procedure:** Command the drone to hold its position, and verify that it remains stable in the designated location relative to the local environment. Ensure that deviations are less than +/- 0.5 m.
f. **Comparison to Ground Truth:**

g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Localization and Odometry Suite, GPS
i. **Validation Owner:** Jackson Wood
j. **Post Test Actions:** reconfirm sensor fusion is functioning

V3.3 Validation Metric for R3.3 (Fly to Set Position):

a. **Purpose:** Ensures drone can store location nodes and execute simple commands
b. **What is Tested:** Flight, Path planning
c. **Significance:** Common command sent to drone
d. **Measurement Method:** Verify the drone's ability to autonomously fly to a set position.
e. **Procedure:** Set a specific destination for the drone, initiate the flight command, and confirm that it reaches the designated autonomously. Ensure that deviations are less than +/- 0.5 m.
f. **Comparison to Ground Truth:** Measure error from origin
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Full Drone System, measuring tape, marker for desired location
i. **Validation Owner:** Jackson Wood
j. **Post Test Actions:** If deviations exceed the acceptable range, investigate potential causes such as sensor inaccuracies or navigation algorithm errors, and make necessary adjustments to improve performance.

V3.3.1 Validation Metric for R3.3.1 (Return to Start):

a. **Purpose:** Ensures drone can store location nodes and execute simple commands
b. **What is Tested:** Flight, Path Planning
c. **Significance:** Common command sent to drone
d. **Measurement Method:** Verify the drone's ability to autonomously return to the starting point.
e. **Procedure:** Initiate the return to start command, and confirm that the drone successfully navigates to the starting point autonomously. Ensure that landing location is within 3m of initial take off.
f. **Comparison to Ground Truth:** Using measuring tape for true position
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Full Drone System, marker for origin, measuring tape
i. **Validation Owner:** Jackson Wood
j. **Post Test Actions:** Investigate potential causes such as sensor inaccuracies, navigation algorithm errors, or environmental factors. Make necessary adjustments to improve performance and repeat the validation test to ensure compliance with the specified tolerance.

V3.4 Validation Metric for R3.4 (Avoid Obstacles):

a. **Purpose:** Ensures drone keeps itself and environment safe during operation
b. **What is Tested:** Obstacle avoidance, sensing, navigation

c. **Significance:** Tests critical component of drone autonomous operation
d. **Measurement Method:** Confirm the drone's capability to autonomously avoid obstacles during flight.
e. **Procedure**: Introduce obstacles in the drone's path and validate its ability to autonomously navigate around them. Ensure that the drone is capable of avoiding obstacles within a projected flight path.
f. **Comparison to Ground Truth:** Track nearest distance from obstacles
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Full Drone System, obstacles space intermittently between destination and origin
i. **Validation Owner:** Jackson Wood
j. **Post Test Actions:** Fix Drone, tune obstacle berth parameter

V4: Validation for R4 (Robot Must Plan)

Objective: Verify that the system meets the specified capabilities for planning and task execution.

V4.1 Validation Metric for R4.1 (Path Planning with Real-Time Obstacle Avoidance):

a. **Purpose:** Ensures drone can plan in real time as obstacles and interferences arise while executing a task
b. **What is Tested:** Path Planning
c. **Significance:** Drone is expected to compute and execute plans as information about its surroundings change
d. **Measurement Method:** Confirm the drone's ability to plan a path in real-time with obstacle avoidance.
e. **Procedure:** Introduce obstacles in the drone's path, initiate the path planning command, and validate that it plans a trajectory in real-time, avoiding obstacles. Ensure that the drone can avoid at least 2 obstacles for a 10m path plan.
f. **Comparison to Ground Truth:** Test without obstacles (ie drone simply follows predetermined plan)
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Full Drone System
i. **Validation Owner:** Tahaseen Shaik
j. **Post Test Actions:** Scale back requirements

V5: Validation for R5 (Robot Must Communicate with Operator)

Objective: Verify that the system meets specified capabilities for communicating with the operator.

V5.1 Validation Metric for R5.1 (Serial Radio Link):

a. **Purpose:** Ensure Workstation and Remote Controller can communicate with drone
b. **What is Tested:** Communication

c. **Significance:** Vital aspect of mid-flight drone operation as users need to be able to override and change drone plans as needed
d. **Measurement Method:** Validate the functionality of the serial radio link.
e. **Procedure:** Test emergency stop, manual flight commands, and the transmission of basic status information over the serial radio link. Ensure latency is less than 500 ms.
f. **Comparison to Ground Truth:** measure communication rates compared to stock values
g. **Meeting Logistics:** 12:00, 3/17/2024, Wightman Park
h. **Equipment:** Drone communication, Workstation
i. **Validation Owner:** Dustin Moss
j. **Post Test Actions:** Unit test with each communication piece

V5.1.1 Validation Metric for R5.1.1 (Emergency Stop):

a. **Purpose:** Ensures baseline mode is functional if all else goes wrong
b. **What is Tested:** Planning, mode switching, communication
c. **Significance:** Tests drone full stopping abilities and ability of users to shut off if necessary
d. **Measurement Method:** Confirm the effectiveness of the emergency stop command over the serial radio link.
e. **Procedure:** Initiate the emergency stop command and verify that the drone halts all operations within 2 seconds.
f. **Comparison to Ground Truth:** measure time to full stop
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Full Drone System, Timer
i. **Validation Owner:** Jacob Hsu
j. **Post Test Actions:** look at communication logs, track Estop

V5.1.2 Validation Metric for R5.1.2 (Manual Flight Commands):

a. **Purpose:** Ensures drone can be flown manually
b. **What is Tested:** Flight, Remote Communication
c. **Significance:** Tests drone's ability to be flown through typical drone control means
d. **Measurement Method:** Validate the responsiveness of the drone to manual flight commands over the serial radio link.
e. **Procedure:** Issue manual flight commands and confirm that the drone executes the commands within 2 seconds.
f. **Comparison to Ground Truth:** Compare responsiveness to that of a typical off the shelf drone
g. **Meeting Logistics:** 12:00, 3/17/2024, Wightman Park
h. **Equipment:** Full Drone System, Remote Control
i. **Validation Owner:** Jackson Wood

j. **Post Test Actions:** Tune flight controller

V5.1.3 Validation Metric for R5.1.3 (Returns Basic Status Information):

a. **Purpose:** Establish communication between user and drone through status updates
b. **What is Tested:** Communication
c. **Significance:** Keeps Operator informed about drone observations and intended actions
d. **Measurement Method:** Confirm the drone's ability to transmit basic status information over the serial radio link.
e. **Procedure:** Request basic status information from the drone and validate the accuracy and completeness of the transmitted data within 4 seconds.
f. **Comparison to Ground Truth:** measure message frequency
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Communication, Workstation
i. **Validation Owner:** Dustin Moss
j. **Post Test Actions:** Tune communication params, identify what logs are valuable for reporting

V5.2 Validation Metric for R5.2 (Wi-Fi Connection):

a. **Purpose:** Ensure major communication method between drone and user is functioning
b. **What is Tested:** Communication
c. **Significance:** Wifi will be the primary method of communication between the drone and the operator
d. **Measurement Method:** Validate the functionality of the Wi-Fi connection.
e. **Procedure:** Test the reception of waypoint coordinates, execution of advanced flight commands, and the transmission of advanced status, maps, and logs over the Wi-Fi connection within 4 seconds.
f. **Comparison to Ground Truth:** time to receive commands
g. **Meeting Logistics:** 12:00, 3/17/2024, Wightman Park
h. **Equipment:** Full drone System
i. **Validation Owner:** Tahaseen Shaik
j. **Post Test Actions:** Upgrade router or wifi receiver, scale back requirements

V5.2.1 Validation Metric for R5.2.1 (Receive Waypoint Coordinates):

a. **Purpose:** Ensure waypoints are received from user to drone
b. **What is Tested:** Communication
c. **Significance:** One of the commands for the user to assign to the drone
d. **Measurement Method:** Confirm the drone's ability to receive waypoint coordinates over the Wi-Fi connection.
e. **Procedure:** Transmit waypoint coordinates to the drone and validate that it accurately receives and acknowledges the coordinates.

f. **Comparison to Ground Truth:** n/A
g. **Meeting Logistics:** 12:00, 3/17/2024, Wightman Park
h. **Equipment:** Full Drone System, Workstation
i. **Validation Owner:** Jacob Hsu
j. **Post Test Actions:** retest wifi communication

V5.2.2 Validation Metric for R5.2.2 (Advanced Flight Commands):

a. **Purpose:** Ensure drone can receive and execute advanced flight commands like path following or respecting a geofence
b. **What is Tested:** Communication, Planning, Flight
c. **Significance:** drone needs to be able to interpret and execute more complex and demanding commands
d. **Measurement Method:** Validate the responsiveness of the drone to advanced flight commands over the Wi-Fi connection.
e. **Procedure:** Issue advanced flight commands and confirm that the drone executes the commands accurately and within 4 seconds.
f. **Comparison to Ground Truth:** Timer
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Full drone System, timer
i. **Validation Owner:** Jacob Hsu
j. **Post Test Actions:** reverify comms and reduce elapsed time for interpretation

V5.2.3 Validation Metric for R5.2.3 (Returns Advanced Status, Maps, Logs):

a. **Purpose:** Ensure drone communicating more complex behavior and decision making if needed
b. **What is Tested:** Communication, Logs
c. **Significance:** Test drone data communication and interpretability
d. **Measurement Method:** Confirm the drone's ability to transmit advanced status, maps, and logs over the Wi-Fi connection.
e. **Procedure:** Request advanced status, maps, and logs from the drone and validate the accuracy and completeness of the transmitted data within 4 seconds.
f. **Comparison to Ground Truth:** timer
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Full Drone System, Timer
i. **Validation Owner:** Jacob Hsu
j. **Post Test Actions:** reverify comms

V5.3 Validation Metric for R5.3 (<1 hr Operator Training Time):

a. **Purpose:** Drone should be quick to learn from someone who has just bought product
b. **What is Tested:** UI, Communication
c. **Significance:** drone should be relatively easy to operate, even by a operator unfamiliar with drones

d. **Measurement Method:** Confirm that operator training can be completed in less than one hour.
e. **Procedure**: Time the training session required for an operator to become familiar with the drone's controls, features, and communication systems, ensuring it is completed within at most 1 hour.
f. **Comparison to Ground Truth:** Timer
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Full Drone System, Remote Workstation, Timer
i. **Validation Owner:** Tahaseen Shaik
j. **Post Test Actions:** Scale back requirement, improve UI

V6: Validation for R6 (Robot Must Respond Quickly)

Objective: Verify that the system meets the specified capabilities for responding quickly.

V6.1 Validation Metric for R6.1 (>30Hz Depth Map Update):

a. **Purpose:** Ensures Depth Map Updates at a regular and usable speed
b. **What is Tested:** Depth Map
c. **Significance:** Depth Map needs to update frequently to detect and avoid obstacles during flight
d. **Measurement Method:** Confirm the drone's ability to update the depth map at a frequency greater than 30Hz.
e. **Procedure:** Initiate depth map updates and measure the update frequency to ensure it exceeds 30Hz.
f. **Comparison to Ground Truth:** Logs to confirm update speeds
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Full Drone System
i. **Validation Owner:** Jackson Wood
j. **Post Test Actions:** reduce computation elsewhere, scale back requirements

V6.2 Validation Metric for R6.4 (>300Mbps Wi-Fi Communication):

a. **Purpose:** Ensures Wifi Updates at a regular and usable speed
b. **What is Tested:** Communication
c. **Significance:** Test Drone ability to quickly and effectively communicate with user
d. **Measurement Method:** Confirm the drone's ability to communicate over Wi-Fi at a speed greater than 300Mbps.
e. **Procedure:** Test Wi-Fi communication and measure the data transfer rate to ensure it exceeds 300Mbps.
f. **Comparison to Ground Truth:** measure with logs
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Full Drone System
i. **Validation Owner:** Jackson Wood

j. **Post Test Actions:** Upgrade Router, Adjust Requirements

V6.3 Validation Metric for R6.5 (>=5Hz GPS Update):

a. **Purpose:** Ensures GPS Updates at a regular and usable speed
b. **What is Tested:** GPS
c. **Significance:** Keep track of global position at a rate consistent with drone flight speeds
d. **Measurement Method:** Confirm the drone's GPS update frequency is greater than or equal to 5Hz.
e. **Procedure:** Monitor the drone's GPS updates and verify that the frequency is greater than or equal 5Hz.
f. **Comparison to Ground Truth:** measure rate with logs
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Full Drone System
i. **Validation Owner:** Jackson Wood
j. **Post Test Actions:** Adjust Requirements, decrease compute power, if frequency is too high

V6.4 Validation Metric for R6.6 (>30mph Flight Speed):

a. **Purpose:** Ensures drone can cover distances quickly
b. **What is Tested:** Flight
c. **Significance:** Tests what drone is capable of, which can later be used to optimize path planning
d. **Measurement Method:** Confirm the drone's ability to achieve a flight speed greater than 30mph.
e. **Procedure:** Measure the drone's flight speed and validate that it exceeds 30mph.
f. **Comparison to Ground Truth:** measure with drone logs + video
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Full Drone System, video recorder
i. **Validation Owner:** Jackson Wood
j. **Post Test Actions:** scale back requirements, check power draw and battery level

V6.5 Validation Metric for R6.7 (<3m Stopping Distance):

a. **Purpose:** Ensures drone stops movement as soon as it receive a command to
b. **What is Tested:** Flight, Task Manager
c. **Significance:** users need to have ability to completely shut down drone operation
d. **Measurement Method:** Confirm the drone's ability to come to a stop within a distance less than 3m.
e. **Procedure:** Release controller input mid flight and measure the distance covered by the drone to ensure it is less than 3m.
f. **Comparison to Ground Truth:** Measure stopping distance
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Full Drone System, Remote Controller/Base Station, Tape Measure

i. **Validation Owner:** Jackson Wood
j. **Post Test Actions:** operate a slower speeds, adjust requirement

V6.6 Validation Metric for R6.8 (>5Hz Path Planning):

a. **Purpose:** Ensures Path Planning Updates at a regular and usable speed
b. **What is Tested:** Path Planning
c. **Significance:** Update path planning depending on detected obstacles and waypoint achievements
d. **Measurement Method:** Confirm the drone's path planning frequency is greater than 5Hz.
e. **Procedure:** Execute path planning commands and validate that the drone plans paths at a frequency exceeding 5Hz.
f. **Comparison to Ground Truth:** measure frequency with logs
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Full Drone System
i. **Validation Owner:** Jackson Wood
j. **Post Test Actions:** Adjust Requirements

V6.7 Validation Metric for R6.9 (>1Hz Map Update):

a. **Purpose:** Ensures Map Updates at a regular and usable speed
b. **What is Tested:** Mapping
c. **Significance:** Global/Local Mapping should update frequently to ensure accurate surroundings
d. **Measurement Method:** Confirm the drone's map update frequency is greater than 1Hz.
e. **Procedure:** Monitor map updates and verify that the drone updates maps at a frequency surpassing 1Hz.
f. **Comparison to Ground Truth:** measure rate with logs
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Full drone system
i. **Validation Owner:** Jackson Wood
j. **Post Test Actions:** adjust requirements, divert more compute power to mapping

V7: Validation for R7 (Robot Must Have Modular Accessory Interface)

Objective: Verify that the system meets the specified capabilities for responding quickly.

V7.1 Validation Metric for R7.1 (Provide Power):

a. **Purpose:** Ensure the drone's modular accessory interface can reliably provide power to connected accessories.
b. **What is Tested:** Power Subsystem, Modules
c. **Significance:** The ability to provide power through the modular accessory interface is crucial for the functionality and versatility of the drone system.

d. **Measurement Method:** Validate the drone's ability to provide power through the modular accessory interface.
e. **Procedure:** Connect various accessories to the interface and confirm that they receive power as intended. Ensure power loss is less than 0.1 V.
f. Comparison to Ground Truth: Using voltage meter to obtain readings.
g. Meeting Logistics: 5:00 3/20/2024 Wightman Park
h. Equipment: Drone system, modules , voltage meter
i. Validation Owner: Dustin Moss
j. Post Test Actions: Investigate potential causes such as interface compatibility issues or power distribution problems. Take necessary actions to address any identified issues and ensure that the modular accessory interface reliably provides power as required.

V7.2 Validation Metric for R7.2 (Provide Control Commands for Specific Actuators):

a. Purpose: Ensure the drone's modular accessory interface can effectively send control commands for specific actuators
b. What is Tested: Modules, Autonomy Computer
c. Significance: The ability to provide control commands for specific actuators through the modular accessory interface is essential for controlling various functionalities of the drone system.
d. **Measurement Method:** Confirm the drone's ability to send control commands for specific actuators through the modular accessory interface.
e. **Procedure**: Connect modules with actuators to the interface, initiate control commands, and verify that the actuators respond within 2 seconds.
f. Comparison to Ground Truth: Use a stopwatch to get true time and benchmark the tests
g. Meeting Logistics: 5:00 3/20/2024 Wightman Park
h. Equipment: Drone System, Modules, Stop Watch
i. Validation Owner: Dustin
j. Post Test Actions: Investigate potential causes such as communication protocols or compatibility issue

V7.3 Validation Metric for R7.3 (Save Log/Scientific Data):

a. Purpose: Ensure the drone can effectively save logs and scientific data through the modular accessory interface.
b. What is Tested: Modules, Autonomy Subsystem
c. Significance: The ability to save logs and scientific data is crucial for recording and analyzing flight information and scientific observations.
d. **Measurement Method:** Validate the drone's capability to save logs and scientific data through the modular accessory interface.

e. **Procedure**: Connect data logging or scientific modules to the interface, record data, and confirm that the drone saves logs or scientific data as required. Ensure that there is less than 1% data loss.
f. Comparison to Ground Truth: Compare the recorded data with the expected data to ensure accuracy and completeness, in this case temperature and humidity.
g. Meeting Logistics: 5:00 3/20/2024 Wightman Park
h. Equipment: Modules, Drone System, Temperature and Humidity Sensors
i. Validation Owner: Dustin
j. Post Test Actions: Investigate potential causes such as storage limitations or communication errors

V7.4 Validation Metric for R7.4 (Provide Mounting Provisions):
a. Purpose: Ensure the drone's modular accessory interface includes adequate mounting provisions.
b. What is Tested: Frame, Modules
c. Significance: Proper mounting provisions are essential for securely attaching accessories to the drone, ensuring stability and functionality during operation.
d. **Measurement Method:** Confirm the presence of adequate mounting provisions in the modular accessory interface.
e. **Procedure:** Attempt to mount various accessories on the drone using the modular accessory interface and ensure secure and stable attachment. Ensure at least 100N of force does not remove modules.
f. Comparison to Ground Truth: Using a force meter
g. Meeting Logistics: 5:00 3/20/2024 Wightman Park
h. Equipment: Force Meter, Modules, Drone
i. Validation Owner: Dustin
j. Post Test Actions: Investigate potential causes such as design flaws or compatibility issues.

V8: Validation for R8 (Robot Must Comply with Federal and Local Regulations)
Objective: Verify that the system complies with federal and local regulations.
a. Purpose: The purpose of this test is to ensure that we are complying with federal and local standards when testing our system.
b. What is Tested: The system against local and federal regulations.
c. Significance: This determines the usability of the overall system.
d. **Measurement Method:** Conduct a comprehensive review of the system's design, features, and operational aspects to ensure alignment with relevant federal and local regulations governing drone usage.
e. **Procedure:** Procedure: Conduct a comprehensive review by compiling a list of relevant federal and local regulations governing drone usage. Cross-reference the drone system's design, features, and operational capabilities against these

regulations, verifying compliance with safety, airspace, communication, and data privacy requirements. Ensure 0 regulations or rules are broken.

f. **Comparison to Ground Truth:** The number of rules broken must be 0.
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Drone
i. **Validation Owner:** Jackson Wood
j. **Post Test Actions:** Change any and all elements of the drone that violate these regulations so that they no longer do.

V9: Validation for R9 (Robot Must Perform in Described Weather Conditions)

Objective: Verify that the system performs in described weather conditions.

V9.1 Validation Metric for R9.1 (Temperature Performance):

a. **Purpose:** The purpose of this test is to ensure the robot can perform in standard temperature conditions.
b. **What is Tested:** Ability of the system to perform is standard temperatures.
c. **Significance:** This determines the usability of the overall system.
d. **Measurement Method**: Assess the robot's performance in temperatures below 50°C by subjecting it to varying temperature levels within the specified range.
e. **Procedure:** Expose the robot to different temperature conditions and verify its functionality to ensure it operates effectively below 40°C and above -5 °C.
f. **Comparison to Ground Truth:** Performance observation can be compared to the average temperature in the area.
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Thermometer, various temperature settings, Drone
i. **Validation Owner:** Tahaseen Shaik
j. **Post Test Actions:** Add some sort of temperature proofing in order to ensure it can perform in a range of temperatures.

V9.2 Validation Metric for R9.2 (Wind Performance):

a. **Purpose:** The purpose of this test is to ensure the robot can perform in standard wind state conditions.
b. **What is Tested:** Ability of the drone to perform despite wind.
c. **Significance:** This determines the usability of the overall system.
d. **Measurement Method:** Evaluate the robot's capabilities in wind speeds below 15 mph by subjecting it to various wind conditions.
e. **Procedure:** Test the robot's performance in different wind speeds and confirm that it operates efficiently in conditions below 15 mph.
f. **Comparison to Ground Truth:** Observations of robot performance can be directly compared to performance at 0 mph wind.
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Drone

i. **Validation Owner:** Tahaseen Shaik
j. **Post Test Actions:** Update position hold code in order to account for wind and move to positions accordingly.

V9.3 Validation Metric for R9.3 (Wet Ground Performance):

a. **Purpose:** The purpose of this test is to ensure that the robot is functional even when the ground is muddy. Essentially ensuring that nothing critical is damaged when landing and/or taking off from wet ground.
b. **What is Tested:** Ability of robot to perform on wet ground.
c. **Significance:** This determines the usability of the overall system.
d. **Measurement Method**: Verify the robot's functionality on wet ground without precipitation by simulating wet environments.
e. **Procedure:** Create wet ground conditions and assess the robot's performance to ensure it operates effectively without precipitation. Ensure performance is not worse by 5% of expected performance.
f. **Comparison to Ground Truth:** Observations of performance can be compared to the robot performance on dry ground.
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Drone, Drone
i. **Validation Owner:** Tahaseen Shaik
j. **Post Test Actions:** Weatherproof landing gear such that it is able to perform on various ground environments.

V9.4 Validation Metric for R9.4 (Lighting Performance):

a. **Purpose:** The purpose of this test is to verify that the robot is able to perform in the various lighting modes of daylight.
b. **What is Tested:** Ability of robot to perform in various lighting settings.
c. **Significance:** This determines the usability of the overall system.
d. **Measurement Method:** Confirm the robot's performance under daylight and external lighting conditions, ensuring its sensors and cameras function effectively.
e. **Procedure:** Test the robot's sensors and cameras in various lighting conditions to ensure optimal performance under daylight and external light. Ensure it is capable of operation with at least 30 lux of light and below 100,000 lux.
f. **Comparison to Ground Truth:** The performance can be directly compared to the standard lighting.
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Different light settings, Drone
i. **Validation Owner:** Jackson Wood
j. **Post Test Actions:** Update cameras and sensors such that they work in all daylight lighting situations. Update requirements if infeasible.

V10: Validation for R10 (Robot Must Not Disturb the Environment)

Objective: Verify that the system does not disturb the environment.

V10.1 Validation Metric for R10.1 (Noise Level):

a. **Purpose:** The purpose of this test is to verify that our system does not contribute to the sound pollution in the area and is minimally destructive.
b. **What is Tested:** The dB of the sound output of the drone
c. **Significance:** Requires us to consider the drone's impacts on the environment
d. **Measurement Method:** Confirm that the robot generates less than 65 decibels of noise at a distance of 20 feet.
e. **Procedure:** Measure the noise produced by the robot at a distance of 20 feet and ensure it remains below the 65 db.
f. **Comparison to Ground Truth:** Measurement can be directly compared against ground truth
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Phone Application, Drone
i. **Validation Owner:** Tahaseen Shaik
j. **Post Test Actions:** No action if passes test, otherwise work to reduce noise by greasing motors and other noise sources.

V10.2 Validation Metric for R10.2 (Environment Impact):

a. **Purpose:** The purpose of this test is to ensure that the drone does not damage the surrounding infrastructure and natural environment by avoiding collisions.
b. **What is Tested:** The drone's ability to avoid obstacles and reduce the damage inflicted on surroundings.
c. **Significance:** Requires us to consider the drone's impacts on the environment
d. **Measurement Method:** Validate that the robot does not damage its surroundings during operation.
e. **Procedure:** Observe the robot's movements and interactions with the environment, ensuring that it does not cause any damage or disruption to its surroundings. Ensure this holds true during a 20 minute operation of the robot.
f. **Comparison to Ground Truth:** Observations can be compared to the ground truth directly.
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Video device, Drone
i. **Validation Owner:** Tahaseen Shaik
j. **Post Test Actions:** Update avoidance algorithm if failed, otherwise account for specific methods of failure.

V11: Validation for R11 (Robot Must Be Portable)

Objective: Verify that the system is portable

V11.1 Validation Metric for R11.1 (Weight):

- **a. Purpose:** The purpose of this test is to verify that the robot can be easily carried by users.
- **b. What is Tested:** The weight of the robot
- **c. Significance:** This helps determine the overall usability of the drone
- d. **Measurement Method:** Confirm that the robot's weight is less than 3000 grams.
- e. **Procedure:** Weigh the robot without batteries and verify that it meets the specified weight requirement of less than 3000 grams.
- **f. Comparison to Ground Truth:** The weight can be directly compared
- **g. Meeting Logistics:** 1:00 3/24/2024, Wightman Park
- **h. Equipment:** Weighing machine, Drone
- **i. Validation Owner:** Dustin Moss
- **j. Post Test Actions:** Reduce weight or update requirement if infeasible to reduce weight further

V11.2 Validation Metric for R11.2 (Size):

- **a. Purpose:** The purpose of this test is to verify that the robot can be easily transported by carrying or in the trunk of a car.
- **b. What is Tested:** The size dimensions of the robot
- **c. Significance:** This helps determine the overall usability of the drone
- d. **Measurement Method:** Validate that the robot's size adheres to the specified dimensions of a diameter smaller than 1 meter and a height less than 0.5 meters.
- e. **Procedure**: Measure the robot's dimensions and ensure it complies with the size constraints of a diameter smaller than 1 meter and a height less than 0.5 meters.
- **f. Comparison to Ground Truth:** Measurement can be directly compared
- **g. Meeting Logistics:** 1:00 3/24/2024, Wightman Park
- **h. Equipment:** Measuring tape, Drone
- **i. Validation Owner:** Dustin Moss
- **j. Post Test Actions:** Reduce size if possible, otherwise update the requirement

V11.3 Validation Metric for R11.3 (Modular Frame):

- **a. Purpose:** The purpose of this test is to verify that the robot can be assembled in a modular fashion to promote transportability.
- **b. What is Tested:** The ability to break the frame down
- **c. Significance:** This helps determine the overall usability of the drone
- d. **Measurement Method:** Confirm that the robot's frame is modular in design.
- e. **Procedure:** Assess the construction of the robot's frame and verify that it follows a modular design, allowing for adaptability and customization. Ensure that the frame can be broken up into greater than 2 pieces.
- **f. Comparison to Ground Truth:** Observations can be directly compared
- **g. Meeting Logistics:** 1:00 3/24/2024, Wightman Park
- **h. Equipment:** Drone
- **i. Validation Owner:** Dustin Moss

j. **Post Test Actions:** Increase modularity by adding more detachment points for assembly.

V11.4 Validation Metric for R11.4 (Swappable Battery):

a. **Purpose:** The purpose of this test is to ensure that the battery is easily swappable by the user.
b. **What is Tested:** The swapability of the battery
c. **Significance:** This helps determine the overall usability of the drone
d. **Measurement Method:** Validate the robot's ability to use swappable batteries.
e. **Procedure:** Swap out the robot's battery with a different one and confirm that it operates seamlessly with the new battery, demonstrating the capability for swappable power sources. Ensure swap takes less than 30 seconds to complete.
f. **Comparison to Ground Truth:** Time measurement can be directly compared.
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Stopwatch, Drone
i. **Validation Owner:** Jacob Hsu
j. **Post Test Actions:** Update mounting mechanism to promote easy use.

V12: Validation for R12 (Robot Must Save Map and Log Information Locally)

Objective: Verify that the system saves map data and logs information locally.

a. **Purpose:** The purpose of this test is to ensure the robot is able to save important information locally for easy access when parsing through data.
b. **What is Tested:** Ability to access map and log information locally.
c. **Significance:** This helps determine the overall usability of the drone
d. **Measurement Method:** Confirm the robot's ability to save map and log information locally.
e. **Procedure:** Procedure: Initiate mapping and logging commands, and verify that the robot successfully saves the generated map and log information in its local storage. Ensure less than 5% data loss.
f. **Comparison to Ground Truth:** Quantity of data loss can be directly compared
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Drone
i. **Validation Owner:** Jackson Wood
j. **Post Test Actions:** Update save location and procedure to avoid greater data loss

ARM Module

V13: Validation for R13 (Robot Must be Directly Controlled by Robot Computer)

Objective: Verify that the system is directly controlled by the robot computer.

a. **Purpose:** The purpose of this test is to ensure that the robot is being controlled by the robot computer.
b. **What is Tested:** The control pipeline of the robot.
c. **Significance:** This determines where computational loads are being taken

d. **Measurement Method:** Confirm that the robot is directly controlled by its onboard computer.
e. **Procedure:** Execute control commands and ensure that they are processed directly by the robot's onboard computer, demonstrating the system's capability for autonomous control. Ensure errors in commands are less than 5% of expected performance.
f. **Comparison to Ground Truth:** Error rate can be directly compared.
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Drone, ARM module
i. **Validation Owner:** Jackson Wood
j. **Post Test Actions:** Change the implemented architecture to meet the specified one in the block diagram.

V14: Validation for R14 (ARM Module Must be Able to Grasp Various Objects)

Objective: Verify that the system is capable of grasping various objects.

V14.1 Validation Metric for R14.1 (Grasping Objects):

a. **Purpose:** The purpose of this test is to verify that the designed claw for the ARM is able to grasp various objects, a perceived use case.
b. **What is Tested:** Ability of ARM to grasp handles.
c. **Significance:** This determines the usability of the ARM module.
d. **Measurement Method:** Confirm the robot's ability to grasp objects.
e. **Procedure:** Introduce objects of varying shapes and sizes, and verify that the robot can successfully grasp and manipulate them. Ensure the robot arm can pick up objects of widths 10 mm to 80 mm.
f. **Comparison to Ground Truth:** The range of handle sizes ARM can pick up can be directly compared.
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** A range of objects widths from 10mm to 80mm, Drone, ARM module
i. **Validation Owner:** Jacob Hsu
j. **Post Test Actions:** Change design of claw in order to encompass the handle and/or grab specified range.

V14.2 Validation Metric for R14.2 (Grasping Cylinders):

a. **Purpose:** The purpose of this test is to ensure that the ARM can grab cylinders up to 80mm in diameter.
b. **What is Tested:** Ability of ARM to grab cylinders.
c. **Significance:** This determines the usability of the ARM module.
d. **Measurement Method:** Validate the robot's capability to grasp cylinders with a diameter of up to 80mm.

e. **Procedure:** Introduce cylinders of different diameters within the specified range, and ensure that the robot can grasp and handle them effectively. Ensure the robot arm can pick up cylinders of sizes 10 mm to 80mm.
f. **Comparison to Ground Truth:** The range of cylinder sizes ARM can pick up can be directly compared.
g. **Meeting Logistics:** 1:00 3/24/2024, Wightman Park
h. **Equipment:** Cylinders of diameters ranging from 10mm - 80mm, Drone, ARM module
i. **Validation Owner:** Jacob Hsu
j. **Post Test Actions:** Change design of claw in order to encompass the cylinders and/or grab specified range.

V14.3 Validation Metric for R14.3 (Weight Handling):

a. **Purpose:** The purpose of this test is to verify the ARM is able to carry a standard weight.
b. **What is Tested:** Ability of ARM to lift weight.
c. **Significance:** This determines the usability of the ARM module.
d. **Measurement Method:** Confirm the robot's ability to grasp and handle objects weighing up to 2 pounds.
e. **Procedure:** Introduce objects with varying weights within the specified range, and verify that the robot can grasp and manipulate them without exceeding the weight limit. Ensure the robot arm can pick up weights up to 2 additional pounds.
f. **Comparison to Ground Truth:** Observations can be directly compared.
g. **Meeting Logistics:** 12:00 3/24/2024, Wightman Park
h. **Equipment:** 5lbs weight, Drone, ARM module
i. **Validation Owner:** Jacob Hsu
j. **Post Test Actions:** Check limiting factors and update the requirements if infeasible to carry beyond a certain base weight.

V15: Validation for R15 (Extension Outside Robot Frame)

Objective: Verify that the ARM module can extend outside the robot frame.

a. **Purpose:** The purpose of this test is to ensure that ARM can extend outside the robot frame to pick up objects.
b. **What is Tested:** Ability of ARM to extend.
c. **Significance:** This determines the usability of the ARM module.
d. **Measurement Method:** Confirm that the robot's manipulator extends 0.5 meters outside of the robot frame.
e. **Procedure:** Activate the manipulator extension command and measure the distance it extends beyond the robot frame, ensuring it reaches the specified 0.5 meters.
f. **Comparison to Ground Truth:** Length of extension can be directly compared.
g. **Meeting Logistics:** 12:00 3/24/2024, Wightman Park

h. **Equipment:** Measuring tape, Drone, ARM module
i. **Validation Owner:** Jacob Hsu
j. **Post Test Actions:** Extend ARM links as needed to ensure that it is able to extend outside the frame.

V16: Validation for R16 (Positioning)

Objective: Verify that the ARM module can position accurately.

V16.1 Validation Metric for R16.1 (**Position Accuracy**)

a. **Purpose:** The purpose of this test is to verify that the ARM is able to accurately position itself in order to grab targets easily.
b. **What is Tested:** Ability of ARM to move to position accurately.
c. **Significance:** This determines the usability of the ARM module.
d. **Measurement Method:** Validate the robot's accuracy in positioning.
e. **Procedure:** Command the robot to assume different positions, and employ measurement tools to verify that it accurately achieves the specified positions relative to its surroundings with less than 10mm error.
f. **Comparison to Ground Truth:** Error can be directly compared.
g. **Meeting Logistics:** 12:00 3/24/2024, Wightman Park
h. **Equipment:** Measuring tape, Drone, ARM module
i. **Validation Owner:** Jacob Hsu
j. **Post Test Actions:** Update positioning control to reduce error as much as possible.

V17: Validation for R17 (Response)

Objective: Verify that the ARM module can respond quickly.

V17.1 Validation Metric for R17.1 (**Quick Response**)

- **a. Purpose:** The purpose of this test is to ensure that the ARM module is able to quickly respond to commands in order to grab targets as efficiently as possible.
- **b. What is Tested:** Drone's response time.
- **c. Significance:** This determines the usability of the ARM module.
- d. **Measurement Method:** Validate the arm's response time to commands.
- e. **Procedure:** Initiate commands for arm movements, measure the time taken for the arm to reach a stable position from the initial command, and ensure it meets the requirement of a response time of less than 2 seconds.
- **f. Comparison to Ground Truth:** The time taken can be directly compared.
- **g. Meeting Logistics:** 1:00 3/24/2024, Wightman Park
- **h. Equipment:** Stopwatch, Drone, ARM module
- **i. Validation Owner:** Jacob Hsu
- **j. Post Test Actions:** Reduce latency as much as possible to increase responsiveness. Potentially also update user interface to be more intuitive.

SkyHook Module

V18: Validation for R18 (Direct Control by Robot Computer)

Objective: Verify that the SkyHook Module can be directly controlled by the robot computer.

- **a. Purpose:** The purpose of this test is to ensure that the SkyHook module is being controlled by the robot computer.
- **b. What is Tested:** The control pipeline of the robot.
- **c. Significance:** This determines the usability of the SkyHook module.
- d. **Measurement Method:** Confirm that the robot is directly controlled by its onboard computer.
- e. **Procedure:** Execute control commands and ensure that they are processed directly by the robot's onboard computer, demonstrating the system's capability for autonomous control. Ensure that commanded action is within 1-meter of expected performance.
- **f. Comparison to Ground Truth:** Performance can be directly compared to tape measure.
- **g. Meeting Logistics:** 12:00 3/24/2024, Wightman Park
- **h. Equipment:** Drone, SkyHook module
- **i. Validation Owner:** Dustin Moss
- **j. Post Test Actions:** Update implementation to follow outlined system architecture for software with correct command pipeline

V19: Validation Metric for R19 (Lowering Capability):

Objective: Confirm that the SkyHook Module can lower objects to a distance of 100 feet.

a. **Purpose:** The purpose of this test is to ensure the SkyHook module is able to lower the hook to at least 100ft to ensure it can carry many objects.
b. **What is Tested:** Length of SkyHook cable drop down.
c. **Significance:** This determines the usability of the SkyHook module.
d. **Measurement Method:** Confirm the actual distance the SkyHook Module descends when executing the lowering command.
e. **Procedure:** Execute the lowering command for the SkyHook Module and measure the distance it descends. Validate that the module can achieve the required lowering capability of 100 feet.
f. **Comparison to Ground Truth:** The dropped length can be directly compared.
g. **Meeting Logistics:** 12:00 3/24/2024, Wightman Park
h. **Equipment:** Measuring tape, Drone, SkyHook module
i. **Validation Owner:** Dustin Moss
j. **Post Test Actions:** Change spool or cable size in response to the measure lengths in the test. Update requirements if infeasible.

V20: Validation Metric for R20 (Positioning Capability):

Objective: Confirm that the SkyHook Module can accurately position itself within a 1m range.

a. **Purpose:** The purpose of this test is to ensure that the SkyHook is able to drop down within a set range from the user in order for interaction to occur.
b. **What is Tested:** Position accuracy of drone.
c. **Significance:** This determines the usability of the SkyHook module.
d. **Measurement Method:** Measure the actual position of the SkyHook Module when commanded to a specific location.
e. **Procedure:** Execute commands to position the SkyHook Module, measure the actual position, and verify that it can achieve accurate positioning within the specified 10-meter range.
f. **Comparison to Ground Truth:** The distance from intended origin can be compared directly.
g. **Meeting Logistics:** 12:00 3/24/2024, Wightman Park
h. **Equipment:** Measuring tape, Drone, SkyHook module
i. **Validation Owner:** Dustin Moss
j. **Post Test Actions:** Update position command code in order to be more accurate or update requirement if infeasible.

V21: Validation Metric for R21 (Object Holding Capacity):

Objective: Confirm that the SkyHook Module can hold objects weighing up to 3.5 pounds.

a. **Purpose:** The purpose of this test is to ensure that the SkyHook is able to carry a set standard weight.
b. **What is Tested:** Ability of the SkyHook to support weight
c. **Significance:** This determines the usability of the SkyHook module.

d. **Measurement Method:** Assess the module's ability to securely hold objects of varying weights, up to 3.5 pounds.
e. **Procedure:** Attach objects of various weights to the SkyHook Module, initiate the holding command, and ensure that the module can securely hold objects up to 3.5 lbs.
f. **Comparison to Ground Truth:** The amount of weight carried can be directly compared.
g. **Meeting Logistics:** 12:00 3/24/2024, Wightman Park
h. **Equipment:** Weights of varying sizes, Drone, SkyHook module
i. **Validation Owner:** Dustin Moss
j. **Post Test Actions:** Increase SkyHook strength in case of component failure and/or update the requirements if infeasible.

V22: Validation Metric for R22 (User Interaction):

Objective: Confirm that the SkyHook Module requires user interaction for clipping the hook.

a. **Purpose:** The purpose of this test is to verify the ease for a user to clip the load onto the SkyHook for use.
b. **What is Tested:** User's ability to clipping onto the hook.
c. **Significance:** This determines the usability of the SkyHook module.
d. **Measurement Method:** Observe and verify user interaction during the hook-clipping process.
e. **Procedure:** Instruct a user to clip the hook onto an object using the SkyHook Module. Observe the interaction and verify that the module necessitates user involvement in the hook-clipping process. Ensure the process takes less than 10 seconds to perform.
f. **Comparison to Ground Truth:** The time taken to perform the task can be directly compared,
g. **Meeting Logistics:** 12:00 3/24/2024, Wightman Park
h. **Equipment:** Stopwatch, Drone, SkyHook module
i. **Validation Owner:** Tahaseen Shaik
j. **Post Test Actions:** If it takes too long, create a more intuitive system or better instruct users on use of SkyHook.

Science Payload

V23: Validation for R23 (Robot Must Not Impair Readings)

Objective: Confirm that the Science Payload Module can measure and record data during flight.

a. **Purpose:** Verify the Science Payload Module's capability to maintain accurate readings despite changes in orientation.
b. **What is Tested:** Science Payload Module
c. **Significance:** Ensuring reliable data collection.

d. **Measurement Method:** Alter flight of drone and orientation of the module and assess its ability to maintain accurate readings.
e. **Procedure:** Request measurements from Science Module during flight, ensure measurements values are consistent with baseline estimates, and that the values vary as expected
f. **Comparison to Ground Truth:** Benchmark Tests
g. **Meeting Logistics:** 12:00 3/24/2024, Wightman Park
h. **Equipment:** Science Payload Module, Drone System, Temperature / Humidity Sensor
i. **Validation Owner:** Jackson Wood
j. **Post Test Actions:** If orientation effects readings, potentially move sensor location or code.

V24: Validation Metric for R24 (Integration with Robot Computer)

Objective: Verify that the Science Payload Module seamlessly integrates with the robot computer for data collection.

a. **Purpose:** Ensure that the Science Payload integrates with the robot computer for control and coordination.
b. **What is Tested:** Science Payload, Autonomy Subsystem
c. **Significance:** This is necessary for our system to correctly work
d. **Measurement Method:** Assess the communication and coordination between the Science Payload Module and the robot computer during operation.
e. **Procedure:** Command drone to record data from science payload while in flight, ensure logs update at an expected frequency, and relevant numbers are reported to the operator.
f. **Comparison to Ground Truth:** Benchmark Tests
g. **Meeting Logistics:** 12:00 3/24/2024, Wightman Park
h. **Equipment:** Science Payload, Full Drone System
i. **Validation Owner:** Jackson Wood
j. **Post Test Actions:** Tweak communication or science module code if needed.

Functionality Verification

V25. Verification Metric for Frame

a. Frame supports all of the drone components [R11]

V26. Verification Metric for Power

a. Each battery continuously discharges at 24 V and 45 amps [R1.6]
   i. Voltmeter
b. PDB adequately powers all components during flight, given ample battery [R1.6]
   i. Voltage Step down (5 V) functions

V27. Verification Metric for Air Mobility

a. Motors have specified ramp up/down and maximum output speeds as defined by specs
    i. Plot motor speeds given input over time, should be included in self-diagnosis procedure
b. IMU/Gyro operate with specified update frequencies [R6]
    i. Measure with ROS
c. ESC relays IMU/Gyro information to autonomy computer + sets motor speeds as expected [R6]
    i. Measure information update frequency with ros

V28. Verification Metric for Module
a. Module arm holds at least 5 lbs [R14.3]
    i. Test arm detached from drone to verify
b. Swappable with other modules from baseplate on drone [R7.4]
c. Module CPU sends/receives commands to/from module/task manager quickly and accurately [R6]
    i. Test Module inverse kinematic performance in isolation.

V29. Verification Metric for Autonomy Computer
a. Communicates between all subsystems well under 100% operation [R3]
    i. ROS monitors and reports all inputs and outputs

V30. Verification Metric for Communication
a. Wifi range and frequency are within specified parameters [R5.2]
    i. Test max wifi range and signals from static drone position

V31. Verification Metric for Sensing
a. Cameras, Lidar, and GPS operate with expected specs [R2]
    i. Measure reporting frequencies with ros

V32. Verification Metric for Mapping
a. Depth map is accurate to ~20 cm from a static/hover position [R6]
    i. Compare depthmap to live observations within a known/confined area.

V33. Verification Metric for Navigation
a. Path avoids obstacles based on depth map [R3.4]
    i. Overlay navigation path onto actual map generated ahead of time

V34. Verification Metric for Task Manager
a. Communication between any ros input/output from/to any sensor/subsystem occurs < 5 ms [R6]
    i. Record message send and receive times to subsystems periodically, and verify they are within bounds, given output frequencies of the subsystems

Performance Verification

V35. Verification Metric for Frame
a. Frame can support payload of 10 lbs [R1.5]

i. Support frame by motor contact points. Load center of frame with progressive weights up until 10 lbs
b. Frame damps oscillations up to motor frequency
i. Perform frequency FEA with load placed at the motor contact points. Analyze vibrations at drone center to ensure <2mm amplitude

V36. Verification Metric for Power
a. Total max power output of 24V 100Amp [R1]
b. Continuous power of 24V 20Amp [R1]
i. Test setup using main power source (Battery) connected to resistor output and measuring voltage and amperage output for increasing loads. Test 100 Amp output for a duration of 10 sec
c. Power distribution board can handle max load for 10 seconds [R1]
i. Setup distribution board with known power supply input. Attach load to all key component outputs. Ensure no failure for 10 sec at max drain.
d. Auxiliary power at 5V 10 Amp [R1]
i. Test voltage step down unit with power supply input. Connect output to a load and ensure output voltage and amperage remains consistent indefinitely.

V37. Verification Metric for Air Mobility
a. IMU and Gyro update frequency >200 hz [R6]
i. Connect the flight controller with the autonomy computer and log the message rate using ros.
b. Stability loop > 1000 hz [R6]
i. Record log of loop cycles and times in order to ensure rate
c. Angle position up to 30° in less than .5s [R6]
i. First simulate stability model and measure settling time given a step input of angle control
ii. Implement on drone and look at logs to ensure performance
d. XYZ position up to 2m in less than 4s [R1]
i. First simulate stability model and measure settling time given a step input of position control
ii. Implement on the drone and look at logs to ensure performance. This will also require connection with autonomy computer to get odometery.

V38. Verification Metric for Module
a. I/O rate of 10hz [R7]
i. Connect module with autonomy computer. Send commands and read responses from the module. Record rates using ros.
b. ARM Holds 5lb [R14]

i. Mount arm in multiple positions (straight down, straight forward, folded). Attach increasing weights in each configuration and ensure no loss of positional accuracy up to 5lbs

c. ARM has accuracy of 10mm [R16]

i. Command a variety of positions to arm. Measure end effector position to ensure each result is within 10 mm of commanded position.

V39. Verification Metric for Autonomy Computer

a. Ensure temperature stays below 90 °C during operation [R5]

i. Run all subsystems on a stationary computer. Record temperature using internal sensors.

b. CPU and GPU load are <100% during operation [R5]

i. Run all subsystems and monitor usage using task manager software. Ensure CPU and GPU are not at max.

V40. Verification Metric for Communication

a. Wifi data rate of 100 mbps [R5]

i. Send image data from autonomy computer to the user until bandwidth is maxed out. Record data rate using an access point to monitor speeds.

b. Wifi range of 400m [R5]

i. Hover drone in air at standard operating height. Walk away with a user computer up to 400m. Ensure connection is not lost

c. Serial input at 100 Hz [R6]

i. Send serial commands to the flight controller at a set rate. Use logging to ensure all messages are received cleanly up to 100 hz.

d. Serial delay < 10ms [R6]

i. Synchronize clocks between flight controller and user. Send message and record receive time on the flight controller. Ensure delay is below threshold.

V41. Verification Metric for Sensing

a. Depth map rate >30hz [R6]

i. Connect the stereo camera and run depth map creation code. Use ROS to check publish frequency

b. Odometry update >30hz [R6]

i. Connect the stereo camera and run the odometry code. Use ROS to check publish frequency

c. Support for multiple stereo cameras [R2, R6]

i. Connect multiple stereo cameras and ensure publish frequencies are maintained.

d. GPS update >5hz [R6]

i. Monitor publish rate using ROS

V42. Verification Metric for Mapping

a. Map creation and update <10 hz [R6]
    i. Use ROS to record monitor output rate
b. Map points accurate to 20 cm relative [R2]
    i. Create a map of a known environment and compare map positions vs real life measurements.

V43. Verification Metric for Navigation

a. Waypoint generation <1 sec [R6]
    i. Given a map and end waypoint record the amount of time it takes this subsystem to create a list of waypoints. Ensure duration is less than specification.

V44. Verification Metric for Task Manager

a. Incurs less than 1ms on any transmission [R6]
    i. Record message timestep when received from the user and record the time when the message reaches its destination subsystem. Ensure the difference in times is less than the requirement.

Failure Modes

Safety Hazards

The safety hazards of the system can be categorized into several key groups.

- Electrical Hazards:
    - Swapping modules may involve electrical connections, increasing the risk of short circuits or electric shocks. There may be additional electrical hazards due to the usage of a battery to power our system.
    - Mitigation: Ensure that the drone's power system is designed with proper insulation, and connectors are designed to prevent accidental contact. Modules should have foolproof connectors, and users should be trained to handle them safely. The mitigation for the batteries is a similar process.
- Mechanical Hazards:
    - Improperly secured or damaged modules could lead to mechanical failures in-flight. Additional mechanical hazards could be present due to the structure of the drone.
    - Mitigation: Implement a robust and standardized module attachment mechanism with clear indicators of proper attachment. Regularly inspect modules for any signs of wear or damage, and establish maintenance procedures for replacement.
- Software Hazards:
    - Incompatibility or software bugs when swapping modules or drone modes may affect the drone's stability and control.

  - Mitigation: Ensure a standardized interface for modules and drone modes and rigorous testing of new modules to verify compatibility. Implement software safeguards to detect anomalies and prevent unsafe operations.
- Module Specific Hazards
  - Each module comes with its own set of hazards in terms of physical and software interfacing. If the module does not integrate properly with the rest of the system, this can cause the module to not work as expected, if at all.
  - Mitigation: Ensure that all modules follow the same rules with regards to the interfacing of the system. Have clear specifications included with the drone so that users are able to easily integrate with our system.
- User Error:
  - Users may not follow proper procedures when swapping modules, leading to accidents.
  - Mitigation: Develop comprehensive user manuals and training programs. Implement safety features that require strict adherence to procedures for module swapping.

Fault Table

In order to be as thorough as possible, we have opted to maintain our FMEA in a spreadsheet format. By doing so, it is possible to frequently and efficiently update the FMEA as needed. This is important since the fault table is a live document and warrants its own maintenance. The most recent version of the fault table will be attached in Table 14 of the Appendix of this document and the live version can be viewed.

Failure Detection Tests

Below are the failure detection tests that will assist in identifying the faults listed in the fault table. They are organized in the same order with the addition of full system testing.

- Electrical Testing:
  - Power System Stress Testing:
    - Test Procedure: Simulate high-demand scenarios by rapidly swapping modules and monitor the power system.
    - Measurements: Record voltage and current fluctuations, temperature changes in the power system.
    - Results: Confirm that the power system can handle rapid module swaps without overheating or experiencing voltage irregularities.
- Mechanical Testing:
  - Shock and Vibration Test:
    - Test Procedure: Subject the system frame to controlled shocks and vibrations to simulate transportation and operational conditions.

    - Measurements: Monitor changes in alignment, structure, and overall system integrity.
    - Results: Assess the system's resilience to shocks and vibrations, identifying any potential points of failure or misalignment.
- Software Testing:
  - Communication Reliability Testing:
    - Test Procedure: Introduce controlled interference and evaluate the drone's communication system.
    - Measurements: Monitor signal strength, latency, and packet loss during interference scenarios.
    - Results: Ensure that the drone can maintain communication integrity and execute fail-safe procedures in the presence of communication disruptions.
  - Failure Recovery Testing:
    - Test Procedure: Introduce simulated faults such as disconnected modules or malfunctioning sensors and assess the drone's response.
    - Measurements: Record the time taken to detect and recover from faults, assess the effectiveness of fail-safe mechanisms.
    - Results: Verify that the drone can quickly identify and recover from failures without compromising safety.
  - Security Testing:
    - Test Procedure: Attempt unauthorized module swaps or tampering.
    - Measurements: Monitor the drone's response to unauthorized access attempts.
    - Results: Confirm the effectiveness of security features in preventing unauthorized modifications to the drone's configuration.
- Module Specific Testing
  - Module Compatibility Testing:
    - Test Procedure: Install each module on the drone and verify compatibility with the existing hardware and software.
    - Measurements: Record any anomalies in communication, power consumption, or flight behavior.
    - Results: Ensure that all modules work seamlessly with the drone's system without causing unexpected failures or errors.
  - SkyHook: Cable Strength and Integrity Test:
    - Test Procedure: Subject the retrieval cable to controlled tension tests.
    - Measurements: Measure the force applied and monitor cable elongation.
    - Results: Determine the breaking point and identify any signs of fraying or weakening, ensuring the cable meets specified strength requirements.
- User Error Testing
  - User Error Simulation:

        - Test Procedure: Instruct users to intentionally swap modules incorrectly and assess the drone's response.
        - Measurements: Record any system errors or deviations from normal operation.
        - Results: Ensure that the drone's design can handle user errors during module swapping without causing critical failures.
- Full System Testing
    - Functional Testing:
        - Test Procedure: Execute a series of predefined maneuvers and operations using each module.
        - Measurements: Monitor the drone's responses, power consumption, and communication signals.
        - Results: Confirm that the drone maintains stability and control during various tasks with different modules.
    - Environmental Testing:
        - Test Procedure: Expose the drone to various environmental conditions (wind, rain, temperature variations) while using different modules.
        - Measurements: Monitor the drone's performance, stability, and sensor accuracy.
        - Results: Validate that the drone can operate safely and effectively in diverse environmental conditions with different modules.
    - Long-Term Reliability Testing:
        - Test Procedure: Operate the drone continuously for an extended period with frequent module swaps.
        - Measurements: Monitor system performance, wear and tear of components, and battery degradation.
        - Results: Assess the long-term reliability of the drone and identify any cumulative issues that may arise with extended use.

Validation Testing

[AIR_TEST]

Robot

T1: Validation Test for R1 (Robot Must Fly)

Objective: Verify that the system meets the specified flying capabilities.

T1.1 Validation Test for R1.1 (Vertical Takeoff):

a. **Tested Subsystems:** Autonomy, Power, Communication, and Sensing
b. **Procedure:** Initiate the vertical takeoff command and measure the time until the drone reaches a stable hover at 2m. Ensure time is under 5 seconds.
c. **Data:** 3.8, 3.57, 3.81 s
d. **Results:** The average time for a 2m stable take off was 3.727 seconds with a standard deviation of 0.136

T1.2 Validation Test for R1.2 (Rotate in Place):

a. **Tested Subsystem:** Autonomy Subsystem, Power System, and Sensing
b. **Procedure:** Command the drone to rotate in place, measuring the angular rotation and ensuring it completes a full circle with less than 5 degrees of error.
c. **Data:** (1,3,4)
d. **Results:** The average rotation for a full commanded rotation was 2.667 degrees with a standard deviation of 1.528 degrees

T1.3 Validation Test for R1.3 (Hover for 20 mins):

a. **Tested Subsystems:** Autonomy Subsystem, Power System, and Sensing
b. **Procedure:** Activate the hover command and record the duration until the drone lands or exhibits any deviations. Ensure time is over 20 minutes..
c. **Data:** 23:50 min:sec
d. **Results:** The drone was able to hover for 23 minutes and 50 seconds in 18 mph winds while running fully autonomy and sensor readings.

T1.4 Validation Test for R1.4 (200 meters):

a. **Tested Subsystems:** Autonomy Subsystem, Power System, Communication, and Sensing
b. **Procedure:** Command the drone to fly in a straight line, recording the gps data of the distance covered until it reaches the maximum range or loses connection. Ensure the range is at least 200 meters.
c. **Data:** (220,225,230) +/- 5m
d. **Results:** The average distance until poor connection 225 meters

T1.5 Validation Test for R1.5 (5lb Payload):

a. **Tested Subsystems:** Frame and Power System
b. **Procedure:** Attach a 5lb weight to the drone and assess its capacity to hover and navigate with the added payload.

c. **Data:** The drone was able to fly with 6.5lb of additional payload
d. **Results:** The drone can successfully carry > 5 lb payload and operate efficiently

T2: Validation Test for R2 (Robot Must Sense)

Objective: Verify that the system meets the specified sensing capabilities.

T2.1 Validation Test for R2.1 (Front, Above, and Below Occupancy Map):

a. **Tested Subsystems:** Mapping Subsystem
b. **Procedure:** Activate sensors to detect obstacles in front, above, and below the drone, ensuring occupancy maps are continuously updated. Ensure update frequency is at least 5 times per second.
c. **Data:** (5Hz, 5Hz, 5Hz)
d. **Results:** The average update frequency of the occupancy map was 5 Hz.

T2.1.1 Validation Test for R2.1.1 (Depth Map Update at 30 fps - 2m Range):

a. **Tested Subsystems:** Mapping Subsystem
b. **Procedure:** Activate sensors within the 2m range and record the update frequency of the depth map. Ensure rate is at a minimum 30 frames per second.
c. **Data:** Took average fps over 40s: 50fps
d. **Results:** The average update frequency of the depth map at 2m was 50 fps

T2.2 Validation Test for R2.2 (360° Obstacle Detection - 10m Range):

a. **Tested Subsystems:** Mapping Subsystem, Sensor layout
b. **Procedure:** Surround the drone with obstacles and confirm that it successfully detects objects 360° in front of the drone within the 10m range.
c. **Data:** Objects were placed around the drone and the RVIZ output of the drone showed its ability to sense obstacles around it over 10m away.
d. **Results:** The drone can successfully sense obstacles over 10m away 360 degrees around the drone

T2.3 Validation Test for R2.3 (Map Creation):

a. **Tested Subsystems:** Mapping Subsystem
b. **Procedure:** Initiate the map creation process and confirm that data from different sensor streams are fused to generate a complete map. Ensure map accuracy of 50%.
c. **Data:** Through many separate tests of obstacle avoidance, path planning, and autonomous navigation, the drone is capable of creating maps with enough resolution to operate.
d. **Results:** Although not as high fidelity as originally desired the drone is capable of creating low resolution maps, enough to successfully path plan with

T2.4 Validation Test for R2.4 (Global Position - 1.5m Accuracy):

a. **Tested Subsystems:** Localization, GPS
b. **Procedure:** Obtain the global position coordinates and validate against a known reference point, ensuring the accuracy falls within 1.5m.

c. **Data:** We placed the drone on home plate of a baseball field and plotted where it thought it was geographically and took the distance from the drone to homeplate.
d. **Results:** The average gps accuracy fell within +/- 1.01m

T3: Validation Test for R3 (Robot Must be Capable of Autonomous Operation)

Objective: Verify that the system meets the specified capabilities for autonomous operation.

T3.1 Validation Test for R3.1 (Autonomously Stabilize with All Accessories):
- a. **Tested Subsystems:** Autonomy Computer, Flight Controller
- b. **Procedure:** Attach various modules to the drone and assess its ability to autonomously stabilize under normal operating conditions. Ensure capable of stabilization in under 5 seconds.
- c. **Data:** 0.453 s, 0.389 s, 0.564 s
- d. **Results:** The average time to stabilization was 0.4687 seconds

T3.2 Validation Test for R3.2 (Autonomously Hold Position):
- a. **Tested Subsystems:** Autonomy Computer, Flight Controller
- b. **Procedure:** Initiate the hold position command and verify that the drone maintains stability in terms of height, orientation, and position relative to the local environment. Ensure the drone can hold position with disturbances for at least 5 seconds.
- c. **Data:** We terminated the test after 3 successful 20+ second holds.
- d. **Results:** The average time of stabilization was over 20+ seconds.

T3.2.1 Validation Test for R3.2.1 (Height):
- a. **Tested Subsystems:** Flight Controller, Localization.
- b. **Procedure:** Command the drone to hold its height, and verify that it remains stable at the designated altitude. Ensure that deviations are less than +/- 0.5 m.
- c. **Data:** +/- 0.0425, 0.25, 0.124 m
- d. **Results:** The average deviation of height during flight was +/- 0.139 m with a standard deviation of 0.105 m

T3.2.2 Validation Test for R3.2.2 (Orientation):
- a. **Tested Subsystems:** Compass
- b. **Procedure:** Command the drone to hold a specific orientation, and verify that it maintains stability in the designated position. Ensure that deviations are less than +/- 10 degrees.
- c. **Data:** Over the duration of a 1:20 minute flight the drones orientation deviated by a max of -3.5 degrees and +5.8 degrees
- d. **Results:** Over the flight duration, the drone deviated on average by +/- 1.150 degrees which is less than the required deviations.

T3.2.3 Validation Test for R3.2.3 (Position Relative to Local Environment):
- a. **Tested Subsystems:** Localization and Sensor Fusion

b. **Procedure:** Command the drone to hold its position, and verify that it remains stable in the designated location relative to the local environment. Ensure that deviations are less than +/- 0.5 m.
c. **Data:** 0.213 ,0.32, 0.404 m
d. **Results:** The average position deviation on a hold was 0.312 m with standard deviation of 0.096 m.

T3.3 Validation Test for R3.3 (Fly to Set Position):

a. **Tested Subsystems:** Flight, Path planning
b. **Procedure:** Set a specific destination for the drone, initiate the flight command, and confirm that it reaches the designated autonomously. Ensure that deviations are less than +/- 0.5 m.
c. **Data:** 0.2659, 0.1483, 0.4267, 0.0450, 0.2347 m
d. **Results:** The average position deviation for a fly to position was 0.224 m with a standard deviation of 0.142 m.

T3.3.1 Validation Test for R3.3.1 (Return to Start):

a. **Tested Subsystems:** Flight, Path Planning
b. **Procedure:** Initiate the return to start command, and confirm that the drone successfully navigates to the starting point autonomously. Ensure that landing location is within 3m of initial take off.
c. **Data:** 0.2268, 0.127 m
d. **Results:** The average return to land position displacement was 0.178 m with a standard deviation of 0.072 m.

T3.4 Validation Test for R3.4 (Avoid Obstacles):

a. **Tested Subsystems:** Obstacle avoidance, sensing, navigation
b. **Procedure**: Introduce obstacles in the drone's path and validate its ability to autonomously navigate around them. Ensure that the drone is capable of avoiding obstacles within a projected flight path.
c. **Data:** We completed this test many separate times with successful avoidances
d. **Results:** The drone is capable of avoiding obstacles that come within 2 meters of its center.

T4: Validation Test for R4 (Robot Must Plan)

Objective: Verify that the system meets the specified capabilities for planning and task execution.

T4.1 Validation Test for R4.1 (Path Planning with Real-Time Obstacle Avoidance):

a. **Tested Subsystems:** Path Planning
b. **Procedure:** Introduce obstacles in the drone's path, initiate the path planning command, and validate that it plans a trajectory in real-time, avoiding obstacles. Ensure that the drone can avoid >1 obstacles for a >10m path plan.
c. **Data:** We completed this test many separate times with successful avoidances

d. **Results:** The drone was capable of avoiding obstacles and reaching a destination however we were unhappy with the speed and efficiency to do so.

T5: Validation Test for R5 (Robot Must Communicate with Operator)

Objective: Verify that the system meets specified capabilities for communicating with the operator.

T5.1 Validation Test for R5.1 (Serial Radio Link):

a. **Tested Subsystems:** Communication
b. **Procedure:** Test emergency stop, manual flight commands, and the transmission of basic status information over the serial radio link. Ensure latency is less than 500 ms.
c. **Data:** Over all of our testing, we have consistently experienced no noticeable latency, thus we can conclude that we are much under the requirement of 500ms.
d. **Results:** There is no noticeable latency between command and response of the drone. Thus it is much less than 500 ms.

T5.1.1 Validation Test for R5.1.1 (Emergency Stop):

a. **Tested Subsystems:** Planning, mode switching, communication
b. **Procedure:** Initiate the emergency stop (L, RTL, & SRTL) command and verify that the drone halts all operations within 2 seconds.
c. **Data:** Switching to an emergency mode like Return to Land is executed via a switch on the remote, and this has the same latency as any controller input which is significantly less than 500 ms and thus passes this requirement.
d. **Results:** The time to switch to manual flight was significantly less than 500 ms and thus is faster than the requirement of 2 seconds.

T5.1.2 Validation Test for R5.1.2 (Manual Flight Commands):

a. **Tested Subsystems:** Flight, Remote Communication
b. **Procedure:** Issue manual flight commands and confirm that the drone executes the commands within 2 seconds.
c. **Data:** Switching to manual flight from autonomous flight is executed via a switch on the remote, and this has the same latency as any controller input which is significantly less than 500 ms and thus passes this requirement.
d. **Results:** The time to switch to manual flight was significantly less than 500 ms and thus is faster than the requirement of 2 seconds.

T5.1.3 Validation Test for R5.1.3 (Returns Basic Status Information):

a. **Tested Subsystems:** Communication
b. **Procedure:** Request basic status information from the drone and validate the accuracy and completeness of the transmitted data within 4 seconds.
c. **Data:** System updates position at 100 Hz and all other status info 1 Hz
d. **Results:** The drone will return status information every second thus meeting the 4 second requirement.

T5.2 Validation Test for R5.2 (Wi-Fi Connection):

a. **Tested Subsystems:** Communication

b. **Procedure:** Test the reception of waypoint coordinates, execution of advanced flight commands, and the transmission of advanced status, maps, and logs over the Wi-Fi connection within 4 seconds.
c. **Data:** 0.56, 0.50 s
d. **Results:** The average response time of the wifi-connection was 0.530 seconds with a standard deviation of 0.042 seconds

T5.2.1 Validation Test for R5.2.1 (Receive Waypoint Coordinates):
a. **Tested Subsystems:** Communication
b. **Procedure:** Transmit waypoint coordinates to the drone and validate that it accurately receives and acknowledges the coordinates.
c. **Data:** 0.56, 0.50 s
d. **Results:** The average response time of the wifi-connection was 0.530 seconds with a standard deviation of 0.042 seconds

T5.2.2 Validation Test for R5.2.2 (Advanced Flight Commands):
a. **Tested Subsystems:** Communication, Planning, Flight
b. **Procedure:** Issue advanced flight commands and confirm that the drone executes the commands accurately and within 4 seconds.
c. **Data:** 0.56, 0.50 s
d. **Results:** The average time for advice flight command execution was x seconds.

T5.2.3 Validation Test for R5.2.3 (Returns Advanced Status, Maps, Logs):
a. **Tested Subsystems:** Communication, Logs
b. **Procedure:** Request advanced status, maps, and logs from the drone and validate the accuracy and completeness of the transmitted data within 4 seconds.
c. **Data:** System updates position at 100 Hz and all other status info 1 Hz
d. **Results:** The drone will return status information every second thus meeting the 4 second requirement.

T5.3 Validation Test for R5.3 (<1 hr Operator Training Time):
a. **Tested Subsystems:** UI, Communication
b. **Procedure**: Time the training session required for an operator to become familiar with the drone's controls, features, and communication systems, ensuring it is completed within at most 1 hour.
c. **Data:** We trained 3 people with various amounts of drone experience on how to operate the drone. Users were comfortable within 15, 32, 20 minutes
d. **Results:** The average time to learn drone operation was 22.333 minutes.

T6: Validation Test for R6 (Robot Must Respond Quickly)

Objective: Verify that the system meets the specified capabilities for responding quickly.

T6.1 Validation Test for R6.1 (>30Hz Depth Map Update):
a. **Tested Subsystems:** Depth Map

b. **Procedure:** Initiate depth map updates and measure the update frequency to ensure it exceeds 30Hz.
c. **Data:** Took average fps of 40s: 50fps
d. **Results:** The average update frequency of the depth map at 2m was 50 fps

T6.2 Validation Test for R6.4 (>50Mbps Wi-Fi Communication):
a. **Tested Subsystems:** Communication
b. **Procedure:** Test Wi-Fi communication and measure the data transfer rate to ensure it exceeds 50Mbps.
c. **Data:** Ran Iperf server and client to test data rate: 92.1, 92.8, 91.6, 85.5, 92.2
d. **Results:** The average Wi-Fi communication data transfer rate was 90.84 Mbps with a standard deviation of 3.015.

T6.3 Validation Test for R6.5 (>=5Hz GPS Update):
a. **Tested Subsystems:** GPS
b. **Procedure:** Monitor the drone's GPS updates and verify that the frequency is at least 5 Hz.
c. **Data:** 5, 5, 5 Hz
d. **Results:** The drone's GPS update frequency was 5 Hz. This is the frequency in which the cube orange pulls the GPS data.

T6.4 Validation Test for R6.6 (>30mph Flight Speed):
a. **Tested Subsystems:** Flight
b. **Procedure:** Measure the drone's flight speed and validate that it exceeds 30mph.
c. **Data:** Measured out 25 feet on the field and recorded in slow motion. Counted frames of video and determined it took 0.564s to cover the distance. 25ft/0.564s computes to 30.22 mph
d. **Results:** The drone was successful in flying over the required 30 mph specification

T6.5 Validation Test for R6.7 (<2m Stopping Distance):
a. **Tested Subsystems:** Flight, Task Manager
b. **Procedure:** Release controller input mid flight and measure the distance covered by the drone to ensure it is less than 3m.
c. **Data:** 2.1, 2.8, 1.75 m
d. **Results:** The average stopping distance was 2.217 m with a standard deviation of 0.535 m.

T6.6 Validation Test for R6.8 (>=1Hz Path Planning):
a. **Tested Subsystems:** Autonomy Computer
b. **Procedure:** Execute path planning commands and validate that the drone plans paths at a frequency exceeding 5Hz.
c. **Data:** 5 Hz
d. **Results:** The frequency in which the Jeston computer recalculates the path planning is 5 times per second.

T6.7 Validation Test for R6.9 (>1Hz Map Update):

a. **Tested Subsystems:** Autonomy Computer
b. **Procedure:** Monitor map updates and verify that the drone updates maps at a frequency surpassing 1Hz.
c. **Data:** 3 Hz
d. **Results:** The update frequency of our map is 3 Hz.

T7: Validation Test for R7 (Robot Must Have Modular Accessory Interface)

Objective: Verify that the system meets the specified capabilities for responding quickly.

T7.1 Validation Test for R7.1 (Provide Power):

a. **Tested Subsystems:** Power Subsystem, Modules
b. **Procedure:** Connect various accessories to the interface and confirm that they receive power as intended. Ensure power loss is less than 0.1 V.
c. **Data:** 12.1, 12.1, 12.1V
d. **Results:** The minimum operating voltage of our 24v-12v buck converter is 18V. We never drop below this during operation and thus the voltage output is consistently 12.1V.

T7.2 Validation Test for R7.2 (Provide Control Commands for Specific Actuators):

a. **Tested Subsystems:** Modules, Autonomy Computer
b. **Procedure**: Connect modules with actuators to the interface, initiate control commands, and verify that the actuators respond within 2 seconds.
c. **Data:** 1.73, 1.61, 1.82, 1.94 s
d. **Results:** The average time for actuator response was 1.775 seconds with a standard deviation of 0.140 seconds.

T7.3 Validation Test for R7.3 (Save Log/Scientific Data):

a. **Tested Subsystems:** Modules, Autonomy Subsystem
b. **Procedure:** Connect data logging or scientific modules to the interface, record data, and confirm that the drone saves logs or scientific data as required. Ensure that there is expected data
c. **Data:** Massive Log, See AIR_TEST_V1
d. **Results:** The drone is able to save module logs.

T7.4 Validation Test for R7.4 (Provide Mounting Provisions):

a. **Tested Subsystems:** Frame, Modules
b. **Procedure:** Attempt to mount various accessories on the drone using the modular accessory interface and ensure secure and stable attachment. Ensure at least 100N of force does not remove modules.
c. **Data:** Using 2 M3 Screws the modules can be secured to the frame without issue,
d. **Results:** The drone's modular accessory interface includes adequate mounting provisions. Dustin, pushing as hard as he could, could not break the mounting provision which we know is over the 100N (22.5lb) of force required.

T8: Validation Test for R8 (Robot Must Comply with Federal and Local Regulations)
Objective: Verify that the system complies with federal and local regulations.

a. **Tested Subsystems:** The system against local and federal regulations.
b. **Procedure:** Conduct a comprehensive review by compiling a list of relevant federal and local regulations governing drone usage. Cross-reference the drone system's design, features, and operational capabilities against these regulations, verifying compliance with safety, airspace, communication, and data privacy requirements. Ensure 0 regulations or rules are broken.
c. **Data:** Yes
d. **Results:** The system meets all required regulations.

T9: Validation Test for R9 (Robot Must Perform in Described Weather Conditions)
Objective: Verify that the system performs in described weather conditions.

T9.1 Validation Test for R9.1 (Temperature Performance):

a. **Tested Subsystems:** Ability of the system to perform in standard temperatures.
b. **Procedure:** Expose the robot to different temperature conditions and verify its functionality to ensure it operates effectively below 40°C.
c. **Data:** Coldest Day: 4°C | Hottest Day: 18.4°C
d. **Results:** Our components are all rated at 40°C. We never had the opportunity to test this hot, however we have tested in values ranging from 4.0-18.4°C.

T9.2 Validation Test for R9.2 (Wind Performance):

a. **Tested Subsystems:** Ability of the drone to perform despite wind.
b. **Procedure:** Test the robot's performance in different wind speeds and confirm that it operates efficiently in conditions below 15 mph.
c. **Data:** Tested position hold in varying wind conditions (0mph, 10mph, 16mph).
    i. **Results:** Drone performed as expected in 16mph winds. Position hold remained accurate and stable.

T9.3 Validation Test for R9.3 (Wet Ground Performance):

a. **Tested Subsystems:** Ability of robot to perform on wet ground.
b. **Procedure:** Create wet ground conditions and assess the robot's performance to ensure it operates effectively without precipitation. Ensure performance is not worse.
c. **Data:** Tested on muddy ground, wet grass and dry fields.
d. **Results:** Drone landed without damage in all scenarios and landing gear was safe.

T9.4 Validation Test for R9.4 (Lighting Performance):

a. **Tested Subsystems:** Ability of robot to perform in various lighting settings.
b. **Procedure:** Test the robot's sensors and cameras in various lighting conditions to ensure optimal performance under daylight and external light. Ensure it is capable of operation with at least 30 lux of light.

c. **Data:** Tested camera and sensor performance in various light environments using a phone lux meter to test. Values ranged from 25 lux - 100,000 lux.
d. **Results:** Our drone was able to perform in light ranges of 25-100,000 lux.

T10: Validation Test for R10 (Robot Must Not Disturb the Environment)

Objective: Verify that the system does not disturb the environment.

T10.1 Validation Test for R10.1 (Noise Level):

a. **Tested Subsystems:** The dB of the sound output of the drone
b. **Procedure:** Measure the noise produced by the robot at a distance of 20 feet and ensure it remains below the 65 dB.
c. **Data:** Average 45 dB with a max of 51.8 dB over a 45 second test flight.
d. **Results:** Through multiple tests the noise produced by the drone did not exceed 65 dB successfully staying quiet enough at 20 feet.

T10.2 Validation Test for R10.2 (Environment Impact):

a. **Tested Subsystems:** Frame
b. **Procedure:** Observe the robot's movements and interactions with the environment, ensuring that it does not cause any damage or disruption to its surroundings. Ensure this holds true during a 20 minute operation of the robot.
c. **Data:** Based on all of our testing which is much greater than 20 minutes of operation and even in crashing, we have not caused any damage or disruption to the surrounding environment.
d. **Results:** The drone successfully has little to no negative environmental impact over the course of operation.

T11: Validation Test for R11 (Robot Must Be Portable)

Objective: Verify that the system is portable

T11.1 Validation Test for R11.1 (Weight):

a. **Tested Subsystems:** Frame
b. **Procedure:** Weigh the robot without batteries and verify that it meets the specified weight requirement of less than 3000 grams.
c. **Data:** 2735.7 g
d. **Results:** Our drone weighs less than the mass requirement by 264.3g

T11.2 Validation Test for R11.2 (Size):

a. **Tested Subsystems:** Frame
b. **Procedure**: Measure the robot's dimensions and ensure it complies with the size constraints of a diameter smaller than 1 meter and a height less than 0.5 meters.
c. **Data:**Width 0.702 m, Height 0.421 m
d. **Results:** Our drone is compact enough to meet these size requirements with a width of 0.702 m and a height of 0.421 m

T11.3 Validation Test for R11.3 (Modular Frame):

a. **Tested Subsystems:** Frame
b. **Procedure:** Assess the construction of the robot's frame and verify that it follows a modular design, allowing for adaptability and customization. Ensure that the frame can be broken up into greater than 2 pieces.
c. **Data:** The robot can be easily split into 4 parts as well as fold to endure easy transportation
d. **Results:** The robot’s frame is modular enough to be easily portable.

T11.4 Validation Test for R11.4 (Swappable Battery):

a. **Tested Subsystems:** Power, Frame
b. **Procedure:** Swap out the robot's battery with a different one and confirm that it operates seamlessly with the new battery, demonstrating the capability for swappable power sources. Ensure swap takes less than 30 seconds to complete.
c. **Data:** (27.05s, 26.10s, 25.31s)
d. **Results:** Battery swap was completed in an average time of 26.15s

T12: Validation Test for R12 (Robot Must Save Map and Log Information Locally)

Objective: Verify that the system saves map data and logs information locally.

a. **Tested Subsystems:** Ability to access map and log information locally.
b. **Procedure:** Initiate mapping and logging commands, and verify that the robot successfully saves the generated map and log information in its local storage. Ensure expected logs
c. **Data:** Massive Log, See AIR_TEST_V1
d. **Results:** The drone is able to save module logs.

ARM Module

T13: Validation Test for R13 (ARM Must be Directly Controlled by Robot Computer)

Objective: Verify that the system is directly controlled by the robot computer.

a. **Tested Subsystems:** Module, Autonomy Computer
b. **Procedure:** Execute control commands and ensure that they are processed directly by the robot's onboard computer, demonstrating the system's capability for autonomous control. Ensure errors in commands are less than 5% of expected performance.
c. **Data:** We commanded multiple positions of the ARM using IK and the autonomy computer was capable of executing the commands with no observable error.
d. **Results:** The ARM module is successfully capable of being commanded by the autonomy computer.

T14: Validation Test for R14 (ARM Module Must be Able to Grasp Various Objects)

Objective: Verify that the system is capable of grasping various objects.

T14.1 Validation Test for R14.1 (Grasping):

a. **Tested Subsystems:** Ability of ARM to grasp handles.

b. **Procedure:** Introduce handles of varying shapes and sizes, and verify that the robot can successfully grasp and manipulate them. Ensure the robot arm can pick up objects of sizes 10 mm to 80mm in width.
c. **Data:** Min tested 2.81 mm, Max tested 80.05 mm
d. **Results:** The ARM was successfully able to grab objects with widths less than 10mm and greater the 80mm meeting the desired requirements

T14.2 Validation Test for R14.2 (Grasping Cylinders):

a. **Tested Subsystems:** Ability of ARM to grab cylinders.
b. **Procedure:** Introduce cylinders of different diameters within the specified range, and ensure that the robot can grasp and handle them effectively. Ensure the robot arm can pick up cylinders of sizes 10 mm to 80mm.
c. **Data:** Min tested 5.54 mm, Max tested 80.74
d. **Results:** The ARM was successfully able to grab cylinders with diameters less than 10mm and greater the 80mm meeting the desired requirements

T14.3 Validation Test for R14.3 (Weight Handling):

a. **Tested Subsystems:** Ability of ARM to lift weight.
b. **Procedure:** Introduce objects with varying weights within the specified range, and verify that the robot can grasp and manipulate them without exceeding the weight limit. Ensure the robot arm can pick up weights up to 2 additional pounds.
c. **Data:** Drone was successful in operating with a 2lb weight attached to the end effector of the ar,
d. **Results:** The maximum weight to hover was a 2 lb payload

T15: Validation Test for R15 (Extension Outside Robot Frame)

Objective: Verify that the ARM module can extend outside the robot frame.

a. **Tested Subsystems:** Ability of ARM to extend.
b. **Procedure:** Activate the manipulator extension command and measure the distance it extends beyond the robot frame, ensuring it reaches the specified 0.5 meters.
c. **Data:** The drone has the ability to reach 0.891 m forward of its center.
d. **Results:** Maximum reaching distance is 0.891 m, Flew comfortably at 0.629m.

T16: Validation Test for R16 (Positioning)

Objective: Verify that the ARM module can position accurately.

T16.1 Validation Test for R16.1

a. **Tested Subsystems:** Ability of ARM to move to position accurately.
b. **Procedure:** Command the robot to assume different positions, and employ measurement tools to verify that it accurately achieves the specified positions relative to its surroundings with less than 10mm error.
c. **Data:** Commanded change in position of (85,0,5) moved (87.3,0,3.9) mm
d. **Results:** The error in final position was 2.550 mm which passes the positioning error required.

T17: Validation Test for R17 (Response)

Objective: Verify that the ARM module can respond quickly.

T17.1 Validation Test for R17.1

a. **Tested Subsystems:** Drone's response time.
b. **Procedure:** Initiate commands for arm movements, measure the time taken for the arm to reach a stable position from the initial command, and ensure it meets the requirement of a response time of less than 2 seconds.
c. **Data:** 1.73, 1.61, 1.82, 1.94 s
d. **Results:** The average time for actuator response was 1.775 seconds with a standard deviation of 0.140 seconds.

SkyHook Module

T18: Validation Test for R18 (Direct Control by Robot Computer)

Objective: Verify that the SkyHook Module can be directly controlled by the robot computer.

a. **Tested Subsystems:** The control pipeline of the robot.
b. **Procedure:** Execute control commands and ensure that they are processed directly by the robot's onboard computer, demonstrating the system's capability for autonomous control. Ensure that commanded action is within 20 centimeters of expected performance.
c. **Data:** -0.09, -0.06, -0.08 m
d. **Results:** Our average error during a commanded positioning was within 1 m

T19: Validation Test for R19 (Lowering Capability):

Objective: Confirm that the SkyHook Module can lower objects to a distance of 100 feet.

a. **Tested Subsystems:** Length of SkyHook cable drop down.
b. **Procedure:** Execute the lowering command for the SkyHook Module and measure the distance it descends. Validate that the module can achieve the required lowering capability of 100 feet.
c. **Data:** We put 150 feet of rope on the spool
d. **Results:** The maximum distance lowered is 150 feet with current configuration.

T20: Validation Test for R20 (Positioning Capability):

Objective: Confirm that the SkyHook Module can accurately position itself within a 1m range.

a. **Tested Subsystems:** Position accuracy of drone.
b. **Procedure:** Execute commands to position the SkyHook Module, measure the actual position, and verify that it can achieve accurate positioning within the specified 20-centimeter range.
c. **Data:** hook position under drone (-0.09, -0.06, -0.08)
d. **Results:** 0.07m of variance

T21: Validation Test for R21 (Object Holding Capacity):

Objective: Confirm that the SkyHook Module can hold objects weighing up to 3.5 pounds.

a. **Tested Subsystems:** Ability of the SkyHook to support weight

b. **Procedure:** Attach objects of various weights to the SkyHook Module, initiate the holding command, and ensure that the module can securely hold objects up to 3.5 lbs.
c. **Data:** Maximum object hover weight for effective flying was 3.5 lbs
d. **Results:** The heaviest object the drone can support was 3.5lbs and still operates effectively.

T22: Validation Test for R22 (User Interaction):

Objective: Confirm that the SkyHook Module requires user interaction for clipping the hook.

a. **Tested Subsystems:** User's ability to clip package onto the hook.
b. **Procedure:** Instruct a user to clip the hook onto an object using the SkyHook Module. Observe the interaction and verify that the module necessitates user involvement in the hook-clipping process. Ensure the process takes less than 10 seconds to perform.
c. **Data:** 3.1, 4, 1.5, 3 s
d. **Results:** Between 4 trails, the average clipping process took 2.9 seconds

Science Payload

T23: Validation Test for R23 (Robot Measure From an External Source)

Objective: Confirm that the Science Payload Module can measure and record data during flight.

a. **Tested Subsystems:** Science Payload Module
b. **Procedure:** Request measurements from Science Module during flight, ensure measurements values are consistent with baseline estimates, and that the values vary as expected
c. **Data:** Science Measurement Logs
d. **Results:**

T24: Validation Test for R24 (Integration with Robot Computer)

Objective: Verify that the Science Payload Module seamlessly integrates with the robot computer for data collection.

a. **Tested Subsystems:** Science Payload, Autonomy Subsystem
b. **Procedure:** Command drone to record data from science payload while in flight, ensure logs update at an expected frequency, and relevant numbers are reported to the operator.
c. **Data:** Science Measurement Logs
d. **Results:** Data is logged on Autonomy Computer

Appendix

Table 14: FMEA version 3.1 (current version of FMEA)

Categ ory #	Faul t #	Fail Mode	Effects of Failure	Severi ty	Cause of Fail	Likelihoo d	Detectabili ty	Critical Characteris tic?	RP N
ELE	1	Electrical connections exposed to the elements	Drone has unexpected behavior, especially if connections are exposed to the elements	3	Careless connection manageme nt	1	1	N	3
ELE	2	Insufficient power for drone to sustain operation due to depleted battery	Drone crashes or is unable to perform the correct operation	5	Bad battery swapped in	2	1	Y	10
ELE	3	Module is not properly electrically connected to the correct logical units	The module will not be operational and the system will be unable to complete the required task	3	Damaged or loose electrical connections	2	2	Y	12
ELE	4	Insufficient power for drone to sustain operation due to improper connections	Drone crashes or is unable to perform the correct operation	5	Damaged or loose electrical connections	2	2	Y	20
ELE	5	Communication errors within system due to improper connections	Drone crashes, is unable to perform the correct operation or has	4	Damaged or loose electrical	1	2	Y	8

			unexpected behavior		connections				
ELE	26	Damaged ESCs due to short midflight	Drone crashes, is unable to perform the correct operation or has unexpected behavior	4	Poor power manageme nt	1	1	Y	4
ELE	28	Weather damage to structural components of the frame	Could potentially cause important elements of the system to face reduced functionality	4	Severe/unpr edictable weather	2	1	N	8
MEC	6	Damaged modular node	The modular feature is unable to be used and the system is reduced to a regular drone	4	Physical damage to the modular node	2	1	Y	8
MEC	7	Motor/Propeller malfunctions	Drone crashes and is unable to reach the necessary waypoint	5	Power issue or jammed propellers	2	2	Y	20
MEC	8	Wear and tear of physical connectors throughout the system	Could potentially cause important elements of the system to face reduced functionality	3	Continuous use	3	2	N	18
MEC	9	Structural failure in the frame	System is unable to complete any task	5	Rough usage or accidents while testing system	2	1	Y	10
MEC	10	Custom module does not physically integrate and reduces functionality of the overall system	System is unable to complete user defined task	3	Improper design of the module or lack of clear specification s in	2	2	N	12

					instructions				
MEC	27	Weather damage to structural components of the frame	Could potentially cause important elements of the system to face reduced functionality	4	Severe/unpredictable weather	2	1	N	8
SW	11	Faulty fail-safe algorithms in unexpected cases	System will have an unexpected reaction or simply crash	4	Poor design and testing of software	2	3	N	24
SW	12	Communication errors within system due to incompatible communication protocols	System will have an unexpected reaction or simply crash because necessary information is not relayed	3	Poor design and testing of software	1	3	N	9
SW	13	Security vulnerabilities in the system allowing unexpected actors to access	System will have unexpected behavior because it is no longer being controlled by the user	5	Poor design of security protection	1	2	N	10
SW	14	Insufficient computing to handle all processing	System will be unable to perform some peripheral tasks and be reduced to base functionality	3	Incorrect hardware choice	1	1	N	3
SW	15	Custom module does not integrate software-wise and reduces functionality of the overall system	User's module will not be functional	3	Unclear instructions on software interfacing or poor user design	2	1	Y	6
SW	29	Slower than expected compute power and latency for cameras	Need to descope parts of product, namely cameras	4	Inefficient computation	3	1	Y	12

		and depth map							
SW	30	Unclear module output	The output of the module will be unclear due to non standard communication protocols and improper user coding practices	4	Nonstandard communication with module	2	1	Y	8
SW	32	Insufficient bandwidth between operator and drone	Operator is not able to send and receive information in a timely manner. Commands could be incorrect. Drone could perform undesirable actions	3	Insufficient hardware selection and/or too much data traffic from bad UI design	3	1	Y	9
SW	33	Obstacle avoidance and odometry calculations are below required rates	Insufficient cpu and gpu capabilities for the software to run	5	Insufficient cpu and gpu capabilities for the software to run	3	1	Y	15
SW	34	Drone can not accurate locate in global frame	Map created will be inaccurate. Path planning may fail. May miss target location	3	Inaccurate GPS. Poor start up sequence.	4	1	Y	12
MOD	16	ARM: Issues calibrating the sensors to detect arm position	The ARM module will not function as expected	3	Damage to sensor hardware	2	2	Y	12
MOD	17	ARM: Payload capacity of the arm exceeded	The ARM module will not function as expected and the flight dynamics will suffer	4	Incorrect usage by user	2	1	N	8

Table 15: Risk Matrix version 3.1 (current version of FMEA)

		Likelihood				
		1	2	3	4	5
Severity	1					
	2					
	3	ELE-1, SW-12, SW-14	ELE-3, SW-15, MOD-16, MOD-22	MEC-8, MEC-10, MOD-19, SW-32	SW-34	
	4	ELE-26, MOD-23, USR-24	ELE-28, MEC-6, MEC-27, SW-11, MOD-17, USR-25, USR-26, SW-30	MOD-21, SW-29		MOD-31
	5	SW-13	ELE-2, ELE-5, MEC-7, MEC-9, MOD-18, MOD-20	SW-33		

Table 16: Requirements Traceability Matrix

	ID	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	R17	R18	R19	R20	R21	R22	R23	R24
ID	Count	5	4	4	1	4	11	3	1	1	1	2	1	1	3	1	2	1	1	1	1	1	1	1	1
V1	1	X																							
V2	1		X																						
V3	1			X																					
V4	1				X																				
V5	1					X																			
V6	1						X																		
V7	1							X																	
V8	1								X																
V9	1									X															
V10	1										X														
V11	1											X													
V12	1												X												

V13	1													X											
V14	1														X										
V15	1															X									
V16	1																X								
V17	1																	X							
V18	1																		X						
V19	1																			X					
V20	1																				X				
V21	1																					X			
V22	1																						X		
V23	1																							X	
V24	1																								X
V25	1											X													
V26	1	X																							
V27	1						X																		

V28	3						X	X							X										
V29	1			X																					
V30	1					X																			
V31	1		X																						
V32	1						X																		
V33	1			X																					
V34	1			X																					
V35	2	X					X																		
V36	1	X																							
V37	2	X					X																		
V38	3							X							X		X								
V39	1					X																			
V40	2					X	X																		
V41	2		X				X																		
V42	2		X				X																		

V43	1						X																		
V44	1						X																		

Full Work Breakdown Structure

For easier viewing, a pdf of the structure is available here.

ICON	KEY	SUMMARY	STATUS [CHILDREN STAT...	ASSIGNEE	EPIC LINK	ORIGINAL ESTIMATE
	AIR-11	Frame	22.2% 22.2% 55.6%	Jackson Wood		1w 4d
	AIR-118	Design	0% 25% 75%	Jackson Wood	FRAME	1w
	AIR-63	Select Frame	DONE	Jackson Wood	FRAME	1d
	AIR-62	Order Frame	DONE	Jackson Wood	FRAME	1d
	AIR-77	Design top computer mo...	DONE	Jackson Wood	FRAME	2d
	AIR-101	design stereo camera m...	IN PROGRESS	Jackson Wood	FRAME	1d
	AIR-119	Build	0% 0% 100%	Dustin Moss	FRAME	2d
	AIR-64	Assemble Body	DONE	Dustin Moss	FRAME	1d
	AIR-108	Assemble Top Plate	DONE	Dustin Moss	FRAME	1d
	AIR-120	Test	66.7% 33.3% 0%	Dustin Moss	FRAME	1d
	AIR-166	Check component interf...	IN PROGRESS	Jackson Wood	FRAME	1d
	AIR-184	Payload mass test	TO DO	Jackson Wood	FRAME	1h
	AIR-183	Total Mass Test	TO DO	Jackson Wood	FRAME	1h
	AIR-12	Power	40% 0% 60%	Jackson Wood		1w
	AIR-127	Design	100% 0% 0%	Jackson Wood	POWER	2h
	AIR-158	Order battery supplies	TO DO	Jackson Wood	POWER	1h
	AIR-128	Build	16.7% 0% 83.3%	Jackson Wood	POWER	1d
	AIR-105	Wire Battery Lead to PDB	DONE	Jackson Wood	POWER	1h
	AIR-104	Wire ESCs power to PDB	DONE	Jackson Wood	POWER	5h
	AIR-106	Wire 24-12 to battery lead	DONE	Jackson Wood	POWER	1h
	AIR-186	Wire cube power to batt...	DONE	Jackson Wood	POWER	1h
	AIR-185	Wire 24-5 to battery lead	DONE	Jackson Wood	POWER	1h
	AIR-212	Build battery	TO DO	Jackson Wood	POWER	2d
	AIR-129	Test	66.7% 0% 33.3%	Jackson Wood	POWER	4h
	AIR-79	Test 5v regulator	DONE	Jackson Wood	POWER	1h
	AIR-187	Test system voltage wit...	TO DO	Jackson Wood	POWER	1h
	AIR-188	Test component amp dr...	TO DO	Jackson Wood	POWER	2h

	AIR-14	Air Mobility	55.6% 0% 44.4%	Jacob Hsu		2w
	AIR-133	Design	100% 0% 0%	Jackson Wood	AIR MOBILITY	2d
	AIR-103	Create Mavlink actuator ...	TO DO	Jackson Wood	AIR MOBILITY	1d
	AIR-71	Create Mavlink sensor re...	TO DO	Jackson Wood	AIR MOBILITY	1d
	AIR-205	Configure flight comma...	TO DO	Jackson Wood	AIR MOBILITY	1d
	AIR-134	Build	0% 0% 100%	Jackson Wood	AIR MOBILITY	1d
	AIR-69	Mount Motors	DONE	Jackson Wood	AIR MOBILITY	3h
	AIR-102	Mount ESC	DONE	Jackson Wood	AIR MOBILITY	2h
	AIR-107	Wire ESC to Motor	DONE	Jackson Wood	AIR MOBILITY	1h
	AIR-75	Wire esc Comm Wires	DONE	Jackson Wood	AIR MOBILITY	1h
	AIR-135	Implement	100% 0% 0%	Jacob Hsu	AIR MOBILITY	1w
	AIR-73	Implement Custom Stab...	TO DO	Jacob Hsu	AIR MOBILITY	1w
	AIR-136	Test	100% 0% 0%	Jacob Hsu	AIR MOBILITY	1w 1d
	AIR-240	Test Motor Speed and R...	TO DO	Jacob Hsu	AIR MOBILITY	
	AIR-20	Task Manager	100% 0% 0%	Tahaseen Shaik		1w
	AIR-121	Design	100% 0% 0%	Tahaseen Shaik	TASK MANAGER	1d
	AIR-87	determine user comman...	TO DO	Tahaseen Shaik	TASK MANAGER	1d
	AIR-122	Implement	100% 0% 0%	Tahaseen Shaik	TASK MANAGER	3d
	AIR-88	write ros node	TO DO	Tahaseen Shaik	TASK MANAGER	2d
	AIR-194	Interface with communi...	TO DO	Tahaseen Shaik	TASK MANAGER	1d
	AIR-123	Test	100% 0% 0%	Tahaseen Shaik	TASK MANAGER	1d
	AIR-195	Test all user commands ...	TO DO	Tahaseen Shaik	TASK MANAGER	1d
	AIR-15	Autonomy Computer	0% 0% 100%	Jackson Wood		3d
	AIR-131	Integrate	0% 0% 100%	Jackson Wood	AUTONOMY COMPUTER	3d
	AIR-171	Implement ROS2	DONE	Jackson Wood	AUTONOMY COMPUTER	2d
	AIR-204	Wire to cube	DONE	Jackson Wood	AUTONOMY COMPUTER	1h
	AIR-209	Install and configure SSD	DONE	Jackson Wood	AUTONOMY COMPUTER	1d
	AIR-132	Test	0% 0% 100%	Jackson Wood	AUTONOMY COMPUTER	2h
	AIR-207	Test SSD	DONE	Jackson Wood	AUTONOMY COMPUTER	1h
	AIR-208	Test ROSCORE	DONE	Jackson Wood	AUTONOMY COMPUTER	1h

	AIR-17	Sensing	44.4% 11.1% 44.4%	Jackson Wood		1w 4d
	AIR-137	Design	100% 0% 0%	Jackson Wood	SENSING	1h
	AIR-81	decided on second came...	TO DO	Jackson Wood	SENSING	1h
	AIR-138	Implement	33.3% 0% 66.7%	Jackson Wood	SENSING	1w 2d
	AIR-80	Configure ISaac ros viso...	DONE	Jackson Wood	SENSING	2d
	AIR-196	Configure coordinate fra...	TO DO	Jackson Wood	SENSING	1d
	AIR-177	publish syncronized ima...	DONE	Jackson Wood	SENSING	1d
	AIR-179	read in accelerometer da...	DONE	Jackson Wood	SENSING	1d
	AIR-197	Read GPS from cube	DONE	Jackson Wood	SENSING	1d
	AIR-198	Fuse visodo with gps	TO DO	Jackson Wood	SENSING	1d
	AIR-139	Test	50% 50% 0%	Jackson Wood	SENSING	2d
	AIR-174	Test Single camera visua...	IN PROGRESS	Jackson Wood	SENSING	1d
	AIR-199	Test gps global accuracy	TO DO	Jackson Wood	SENSING	1d
	AIR-16	Communication	100% 0% 0%	Dustin Moss		1w 3d
	AIR-140	Design	100% 0% 0%	Dustin Moss	COMMUNICATION	3d
	AIR-84	Decide on primary user i...	TO DO	Dustin Moss	COMMUNICATION	1d
	AIR-86	Decide on better radio/r...	TO DO	Dustin Moss	COMMUNICATION	1d
	AIR-85	Decide on using wifi	TO DO	Dustin Moss	COMMUNICATION	1d
	AIR-189	Order Components	TO DO	Dustin Moss	COMMUNICATION	1h
	AIR-141	Implement	100% 0% 0%	Dustin Moss	COMMUNICATION	4d
	AIR-192	Configure Software inter...	TO DO	Dustin Moss	COMMUNICATION	2d
	AIR-190	Assemble Hardware	TO DO	Dustin Moss	COMMUNICATION	1d
	AIR-191	Assemble electrical	TO DO	Dustin Moss	COMMUNICATION	1d
	AIR-142	Test	100% 0% 0%	Dustin Moss	COMMUNICATION	1d
	AIR-193	Test all message types a...	TO DO	Dustin Moss	COMMUNICATION	1d

	AIR-19	Navigation	100% 0% 0%	Tahaseen Shaik		1w 1d
	AIR-143	Implement	100% 0% 0%	Tahaseen Shaik	NAVIGATION	1w
	AIR-83	Implement ros2 nav2	TO DO	Tahaseen Shaik	NAVIGATION	2d
	AIR-201	integrate with global an...	TO DO	Tahaseen Shaik	NAVIGATION	1d
	AIR-200	Configure launch file	TO DO	Tahaseen Shaik	NAVIGATION	1d
	AIR-203	Integrate motor comma...	TO DO	Tahaseen Shaik	NAVIGATION	1d
	AIR-144	Test	100% 0% 0%	Tahaseen Shaik	NAVIGATION	1d
	AIR-206	Test navigation planned ...	TO DO	Tahaseen Shaik	NAVIGATION	1d
	AIR-18	Mapping	100% 0% 0%	Jacob Hsu		2w
	AIR-145	Implement	100% 0% 0%	Jacob Hsu	MAPPING	1w
	AIR-159	Install Isaac ROS vslam	TO DO	Jacob Hsu	MAPPING	1d
	AIR-160	Create Vslam launch file	TO DO	Jacob Hsu	MAPPING	2d
	AIR-161	Interface vslam node wit...	TO DO	Jacob Hsu	MAPPING	1d
	AIR-202	Create global map	TO DO	Jacob Hsu	MAPPING	1d
	AIR-146	Test	100% 0% 0%	Jacob Hsu	MAPPING	1w
	AIR-163	Test Vslam quality with ...	TO DO	Jacob Hsu	MAPPING	1w
	AIR-10	Sky Hook Module	75% 25% 0%	Jacob Hsu		1w
	AIR-147	Design	33.3% 66.7% 0%	Jacob Hsu	SKY HOOK MODULE	2d 2h
	AIR-96	cad	IN PROGRESS	Jacob Hsu	SKY HOOK MODULE	2d
	AIR-95	Determine motor	IN PROGRESS	Jacob Hsu	SKY HOOK MODULE	1h
	AIR-97	order parts	TO DO	Jacob Hsu	SKY HOOK MODULE	1h
	AIR-148	Build	100% 0% 0%	Jacob Hsu	SKY HOOK MODULE	2d 3h
	AIR-164	Write controller software	TO DO	Jacob Hsu	SKY HOOK MODULE	2d
	AIR-98	Assemble	TO DO	Jacob Hsu	SKY HOOK MODULE	1h
	AIR-99	Mount and wire to frame	TO DO	Jacob Hsu	SKY HOOK MODULE	2h
	AIR-149	Test	100% 0% 0%	Jacob Hsu	SKY HOOK MODULE	1d
	AIR-210	Test max payload	TO DO	Jacob Hsu	SKY HOOK MODULE	1h
	AIR-211	Test reeling speed	TO DO	Jacob Hsu	SKY HOOK MODULE	1h

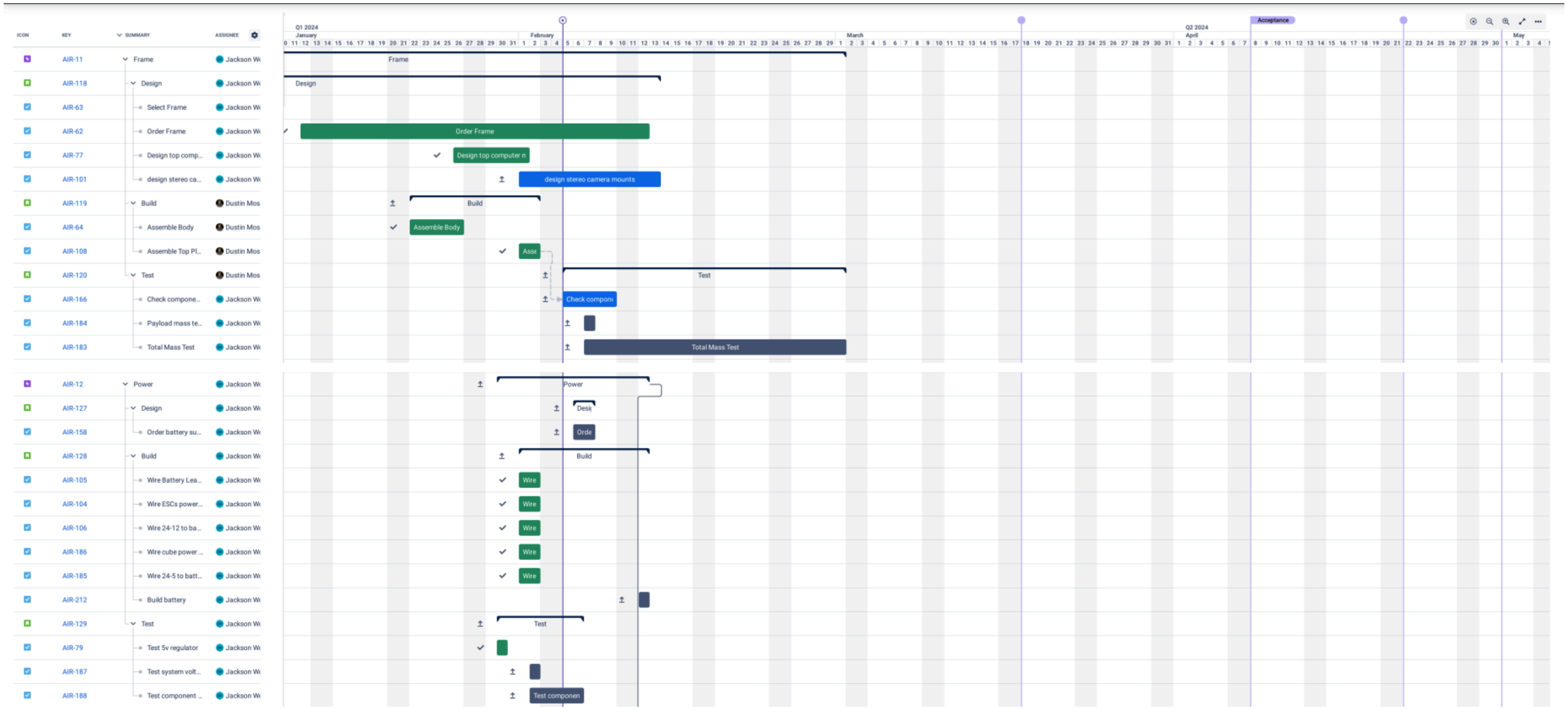
	AIR-7	ARM Module	38.5% 53.8% 7.7%	Jackson Wood		3w
	AIR-150	Design Hardware	0% 80% 20%	Jackson Wood	ARM MODULE	2w 3d
	AIR-50	Design Link 1	IN PROGRESS	Jackson Wood	ARM MODULE	1w
	AIR-51	Design Link 2	IN PROGRESS	Jackson Wood	ARM MODULE	1h
	AIR-52	Design Link 3	IN PROGRESS	Jackson Wood	ARM MODULE	1h
	AIR-53	Design End Effector	IN PROGRESS	Jackson Wood	ARM MODULE	1w
	AIR-110	Design Modular Interface	DONE	Jackson Wood	ARM MODULE	1d
	AIR-151	Design software	50% 50% 0%	Jackson Wood	ARM MODULE	3d
	AIR-55	Inverse Kinematics	TO DO	Dustin Moss	ARM MODULE	2d
	AIR-59	Motor Control Software	IN PROGRESS	Jackson Wood	ARM MODULE	1d
	AIR-152	Build	50% 50% 0%	Jackson Wood	ARM MODULE	2d
	AIR-60	Mount to frame	IN PROGRESS	Jackson Wood	ARM MODULE	1d
	AIR-61	Wire to frame	TO DO	Jackson Wood	ARM MODULE	1d
	AIR-153	Test	75% 25% 0%	Jackson Wood	ARM MODULE	4w
	AIR-109	Test end link and gripper	IN PROGRESS	Jackson Wood	ARM MODULE	1d
	AIR-180	flight with gripper	TO DO	Jackson Wood		2d
	AIR-181	stability + powerdraw wi...	TO DO	Jackson Wood		2d
	AIR-182	user swappability	TO DO	Jackson Wood		1w
	AIR-13	Science Module	100% 0% 0%	Dustin Moss		1w 3d
	AIR-124	Design	100% 0% 0%	Dustin Moss	SCIENCE MODULE	3d
	AIR-89	decided on sensors to use	TO DO	Dustin Moss	SCIENCE MODULE	1d
	AIR-90	cad	TO DO	Dustin Moss	SCIENCE MODULE	2d
	AIR-125	Build	100% 0% 0%	Dustin Moss	SCIENCE MODULE	3d
	AIR-94	data read software	TO DO	Dustin Moss	SCIENCE MODULE	2d
	AIR-93	wire to frame	TO DO	Dustin Moss	SCIENCE MODULE	1h
	AIR-92	Assemble/ Mount to fra...	TO DO	Dustin Moss	SCIENCE MODULE	1d
	AIR-91	3d print	TO DO	Dustin Moss	SCIENCE MODULE	4h
	AIR-126	Test	100% 0% 0%	Dustin Moss	SCIENCE MODULE	2d
	AIR-175	test sensor speeds	TO DO	Dustin Moss		1h
	AIR-176	test weather effects	TO DO	Dustin Moss		1d
	AIR-178	user feedback on physic...	TO DO	Dustin Moss		1d

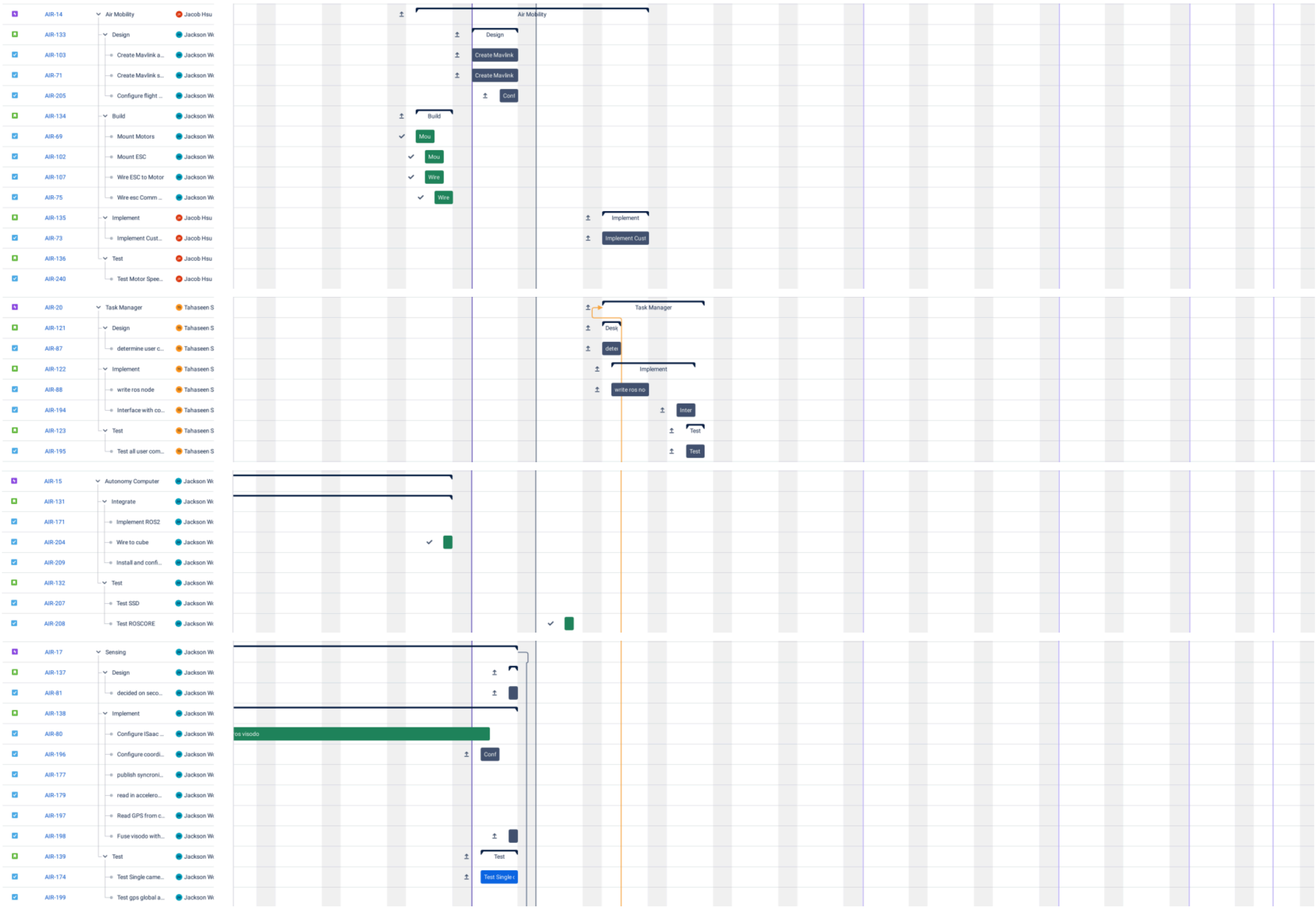
AIR-216	Requirements Validation	100% 0% 0%	Jacob Hsu		3w 3d
AIR-225	Flight	100% 0% 0%	Jacob Hsu	REQUIREMENTS VALIDATI	1w 4d
AIR-167	Test Manual Flight. Defu...	TO DO	Jacob Hsu	REQUIREMENTS VALIDATI	1d
AIR-169	Test manual flight custo...	TO DO	Jacob Hsu	REQUIREMENTS VALIDATI	2d
AIR-168	Test auton flight default ...	TO DO	Jacob Hsu	REQUIREMENTS VALIDATI	1d
AIR-170	test auton flight custom ...	TO DO	Jacob Hsu	REQUIREMENTS VALIDATI	2d
AIR-233	Meaure Loudness from ...	TO DO	Jacob Hsu	REQUIREMENTS VALIDATI	1d
AIR-249	Flight Tests in Nonideal ...	TO DO	Jacob Hsu	REQUIREMENTS VALIDATI	2d
AIR-236	Module Tests	100% 0% 0%	Dustin Moss	REQUIREMENTS VALIDATI	2d
AIR-250	I/O w/ Autonomy CPU	TO DO	Dustin Moss	REQUIREMENTS VALIDATI	1d
AIR-237	Test Max Payloads (sky...	TO DO	Dustin Moss	REQUIREMENTS VALIDATI	1d
AIR-251	Misc	100% 0% 0%	Tahaseen Shaik	REQUIREMENTS VALIDATI	1w 2d
AIR-246	User Interface Tests	TO DO	Tahaseen Shaik	REQUIREMENTS VALIDATI	1w
AIR-231	Verify Local/Federal Co...	TO DO	Tahaseen Shaik	REQUIREMENTS VALIDATI	1d
AIR-248	Modularity + Swapping ...	TO DO	Tahaseen Shaik	REQUIREMENTS VALIDATI	1d

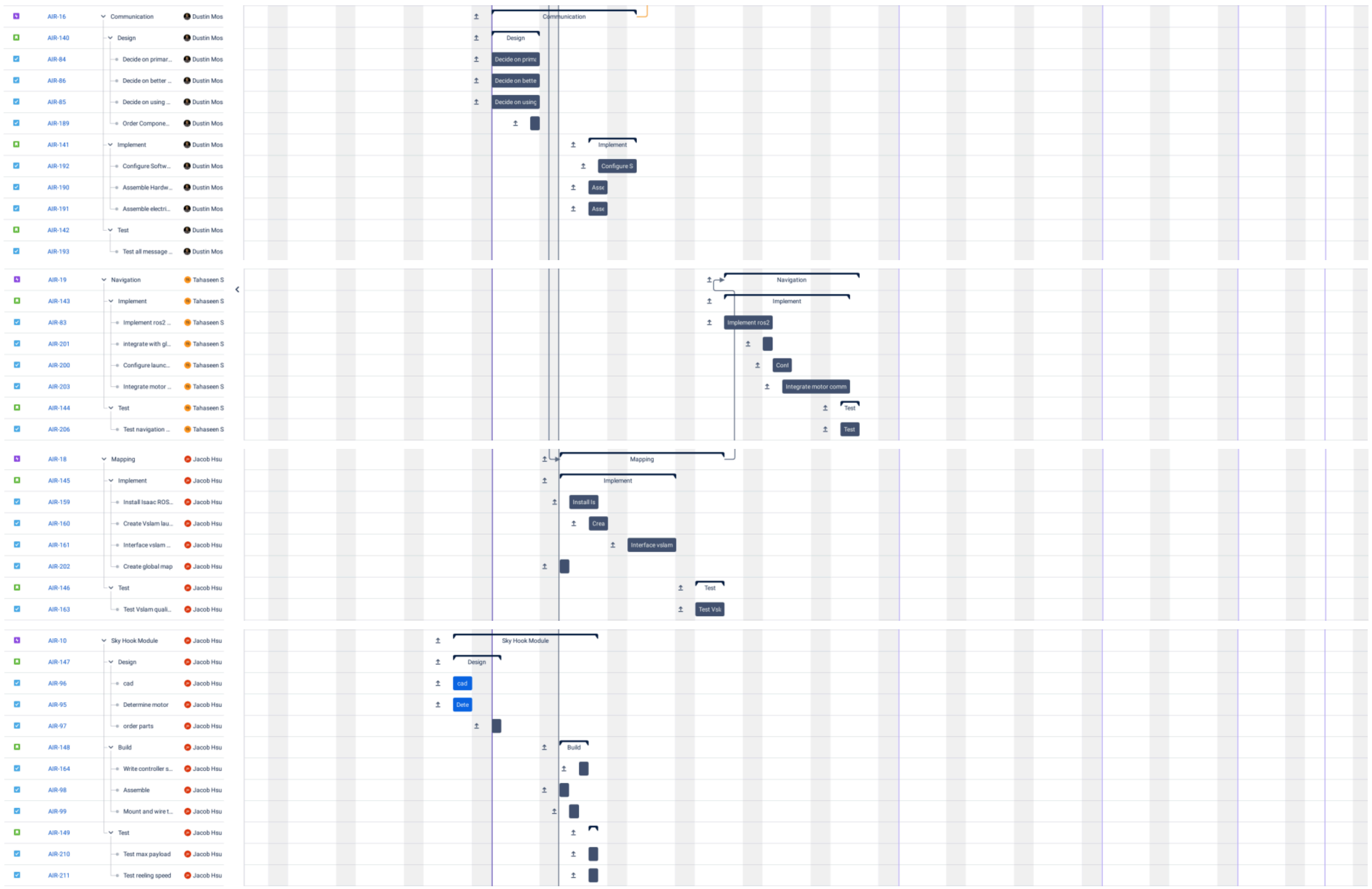
AIR-114	Managment	100% 0% 0%	Jacob Hsu		1w 1d
AIR-115	Documentation	100% 0% 0%	Dustin Moss	MANAGMENT	1d
AIR-116	SDD	TO DO	Dustin Moss		1d
AIR-213	SRD	TO DO	Dustin Moss	MANAGMENT	1d
AIR-214	SAD	TO DO	Dustin Moss	MANAGMENT	1d
AIR-215	Demo Doc	TO DO	Dustin Moss	MANAGMENT	1d
AIR-154	Work	TO DO	Jacob Hsu	MANAGMENT	1d
AIR-155	Finances	TO DO	Jackson Wood	MANAGMENT	1d
AIR-156	Risk	TO DO	Tahaseen Shaik	MANAGMENT	2d
AIR-157	Schedule	TO DO	Jackson Wood	MANAGMENT	1d

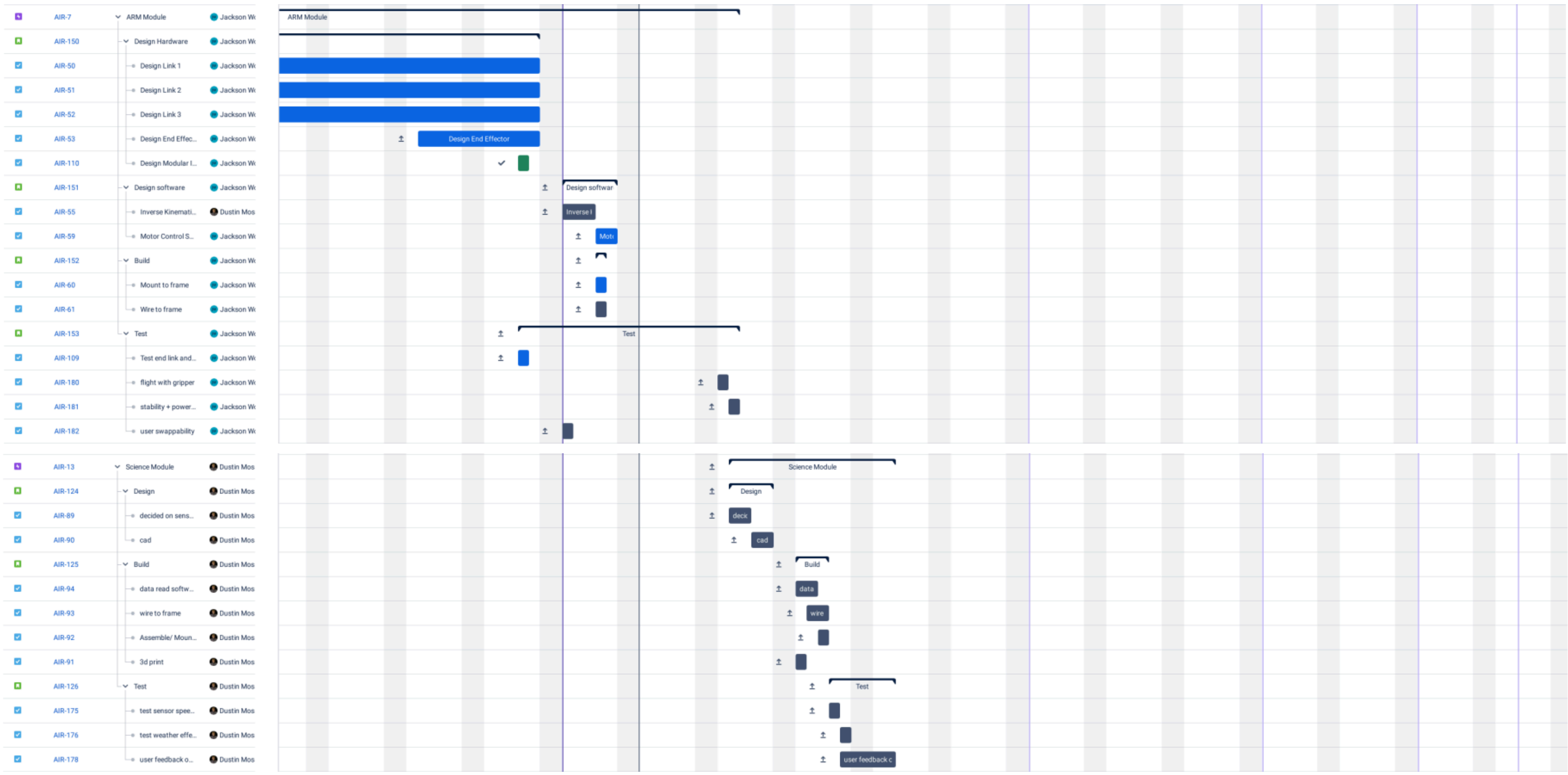
Full Schedule

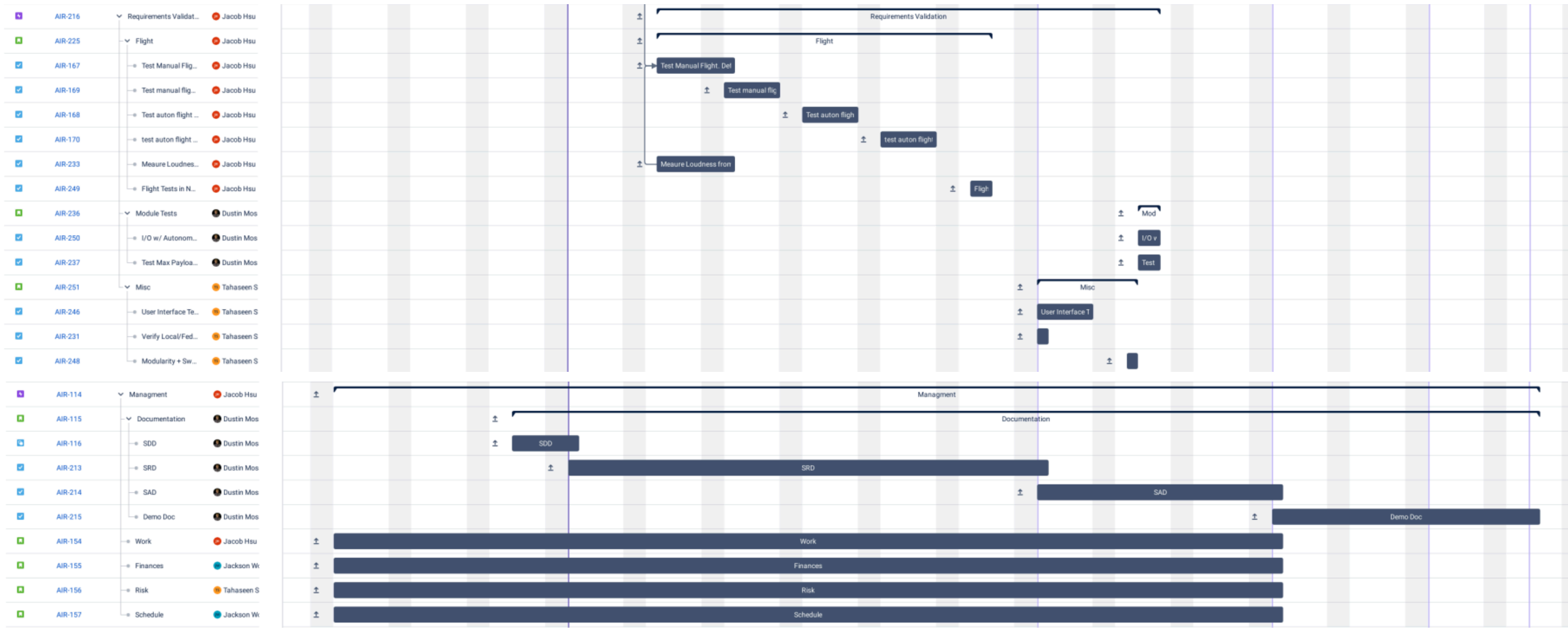
For easier viewing, a PDF of the schedule is available here.




AIR-14 Air Mobility Jacob Hsu
AIR-133 Design Jackson W
AIR-103 Create Mavlink a... Jackson W
AIR-71 Create Mavlink s... Jackson W
AIR-205 Configure flight ... Jackson W
AIR-134 Build Jackson W
AIR-69 Mount Motors Jackson W
AIR-102 Mount ESC Jackson W
AIR-107 Wire ESC to Motor Jackson W
AIR-75 Wire esc Comm ... Jackson W
AIR-135 Implement Jacob Hsu
AIR-73 Implement Cust... Jacob Hsu
AIR-136 Test Jacob Hsu
AIR-240 Test Motor Spee... Jacob Hsu
AIR-20 Task Manager Tahaseen S
AIR-121 Design Tahaseen S
AIR-87 determine user c... Tahaseen S
AIR-122 Implement Tahaseen S
AIR-88 write ros node Tahaseen S
AIR-194 Interface with co... Tahaseen S
AIR-123 Test Tahaseen S
AIR-195 Test all user com... Tahaseen S
AIR-15 Autonomy Computer Jackson W
AIR-131 Integrate Jackson W
AIR-171 Implement ROS2 Jackson W
AIR-204 Wire to cube Jackson W
AIR-209 Install and confi... Jackson W
AIR-132 Test Jackson W
AIR-207 Test SSD Jackson W
AIR-208 Test ROSCORE Jackson W
AIR-17 Sensing Jackson W
AIR-137 Design Jackson W
AIR-81 decided on seco... Jackson W
AIR-138 Implement Jackson W
AIR-80 Configure ISaac ... Jackson W
AIR-196 Configure coordi... Jackson W
AIR-177 publish syncroni... Jackson W
AIR-179 read in accelero... Jackson W
AIR-197 Read GPS from c... Jackson W
AIR-198 Fuse visodo with... Jackson W
AIR-139 Test Jackson W
AIR-174 Test Single came... Jackson W
AIR-199 Test gps global a... Jackson W
Air Mobility
Design
Create Mavlink
Create Mavlink
Conf
Build
Mou
Mou
Wire
Wire
Implement
Implement Cust
Task Manager
Desi
deter
Implement
write ros no
Inter
Test
Test
os visodo
Conf
Test
Test Single c


AIR-16 Communication Dustin Mos
AIR-140 Design Dustin Mos
AIR-84 Decide on primar... Dustin Mos
AIR-86 Decide on better ... Dustin Mos
AIR-85 Decide on using ... Dustin Mos
AIR-189 Order Compone... Dustin Mos
AIR-141 Implement Dustin Mos
AIR-192 Configure Softw... Dustin Mos
AIR-190 Assemble Hardw... Dustin Mos
AIR-191 Assemble electri... Dustin Mos
AIR-142 Test Dustin Mos
AIR-193 Test all message ... Dustin Mos
AIR-19 Navigation Tahaseen S
AIR-143 Implement Tahaseen S
AIR-83 Implement ros2 ... Tahaseen S
AIR-201 integrate with gl... Tahaseen S
AIR-200 Configure launc... Tahaseen S
AIR-203 Integrate motor ... Tahaseen S
AIR-144 Test Tahaseen S
AIR-206 Test navigation ... Tahaseen S
AIR-18 Mapping Jacob Hsu
AIR-145 Implement Jacob Hsu
AIR-159 Install Isaac ROS... Jacob Hsu
AIR-160 Create Vslam lau... Jacob Hsu
AIR-161 Interface vslam ... Jacob Hsu
AIR-202 Create global map Jacob Hsu
AIR-146 Test Jacob Hsu
AIR-163 Test Vslam quali... Jacob Hsu
AIR-10 Sky Hook Module Jacob Hsu
AIR-147 Design Jacob Hsu
AIR-96 cad Jacob Hsu
AIR-95 Determine motor Jacob Hsu
AIR-97 order parts Jacob Hsu
AIR-148 Build Jacob Hsu
AIR-164 Write controller s... Jacob Hsu
AIR-98 Assemble Jacob Hsu
AIR-99 Mount and wire t... Jacob Hsu
AIR-149 Test Jacob Hsu
AIR-210 Test max payload Jacob Hsu
AIR-211 Test reeling speed Jacob Hsu
Communication
Design
Decide on prim
Decide on bette
Decide on using
Implement
Configure S
Navigation
Implement
Implement ros2
Cont
Integrate motor comm
Test
Test
Mapping
Implement
Install Is
Crea
Interface vslam
Test
Test Vsl
Sky Hook Module
Design
cad
Dete
Build


AIR-7 ARM Module Jackson W
AIR-150 Design Hardware Jackson W
AIR-50 Design Link 1 Jackson W
AIR-51 Design Link 2 Jackson W
AIR-52 Design Link 3 Jackson W
AIR-53 Design End Effec... Jackson W
AIR-110 Design Modular I... Jackson W
AIR-151 Design software Jackson W
AIR-55 Inverse Kinemati... Dustin Mos
AIR-59 Motor Control S... Jackson W
AIR-152 Build Jackson W
AIR-60 Mount to frame Jackson W
AIR-61 Wire to frame Jackson W
AIR-153 Test Jackson W
AIR-109 Test end link and... Jackson W
AIR-180 flight with gripper Jackson W
AIR-181 stability + power... Jackson W
AIR-182 user swappability Jackson W
ARM Module
Design End Effector
Design softwar
Inverse I
Moti
Test
AIR-13 Science Module Dustin Mos
AIR-124 Design Dustin Mos
AIR-89 decided on sens... Dustin Mos
AIR-90 cad Dustin Mos
AIR-125 Build Dustin Mos
AIR-94 data read softw... Dustin Mos
AIR-93 wire to frame Dustin Mos
AIR-92 Assemble/ Moun... Dustin Mos
AIR-91 3d print Dustin Mos
AIR-126 Test Dustin Mos
AIR-175 test sensor spee... Dustin Mos
AIR-176 test weather effe... Dustin Mos
AIR-178 user feedback o... Dustin Mos
Science Module
Design
deck
cad
Build
data
wire
Test
user feedback c


AIR-216 Requirements Validat... Jacob Hsu
AIR-225 Flight Jacob Hsu
AIR-167 Test Manual Flig... Jacob Hsu
AIR-169 Test manual flig... Jacob Hsu
AIR-168 Test auton flight ... Jacob Hsu
AIR-170 test auton flight ... Jacob Hsu
AIR-233 Meaure Loudnes... Jacob Hsu
AIR-249 Flight Tests in N... Jacob Hsu
AIR-236 Module Tests Dustin Mos
AIR-250 I/O w/ Autonom... Dustin Mos
AIR-237 Test Max Payloa... Dustin Mos
AIR-251 Misc Tahaseen S
AIR-246 User Interface Te... Tahaseen S
AIR-231 Verify Local/Fed... Tahaseen S
AIR-248 Modularity + Sw... Tahaseen S
Requirements Validation
Flight
Test Manual Flight. Def
Test manual flig
Test auton fligh
test auton flight
Meaure Loudness fron
Fligh
Mod
I/O v
Test
Misc
User Interface T
AIR-114 Managment Jacob Hsu
AIR-115 Documentation Dustin Mos
AIR-116 SDD Dustin Mos
AIR-213 SRD Dustin Mos
AIR-214 SAD Dustin Mos
AIR-215 Demo Doc Dustin Mos
AIR-154 Work Jacob Hsu
AIR-155 Finances Jackson W
AIR-156 Risk Tahaseen S
AIR-157 Schedule Jackson W
Managment
Documentation
SDD
SRD
SAD
Demo Doc
Work
Finances
Risk
Schedule

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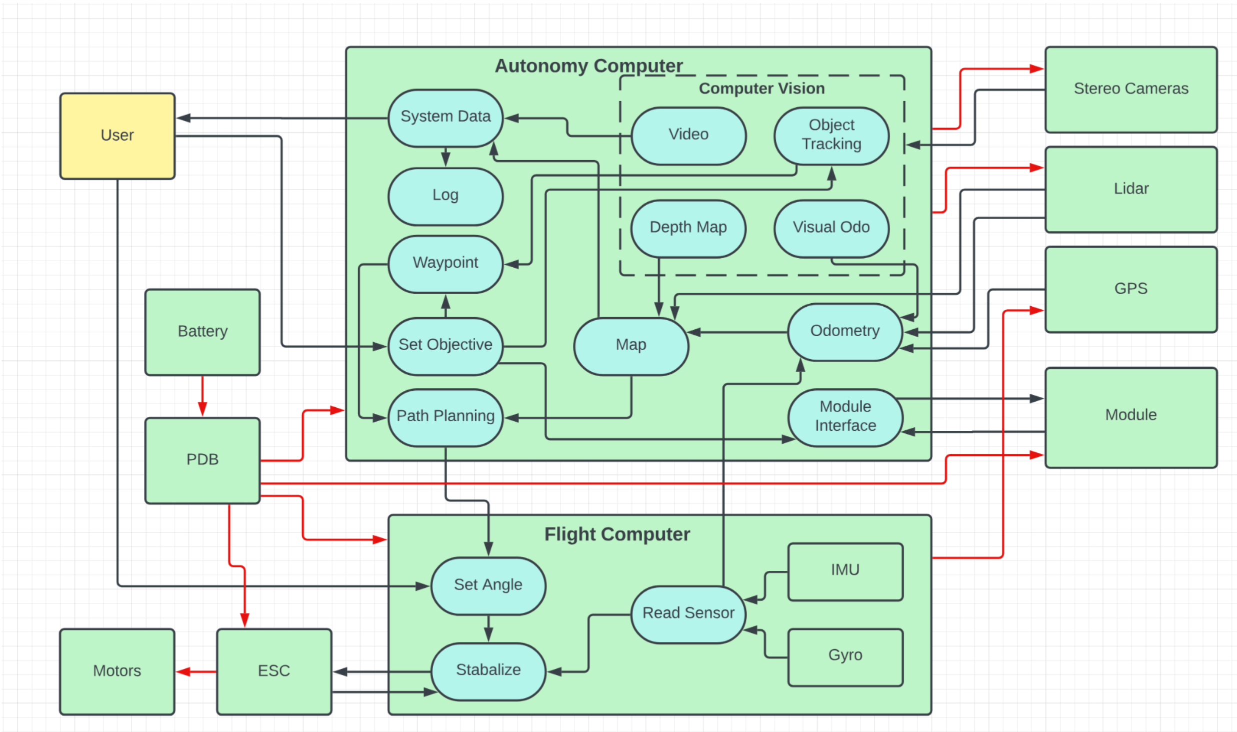
History



Early System Architecture